

STAT 618
Bayesian Statistics
Lecture Notes

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Chapter 1

Introduction to Decision Theory and Bayesian Philosophy

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Welcome to the new semester! And to this course, where we discuss the main concepts and methods of

- Statistical decision theory
- Bayesian modeling
- Bayesian decision making - estimation, hypothesis testing, and forecasting
- Bayesian modeling and computation using R and BUGS
- Minimax approach and game theory

First, let us introduce the main concepts and discuss how this course is different from other Statistics courses that you took, will take, or ... will not take.

1.1 Decisions and Action Spaces

Practically, in all Statistics courses we learn *how to make decisions under uncertainty*. Formally, we are looking for a **decision** δ that belongs to an **action space** $\mathcal{A}$ - a set of all possible decisions that we are allowed to take.

Example 1.1 (ESTIMATION). For example, in *estimation of parameters*, our decision is an estimator $\delta = \hat{\theta}$ of some unknown parameter θ , and the action space $\mathcal{A}$ consists of all possible values of this parameter. If we are estimating the mean of a Normal distribution,

$\mathcal{A} = (-\infty, \infty)$. But if we are estimating its variance, then $\mathcal{A} = (0, \infty)$. To estimate the proportion of voters supporting a certain candidate, we should take $\mathcal{A} = [0, 1]$. $\diamond$

Example 1.2 (TESTING). When we *test a hypothesis*, in the end of the day we have to either accept or reject it. Then the action space consists of just two elements,

$$\mathcal{A} = \left\{ \begin{array}{l} \text{accept the null hypothesis } H_0, \\ \text{reject the hypothesis } H_0 \text{ in favor of the alternative } H_A \end{array} \right\}.$$

$\diamond$

Example 1.3 (INVESTMENTS). There are other kinds of problems. When making an *investment*, we have to decide when to invest, where to invest, and how much. A combination of these will be our decision. $\diamond$

<u>NOTATION</u>	δ	=	decision
	$\mathcal{A}$	=	action space; $\delta \in \mathcal{A}$
	θ	=	parameter
	H_0	=	null hypothesis
	H_A	=	alternative hypothesis

1.2 Frequentist Approach

Now, what kind of information do we use when making these decisions?

Is it a silly question? We know that statisticians collect random samples of **data** and do their statistics based on them. So, their decisions are functions of data,

$$\delta = \delta(\text{data}) = \delta(X_1, \dots, X_n).$$

This is the **frequentist approach**. According to it, **uncertainty** comes from a random sample and its distribution. The only considered distributions, expectations, and variances are distributions, expectations, and variances of data and various statistics computed from data. Population parameters are considered fixed. Statistical procedures are based on the distribution of data given these parameters,

$$f(\mathbf{x} \mid \theta) = f(X_1, \dots, X_n \mid \theta).$$

Properties of these procedures can be stated in terms of **long-run frequencies**. For example (review Chapter 3, if needed),

- an estimator $\hat{\theta}$ is *unbiased* if in a long run of random samples, it averages to the parameter θ ;
- a test has significance level α if in a long run of random samples, $100\alpha\%$ of times the true hypothesis is rejected;
- an interval has confidence level $(1 - \alpha)$ if in a long run of random samples, $(1 - \alpha)100\%$ of obtained confidence intervals contain the parameter, as shown in Figure 3.5, p. 38;
- and so on.

However, there are many situations when using only the data is not sufficient for meaningful decisions.

Example 1.4 (SAME DATA, DIFFERENT CONCLUSIONS (L. SAVAGE)). Imagine three situations:

1. A music expert claims that she can easily distinguish between a line from Mozart and a line from Haydn. She listens to 10 extracts from Mozart's and Haydn's symphonies, and all 10 times she determines the composer correctly.
2. A lady drinks coffee every morning, and she claims she can always tell whether milk was stirred in it before sugar or sugar before milk. Let's test her on 10 cups of coffee. Apparently, all 10 times she answers correctly.
3. A drunken traveler claims that he can always predict results of a coin toss. Well, ten coins are tossed, and all ten times he guesses right.

Notice that the data are identical, 10 out of 10. Let δ_1 , δ_2 , and δ_3 be our decisions about the probability of a correct answer in all three cases. What would they be? Should $\delta_1 = \delta_2 = \delta_3$?

... ..

If not, then we cannot say that $\delta = \delta(\text{data})$, as in the frequentist decision making. $\diamond$

Also, the frequentist concept of *a long run* may inadequately reflect performance of statistical procedures.

Example 1.5 (FREQUENTIST APPROACH: MISLEADING ASSESSMENT OF ACCURACY; $\approx$ BERGER, #1.14). Let X be a Binomial variable with $n = 100$ and unknown parameter θ . For example, θ is a proportion of voters who support our candidate, and we'd like to estimate it based on a quick survey of randomly selected $n = 100$ voters. X participants voted for that candidate.

Suppose we are testing $H_0 : \theta = 1/3$ vs $H_A : \theta = 2/3$. If $X < 50$, we'll accept H_0 . If $X > 50$, reject H_0 in favor of H_A . However, if $X = 50$, we randomly choose between H_0 and H_A by tossing a fair coin.

- (a) Calculate the probabilities of Type I and Type II errors for this test (some review of hypothesis testing may be needed for this).

... ..

(Review Section 3.3, if needed.)

- (b) If $X = 50$ is actually observed, what is the *intuitive* (conditional) probability of actually making an error in use of the test?

... ..

- (c) For the observed value of $X = 50$, calculate the p-value (a frequentist approach).

... ..

Conclusion: the “long-run” frequentist probabilities of Type I and Type II errors as well as the p-value are entirely misleading when $X = 50$. We do not necessarily need a concept of a “long run” because we know the actually observed value of X (adding to an already hot continuing discussion about p-values and significance testing). $\diamond$

Example 1.6 (FREQUENTIST APPROACH: LONG RUN IS MISLEADING;). Suppose we need to run some test. Two labs can conduct the needed experiment (say, a clinical trial) and collect data. Say, one lab at AU and another at UMD. At AU, devices are accurate, and the probability of Type I error is only 2% whereas it is 10% at UMD. The lab is chosen at random, and it happened to be the AU lab. Next, the experiment was conducted, the data were collected, and the null hypothesis was rejected.

So, when we report results of this test, what significance level must we state, to be honest?

A **frequentist** will measure all probabilities over a long run, and in a long run, 50% of the time the test was done at UMD. Then, a possibility that a test could have been done at UMD will be included.

A **Bayesian**, however, will argue that we **know** that the test was conducted at AU, and we **know** that the used devices were AU-accurate, so there is no reason to include UMD-accuracy into the report. $\diamond$

Example 1.7 (FREQUENTIST APPROACH: PROBLEMS WITH LARGE SAMPLES AND SMALL SAMPLES; BERGER, #1.10). Suppose θ is the percentage change in the yield of a process, under the effect of a very expensive new treatment. To obtain information about θ , we can observe i.i.d. Normal(θ , 2500) observations, $X_1, \dots, X_n$.

- (a) If $n = 25,000,000$ and $\bar{X} = 0.02$, show that there is statistically significant evidence at the $\alpha = 0.05$ level (one-tailed) that $\theta > 0$. Would it be wise to adopt the new treatment because of the strong evidence that it increases the yield?

... ..

- (b) If $n = 4$ and $\bar{X} = 30$, show that there is not statistically significant evidence at the $\alpha = 0.1$ level that $\theta > 0$. Does it imply that it would be wise not to adopt the treatment?

... ..

◇

Often the null hypothesis $\theta = \theta_0$ is not really understood literally as the exact equality because that would have been too much of a coincidence. Then, in the previous example, a practically non-significant 0.02% change in yield could be a part of H_0 . The main reason of stating H_0 as an equality is to have only one null distribution and to make sure the probability of Type I error α is given as just one number.

Example 1.8 (MISLEADING RELIABILITY; SHOULDN'T WE CONDITION ON THE KNOWN RESULT? MANY SIMILAR EXAMPLES IN CHRISTENSEN, SEC. 2.1). There exists a test for a certain viral infection. It is 95% reliable for infected patients and 99% reliable for the healthy ones. That is, if a patient has the virus (event V), the test shows that (event S) with probability $P\{S|V\} = 0.95$, and if the patient does not have the virus, the test shows that with probability $P\{\bar{S}|\bar{V}\} = 0.99$.

Consider a patient whose test result is positive (i.e., the test shows that the patient has the virus). Knowing that sometimes the test is wrong, naturally, the patient is eager to know the probability that he or she indeed has the virus. However, [this conditional probability, \$P\{V|S\}\$, is not stated among the given characteristics of this test.](#)

Suppose that 2% of all the patients are infected with the virus, $P\{V\} = 0.02$. If the test shows positive results, the (conditional) probability that a patient has the virus equals

$$P\{V|S\} = \frac{P\{S|V\}P\{V\}}{P\{S|V\}P\{V\} + P\{S|\bar{V}\}P\{\bar{V}\}} = \frac{(0.95)(0.02)}{(0.95)(0.02) + (1 - 0.99)(1 - 0.02)} = \underline{0.66}.$$

Here, characteristics of the test seem very strong (probabilities 0.95 and 0.99 of a correct decision), but in fact, the probability of interest is only 0.66. This is a *conditional probability* of a correct decision given the data, that the test result is positive. ◇

Résumé: Frequentist statistical decision making takes into account only the uncertainty of data. Statistical decisions are based on data only, and their performance is evaluated in terms of a “long run”. However, there are situations where such approach is deficient, unnatural, or misleading in various ways.

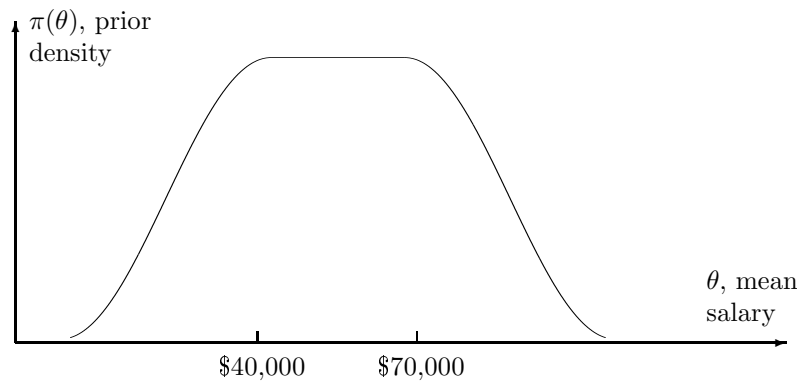


FIGURE 1.1: Our prior distribution for the average starting salary.

1.3 Bayesian approach

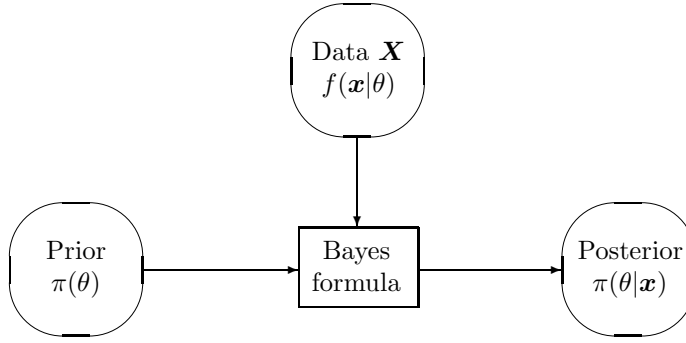
However, there is another approach - the **Bayesian approach**. According to it, uncertainty is attributed not only to the data but also to the unknown parameter θ . Some values of θ are more **likely** than others. Then, as long as we talk about the **likelihood**, we can define a whole distribution of values of θ . We call it a **prior distribution**, and it reflects our ideas, beliefs, and past experiences about the parameter before we collect and use the data.

Example 1.9 (SALARIES). What do you think is the average starting annual salary of an AU graduate? Is it \$20,000 per year? Unlikely, that's too low. Perhaps, \$200,000 per year? No, that's too high for a fresh graduate. Between \$40,000 and \$70,000 sounds like a reasonable range. We can certainly collect data on 100 recent graduates, compute their average salary and use it as an estimate, but before that, we already have our beliefs on what the mean salary may be. We can express it as some distribution with the most likely range between \$40,000 and \$70,000 (Figure 1.1). $\diamond$

One benefit of this approach is that **we no longer have to explain our results in terms of a “long run”**. Often we collect just one sample for our analysis and don't experience any long runs of samples. Instead, with the Bayesian approach, **we can state the result in terms of the distribution of parameter θ** . For example, we can clearly state the probability for a parameter to belong to a certain interval, or the probability that the hypothesis is true. This would have been impossible under the frequentist approach.

Another benefit is that we can use both pieces of information, the data and the prior, to make better decisions. **In Bayesian statistics, decisions are**

$$\delta = \delta(\text{data, prior distribution}).$$

FIGURE 1.2: Two sources of information about the parameter θ .

1.3.1 Prior and posterior

Now we have two sources of information to use in our Bayesian inference:

1. collected and observed data;
2. prior distribution of the parameter.

These two pieces are combined via the **Bayes formula** (p. 17).

Prior to the experiment, our knowledge about the parameter θ is expressed in terms of the **prior distribution** (prior pmf or pdf)

$$\pi(\theta).$$

The observed sample of data $\mathbf{X} = (X_1, \dots, X_n)$ has distribution (pmf or pdf)

$$f(\mathbf{x}|\theta) = f(x_1, \dots, x_n|\theta).$$

This distribution is conditional on θ . That is, different values of the parameter θ generate different distributions of data, and thus, conditional probabilities about $\mathbf{X}$ generally depend on the condition, θ .

Observed data add information about the parameter. The updated knowledge about θ can be expressed as the **posterior distribution**.

**Posterior
distribution**

$$\pi(\theta|\mathbf{x}) = \pi(\theta|\mathbf{X} = \mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{m(\mathbf{x})}.$$

(1.1)

Posterior distribution of the parameter θ is now conditioned on data $\mathbf{X} = \mathbf{x}$. Naturally, conditional distributions $f(\mathbf{x}|\theta)$ and $\pi(\theta|\mathbf{x})$ are related via the Bayes Rule (2.4).

According to the Bayes Rule, the denominator of (1.1), $m(\mathbf{x})$, represents the unconditional distribution of data $\mathbf{X}$. This is the **marginal distribution** (pmf or pdf) of the sample $\mathbf{X}$. Being unconditional means that it is constant for different values of the parameter θ . It can be computed by the **Law of Total Probability** (see p. 17) or its continuous-case version.

**Marginal
distribution
of data**

$$m(\mathbf{x}) = \sum_{\theta} f(\mathbf{x}|\theta)\pi(\theta)$$

for discrete prior distributions π

$$m(\mathbf{x}) = \int_{\theta} f(\mathbf{x}|\theta)\pi(\theta)d\theta$$

for continuous prior distributions π

(1.2)

Example 1.10 (QUALITY INSPECTION). A manufacturer claims that the shipment contains only 5% of defective items, but the inspector feels that in fact it is 10%. We have to decide whether to accept or to reject the shipment based on θ , the proportion of defective parts.

Before we see the real data, let's assign a 50-50 chance to both suggested values of θ , i.e.,

$$\pi(0.05) = \pi(0.10) = 0.5.$$

A random sample of 20 parts has 3 defective ones. Calculate the posterior distribution of θ .

Solution. Apply the Bayes formula (1.1). Given θ , the distribution of the number of defective parts X is Binomial($n = 20, \theta$). For $x = 3$, from Table A2, we have

$$f(x | \theta = 0.05) = F(3 | \theta = 0.05) - F(2 | \theta = 0.05) = 0.9841 - 0.9245 = 0.0596$$

and

$$f(x | \theta = 0.10) = F(3 | \theta = 0.10) - F(2 | \theta = 0.10) = 0.8670 - 0.6769 = 0.1901.$$

The **marginal distribution** of X (for $x = 3$) is

$$\begin{aligned} m(3) &= f(x | 0.05)\pi(0.05) + f(x | 0.10)\pi(0.10) \\ &= (0.0596)(0.5) + (0.1901)(0.5) = 0.12485. \end{aligned}$$

Posterior probabilities of $\theta = 0.05$ and $\theta = 0.10$ are now computed as

$$\pi(0.05 | X = 3) = \frac{f(x | 0.05)\pi(0.05)}{m(3)} = \frac{(0.0596)(0.5)}{0.1248} = 0.2387;$$

$$\pi(0.10 | X = 3) = \frac{f(x | 0.10)\pi(0.10)}{m(3)} = \frac{(0.1901)(0.5)}{0.1248} = 0.7613.$$

Conclusion. In the beginning, we had no preference between the two suggested values of θ . Then we observed a rather high proportion of defective parts, $3/20=15\%$. Taking this into account, $\theta = 0.10$ is now about three times as likely than $\theta = 0.05$. $\diamond$

<u>NOTATION</u>	$\pi(\theta)$	=	prior distribution
	$\pi(\theta \mathbf{x})$	=	posterior distribution
	$f(x \theta)$	=	distribution of data (model)
	$m(\mathbf{x})$	=	marginal distribution of data
	$\mathbf{X}$	=	$(X_1, \dots, X_n)$, sample of data
	$\mathbf{x}$	=	$(x_1, \dots, x_n)$, observed values of $X_1, \dots, X_n$.

Résumé: Bayesian approach presumes a prior distribution of the unknown parameter. Adding the observed data, the Bayes Theorem converts the prior distribution into the posterior which summarizes all we know about the parameter after seeing the data. Bayesian decisions are based on this posterior, and thus, they utilize both the data and the prior.

1.4 Loss function

Besides the data and the prior distribution, is there any other information that can appear useful in our decision making?

How about anticipating possible consequences of making an error? We know that under uncertainty, there is always a chance of making inaccurate decisions.

The third component of the *Bayesian Decision Theory* is the **loss function**

$$L(\theta, \delta) : \Theta \times \mathcal{A} \longrightarrow \mathbf{R} \text{ or } [0, \infty)$$

which tells the amount of “penalty” for taking decision $\delta \in \mathcal{A}$ when the actual parameter equals $\theta \in \Theta$. This penalty may be 0 if the decision is *perfect*, for example, if we accept the true null hypothesis or estimate a parameter θ with no error.

Example 1.11 (LOSS FUNCTION MATTERS). Let’s anticipate consequences of:

- underestimation or overestimation of the value of some house;
- underestimation or overestimation of the expected lifetime of some device;
- underestimation or overestimation of some child’s IQ;
- underestimation or overestimation of the *pure premium*, which is the amount an insurance company should expect to pay to cover its customer’s claims.

... ..

(By the way, insurance is a very popular application of the Bayesian approach. Many customers have little data, but there is rich prior information from so-called risk groups.) $\diamond$

Example 1.12 (LOSS FUNCTION IN TESTING HYPOTHESES). Hypothesis testing is a rather simple situation. There are two possibilities, either the null H_0 is true or the alternative H_A is true. Also, there are two possible decisions - either accept H_0 or reject it:

Testing hypotheses		Action space $\mathcal{A}$	
		Reject H_0	Accept H_0
Parameter space Θ	H_0 is true	Type I error	correct
	H_0 is false	correct	Type II error

What would be reasonable losses for these 4 situations? Naturally, no penalty for correct decisions. The other two situations are dramatically different.

In [clinical trials for efficacy](#), a Type I error results in marketing a useless treatment, and a Type II error results in stopping an efficient medication.

In [clinical trials for toxicity](#), Type I error results in labeling a toxic treatment as safe, and Type II error is not noticing a real danger (as happened with Merck and its product Vioxx).

In [criminal trials](#), Type I error means convicting an innocent defendant, and Type II error means acquitting a real criminal.

As we see, consequences are entirely different, and so are the losses.

Loss function $L(\theta, \delta)$		δ	
		Reject H_0	Accept H_0
θ	H_0 is true	w_0	0
	H_0 is false	0	w_1

$\diamond$

1.4.1 Risks and optimal statistical decisions

Equipped with the loss function, we are looking for **optimal statistical decisions**. Those that [minimize the loss](#). Minimize in what sense? The loss function $L(\theta, \delta) = L(\theta, \delta(\mathbf{X}))$ has uncertainty - unknown parameter θ and data $\mathbf{X} = (X_1, \dots, X_n)$.

DEFINITION 1.1

Risk (frequentist risk) is the expected loss over all possible samples, given a fixed parameter,

$$R(\theta, \delta) = \mathbf{E}_{\theta}^{\mathbf{X}} L(\theta, \delta(\mathbf{X}))$$

NOTATION

$\mathbf{E}_{\theta}^{\mathbf{X}} L(\theta, \delta(\mathbf{X})) = \sum_{\mathbf{x}} L(\theta, \delta(\mathbf{x}))P(\mathbf{x})$ or $\int_{-\infty}^{\infty} L(\theta, \delta(\mathbf{x}))f(\mathbf{x})d\mathbf{x}$
 the expected value for a fixed θ (subscript),
 taken over all possible $\mathbf{x}$ (superscript)

$\mathbf{X} = (X_1, \dots, X_n)$, random sample
 $\mathbf{x} = (x_1, \dots, x_n)$, a value of such sample

With respect to the **risk**, the optimal decisions are still not clear because $R(\theta, \delta)$ depends on the unknown parameter θ . However, it is clear which rules we should *not* use...

DEFINITION 1.2

Decision δ_1 is **R-better** than decision δ_2 if

$$\begin{aligned} R(\theta, \delta_1) &\leq R(\theta, \delta_2) \text{ for all } \theta \\ R(\theta, \delta_1) &< R(\theta, \delta_2) \text{ for some } \theta \end{aligned}$$

Decision δ is **inadmissible** if there exists a decision R-better than δ .

Decision δ is **admissible** if it is not inadmissible.

Decision δ is **minimax** if it minimizes $\sup_{\theta} R(\theta, \delta)$, the worst possible risk over all $\theta \in \Theta$. That is,

$$\sup_{\theta} R(\theta, \delta_{\text{minimax}}) = \inf_{\delta \in \mathcal{A}} \sup_{\theta} R(\theta, \delta)$$

(its max risk is min-max risk).

Minimax decisions are conservative because they protect against the worst situation where the risk is maximized. They are the best decisions in a game against an *intelligent opponent* who will always like to give you the worst case. In **statistical games**, the players know that they are acting against intelligent opponents, and therefore, they devise minimax strategies. We'll learn them in **game theory**.

So far, we have not used the **prior distribution**.

DEFINITION 1.3

Bayes risk is the expected frequentist risk

$$r(\pi, \delta) = \mathbf{E}^{\pi(\theta)} R(\theta, \delta) = \mathbf{E}_{\theta}^{\mathbf{X}} L(\theta, \delta)$$

where the expectation is taken over the prior distribution $\pi(\theta)$. So it is the loss function averaged over all possible samples of data and all possible parameters.

As we have already seen, the **Bayes decisions** are based on the posterior distribution, that is, conditioned on the known data $\mathbf{X}$.

DEFINITION 1.4

Posterior risk is the expected loss, where the expectation is taken over the posterior distribution of parameter θ ,

$$\rho(\pi, \delta | \mathbf{X}) = \mathbf{E}_{\mathbf{X}}^{\pi(\theta|\mathbf{X})} L(\theta, \delta) = \mathbf{E} \{L(\theta, \delta) | \mathbf{X}\}.$$

So, the posterior risk is the loss function averaged over parameters θ , given known data $\mathbf{X}$.

Bayes decision rules minimize the **Bayes risk** and, as we'll see pretty soon, they also minimize the **posterior risk**. That is,

$$\rho(\pi, \delta_{\text{Bayes}} | \mathbf{X}) = \inf_{\delta \in \mathcal{A}} \rho(\pi, \delta | \mathbf{X})$$

for every sample $\mathbf{X}$, and

$$r(\pi, \delta) = \inf_{\delta \in \mathcal{A}} r(\pi, \delta).$$

Résumé: Simple frequentist statistics are based on just the observed data. Decision theory takes into account the data and the loss function. Bayesian statistics is based on the data and the prior distribution. Thus, the Bayes decision rules are based on three components, the data, the prior, and the loss. Bayes rules minimize the posterior risk, given the observed data. Minimax rules minimize the largest or the worst possible risk.

Let's now have a little practice with the new concepts.

Example 1.13. A usual situation. There is an exam tomorrow, but a student does not have enough time to prepare. In fact, he has only 70% of the time he needs to be fully prepared. What should he do then? For example, he can study 70% of all topics. Or 70% of each topic.

Suppose for simplicity that the exam will consist of one problem covering one topic. Describe **minimax**, **Bayes**, **admissible**, and **inadmissible** rules for this student. $\diamond$

Example 1.14 (BERGER, #1.3). A company has to decide whether to accept or reject a lot of incoming parts (label these decisions as δ_1 and δ_2 , respectively). The lots are of three types: θ_1 (very good), θ_2 (acceptable), and θ_3 (bad). The loss $L(\theta, \delta)$ incurred in making

the decision is given in the following table.

	δ_1	δ_2
θ_1	0	3
θ_2	1	2
θ_3	3	0

The prior belief is that $\pi(\theta_1) = \pi(\theta_2) = \pi(\theta_3) = 1/3$.

- (a) What is the **Bayes** decision?
- (b) What is the **minimax** decision?

... ..

◇

Example 1.15. We observed a Normal random variable X with unknown mean θ and known variance σ^2 . Estimate θ under the **squared-error loss function**

$$L(\theta, \delta) = (\delta - \theta)^2.$$

Consider the action space of linear functions $\mathcal{A} = \{\delta_c(X) = cX \text{ for } c \in \mathbf{R}\}$.

- (a) Graph the **risks** of δ_0 , δ_1 , and $\delta_{1/2}$.
- (b) Show that δ_c is **inadmissible** if $c < 0$ or $c > 1$.
- (c) Find the **minimax decision**.

... ..

◇

Example 1.16 (BERGER, #1.6). The owner of a ski shop must order skis for the upcoming season. Orders must be placed in quantities of 25 pairs of skis. The cost per pair of skis is \$50 if 25 pairs are ordered, \$45 if 50 are ordered, and \$40 if 75 are ordered. The skis will be sold at \$75 per pair. Any skis left over at the end of the year can be sold (for sure) at \$25 per pair. If the owner runs out of skis during the season, he will suffer a loss of “goodwill” among unsatisfied customers. He rates this loss as \$5 per unsatisfied customer. For simplicity, the owner feels that demand for the skis will be 30, 40, 50, or 60 pairs of skis, with probabilities 0.2, 0.4, 0.2, and 0.2, respectively.

- (a) Describe $\mathcal{A}$, Θ , the **loss matrix**, and the **prior distribution**.
- (b) Which decisions are **admissible**?
- (c) What is the **Bayes** decision?
- (d) What is the **minimax** decision?

... ..

◇

References: Berger 1.1-1.3, 1.5, 1.6.1-1.6.3; Christensen 1, 2.1.

Chapter 2

Review of Some Probability

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Here we review some basic probability concepts and statistical methods that you have probably seen in other courses.

2.1 Rules of Probability

Probability of any event is a number between 0 and 1. The empty event has probability 0, and the sample space has probability 1. There are rules for computing probabilities of unions, intersections, and complements. For a union of disjoint events, probabilities are added. For an intersection of independent events, probabilities are multiplied.

**Probability
of a union**

$$P\{A \cup B\} = P\{A\} + P\{B\} - P\{A \cap B\}$$

For mutually exclusive events,

$$P\{A \cup B\} = P\{A\} + P\{B\}$$

(2.1)

For 3 events,

$$\begin{aligned} P\{A \cup B \cup C\} &= P\{A\} + P\{B\} + P\{C\} - P\{A \cap B\} - P\{A \cap C\} \\ &\quad - P\{B \cap C\} + P\{A \cap B \cap C\}. \end{aligned}$$

Complement rule

$$P\{\overline{A}\} = 1 - P\{A\}$$

**Independent
events**

$$P\{E_1 \cap \dots \cap E_n\} = P\{E_1\} \cdot \dots \cdot P\{E_n\}$$

Given occurrence of event B , one can compute conditional probability of event A . Unconditional probability of A can be computed from its conditional probabilities by the Law of Total Probability. The Bayes Rule, often used in testing and diagnostics, relates conditional probabilities of A given B and of B given A .

DEFINITION 2.1

Conditional probability of event A given event B is the probability that A occurs when B is *known to occur*.

NOTATION $\parallel P\{A|B\} =$ conditional probability of A given B $\parallel$

**Conditional
probability**

$$P\{A|B\} = \frac{P\{A \cap B\}}{P\{B\}} \quad (2.2)$$

Rewriting (2.2) in a different way, we obtain the general formula for the probability of intersection.

**Intersection,
general case**

$$P\{A \cap B\} = P\{B\} P\{A|B\} \quad (2.3)$$

Independence

Now we can give an intuitively very clear definition of *independence*.

DEFINITION 2.2

Events A and B are **independent** if occurrence of B does not affect the probability of A , i.e.,

$$P\{A|B\} = P\{A\}.$$

Substituting this into (2.3) yields

$$P\{A \cap B\} = P\{A\} P\{B\}.$$

This is our old formula for independent events.

**Bayes
Rule**

$$P\{B|A\} = \frac{P\{A|B\}P\{B\}}{P\{A\}} \quad (2.4)$$

**Law of Total
Probability**

$$P\{A\} = \sum_{j=1}^k P\{A|B_j\}P\{B_j\}$$

In case of two events ($k = 2$),

$$P\{A\} = P\{A|B\}P\{B\} + P\{A|\overline{B}\}P\{\overline{B}\} \quad (2.5)$$

Together with the Bayes Rule, it makes the following popular formula

**Bayes Rule
for two events**

$$P\{B|A\} = \frac{P\{A|B\}P\{B\}}{P\{A|B\}P\{B\} + P\{A|\overline{B}\}P\{\overline{B}\}}$$

2.2 Discrete random variables

Discrete random variables can take a finite or countable number of isolated values with different probabilities. Collection of all such probabilities is a distribution, which describes the behavior of a random variable. Random vectors are sets of random variables; their behavior is described by the joint distribution. Marginal probabilities can be computed from the joint distribution by the Addition Rule.

DEFINITION 2.3

A **random variable** is a function of an outcome,

$$X = f(\omega).$$

In other words, it is a quantity that depends on chance.

DEFINITION 2.4

Collection of all the probabilities related to X is the **distribution** of X . The function

$$P(x) = \mathbf{P}\{X = x\}$$

is the **probability mass function**, or **pmf**. The **cumulative distribution function**, or **cdf** is defined as

$$F(x) = \mathbf{P}\{X \leq x\} = \sum_{y \leq x} \mathbf{P}(y). \quad (2.6)$$

The set of possible values of X is called the **support** of the distribution F .

Discrete random variables take values in a finite or countable set. In particular, it means that their values can be listed, or arranged in a sequence. **Continuous random variables** assume a whole interval of values. Their distribution is described by a probability density function.

2.2.1 Joint distribution and marginal distributions

DEFINITION 2.5

If X and Y are random variables, then the pair (X, Y) is a **random vector**. Its distribution is called the **joint distribution** of X and Y . Individual distributions of X and Y are then called the **marginal distributions**.

The **joint distribution** of a vector is a collection of probabilities for a vector (X, Y) to take a value (x, y) . The **joint probability mass function** of X and Y is the probability of intersection,

$$P(x, y) = \mathbf{P}\{(X, Y) = (x, y)\} = \mathbf{P}\{X = x \cap Y = y\}.$$

Addition Rule

$$\begin{aligned} P_X(x) &= \mathbf{P}\{X = x\} = \sum_y P_{(X,Y)}(x, y) \\ P_Y(y) &= \mathbf{P}\{Y = y\} = \sum_x P_{(X,Y)}(x, y) \end{aligned} \quad (2.7)$$

That is, to get the marginal pmf of one variable, we add the joint probabilities over all values of the other variable.

DEFINITION 2.6

Random variables X and Y are **independent** if

$$P_{(X,Y)}(x,y) = P_X(x)P_Y(y)$$

for all values of x and y . This means, events $\{X = x\}$ and $\{Y = y\}$ are independent for all x and y ; in other words, variables X and Y take their values independently of each other.

2.2.2 Expectation, variance, and covariance

The average value of a random variable is its expectation. Variability around the expectation is measured by the variance and the standard deviation. Covariance and correlation measure association of two random variables.

DEFINITION 2.7

Expectation or **expected value** of a random variable X is its mean, the average value.

**Expectation,
discrete case**

$$\mu = \mathbf{E}(X) = \sum_x xP(x) \quad (2.8)$$

Expectation of a function $Y = g(X)$ is computed by a similar formula,

$$\mathbf{E}\{g(X)\} = \sum_x g(x)P(x). \quad (2.9)$$

**Properties
of
expectations**

$$\begin{aligned} \mathbf{E}(aX + bY + c) &= a\mathbf{E}(X) + b\mathbf{E}(Y) + c \\ \text{In particular,} \\ \mathbf{E}(X + Y) &= \mathbf{E}(X) + \mathbf{E}(Y) \\ \mathbf{E}(aX) &= a\mathbf{E}(X) \\ \mathbf{E}(c) &= c \end{aligned} \quad (2.10)$$

For **independent** X and Y ,

$$\mathbf{E}(XY) = \mathbf{E}(X)\mathbf{E}(Y)$$

DEFINITION 2.8

Variance of a random variable is defined as the expected squared deviation from the mean. For discrete random variables, variance is

$$\sigma^2 = \text{Var}(X) = \mathbf{E}(X - \mathbf{E}X)^2 = \sum_x (x - \mu)^2 P(x)$$

DEFINITION 2.9

Standard deviation is a square root of variance,

$$\sigma = \text{Std}(X) = \sqrt{\text{Var}(X)}$$

DEFINITION 2.10

Covariance $\sigma_{XY} = \text{Cov}(X, Y)$ is defined as

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbf{E}\{(X - \mathbf{E}X)(Y - \mathbf{E}Y)\} \\ &= \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y) \end{aligned}$$

It summarizes interrelation of two random variables.

DEFINITION 2.11

Correlation coefficient between variables X and Y is defined as

$$\rho = \frac{\text{Cov}(X, Y)}{(\text{Std}X)(\text{Std}Y)}$$

Properties of variances and covariances

$$\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

$$\text{Cov}(aX + bY, cZ + dW)$$

$$= ac \text{Cov}(X, Z) + ad \text{Cov}(X, W) + bc \text{Cov}(Y, Z) + bd \text{Cov}(Y, W)$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\rho(X, Y) = \rho(Y, X)$$

In particular,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

$$\rho(aX + b, cY + d) = \rho(X, Y)$$

For independent X and Y ,

$$\text{Cov}(X, Y) = 0$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

(2.11)

<u>NOTATION</u>	μ or $\mathbf{E}(X)$	=	expectation
	σ_X^2 or $\text{Var}(X)$	=	variance
	σ_X or $\text{Std}(X)$	=	standard deviation
	σ_{XY} or $\text{Cov}(X, Y)$	=	covariance
	ρ_{XY}	=	correlation coefficient

2.3 Discrete distribution families

Different phenomena can often be described by the same probabilistic model, or a family of distributions. The most commonly used discrete families are Binomial, including Bernoulli, Negative Binomial, including Geometric, and Poisson. Each family of distributions is used for a certain general type of situations; it has its parameters and a clear formula and/or a table for computing probabilities. These families are summarized in Section 13.1.1.

DEFINITION 2.12

A random variable with two possible values, 0 and 1, is called a **Bernoulli variable**, its distribution is **Bernoulli distribution**, and any experiment with a *binary outcome* is called a **Bernoulli trial**.

**Bernoulli
distribution**

$$\begin{aligned}
 p &= \text{probability of success} \\
 P(x) &= \begin{cases} q = 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \end{cases} \\
 \mathbf{E}(X) &= p \\
 \text{Var}(X) &= pq
 \end{aligned}$$

DEFINITION 2.13

A variable described as the number of successes in a sequence of independent Bernoulli trials has **Binomial distribution**. Its parameters are n , the number of trials, and p , the probability of success.

Remark: “Binomial” can be translated as “two numbers,” *bi* meaning “two” and *nom* meaning “a number,” thus reflecting the concept of binary outcomes.

**Binomial
distribution**

$$\begin{aligned}
 n &= \text{number of trials} \\
 p &= \text{probability of success} \\
 P(x) &= \binom{n}{x} p^x q^{n-x} \quad \text{Table A2} \\
 \mathbf{E}(X) &= np \\
 \text{Var}(X) &= npq
 \end{aligned}$$

DEFINITION 2.14

The number of Bernoulli trials needed to get the first success has **Geometric distribution**.

**Geometric
distribution**

$$\begin{aligned}
 p &= \text{probability of success} \\
 P(x) &= (1-p)^{x-1}p, \quad x = 1, 2, \dots \\
 \mathbf{E}(X) &= \frac{1}{p} \\
 \text{Var}(X) &= \frac{1-p}{p^2}
 \end{aligned} \tag{2.12}$$

DEFINITION 2.15

In a sequence of independent Bernoulli trials, the number of trials needed to obtain k successes has **Negative Binomial distribution**.

**Negative
Binomial
distribution**

$$\begin{aligned}
 k &= \text{number of successes} \\
 p &= \text{probability of success} \\
 P(x) &= \binom{x-1}{k-1} (1-p)^{x-k} p^k, \quad x = k, k+1, \dots \\
 \mathbf{E}(X) &= \frac{k}{p} \\
 \text{Var}(X) &= \frac{k(1-p)}{p^2}
 \end{aligned} \tag{2.13}$$

DEFINITION 2.16

The number of rare events occurring within a fixed period of time has **Poisson distribution**.

**Poisson
distribution**

λ	=	frequency, average number of events
$P(x)$	=	$e^{-\lambda} \frac{\lambda^x}{x!}$, $x = 0, 1, 2, \dots$ Table A3
$\mathbf{E}(X)$	=	λ
$\text{Var}(X)$	=	λ

**Poisson
approximation
to Binomial**

$$\begin{aligned} \text{Binomial}(n, p) &\approx \text{Poisson}(\lambda) \\ \text{where } n &\geq 30, p \leq 0.05, np = \lambda \end{aligned}$$

(2.14)

Relations

$\text{Bern}(p)$	=	$\text{Bin}(1, p)$
$\text{Bin}(n, p)$	=	sum of n independent $\text{Bern}(p)$
$\text{Geom}(p)$	=	$\text{NegBin}(1, p)$
$\text{NegBin}(k, p)$	=	sum of k independent $\text{Geom}(p)$
$\text{Bin}(n, p)$	$\rightarrow$	$\text{Poisson}(\lambda)$ as $n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$

2.4 Continuous random variables

Continuous distributions are used to model various times, sizes, measurements, and all other random variables that assume an entire interval of possible values.

Continuous distributions are described by their densities that play a role analogous to probability mass functions of discrete variables. Computing probabilities essentially reduces to integrating a density over the given set. Expectations and variances are defined similarly to the discrete case, replacing a probability mass function by a density and summation by integration.

Continuous variables can take any value of an interval. For these variables, the *probability mass function* (pmf) is

$$P(x) = 0 \quad \text{for all } x,$$

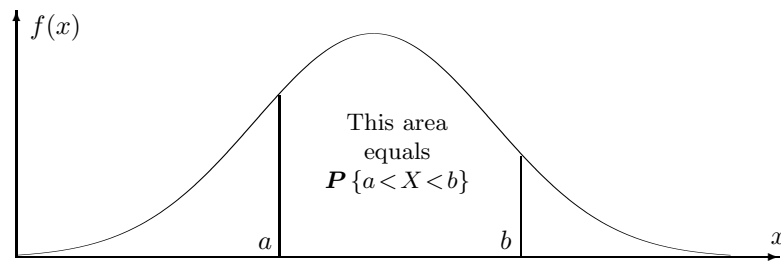


FIGURE 2.1: Probabilities are areas under the density curve.

and the *cumulative distribution function* (cdf) equals

$$F(x) = \mathbf{P}\{X \leq x\} = \mathbf{P}\{X < x\}.$$

DEFINITION 2.17

Probability density function (pdf, density) is the derivative of the cdf, $f(x) = F'(x)$. The distribution is called **continuous** if it has a density.

**Probability density
function**

$$f(x) = F'(x)$$

$$\mathbf{P}\{a < X < b\} = \int_a^b f(x)dx$$

Distribution	Discrete	Continuous
Definition	$P(x) = \mathbf{P}\{X = x\}$ (pmf)	$f(x) = F'(x)$ (pdf)
Computing probabilities	$\mathbf{P}\{X \in A\} = \sum_{x \in A} P(x)$	$\mathbf{P}\{X \in A\} = \int_A f(x)dx$
Cumulative distribution function	$F(x) = \mathbf{P}\{X \leq x\} = \sum_{y \leq x} P(y)$	$F(x) = \mathbf{P}\{X \leq x\} = \int_{-\infty}^x f(y)dy$
Total probability	$\sum_x P(x) = 1$	$\int_{-\infty}^{\infty} f(x)dx = 1$

TABLE 2.1: Pmf $P(x)$ versus pdf $f(x)$.

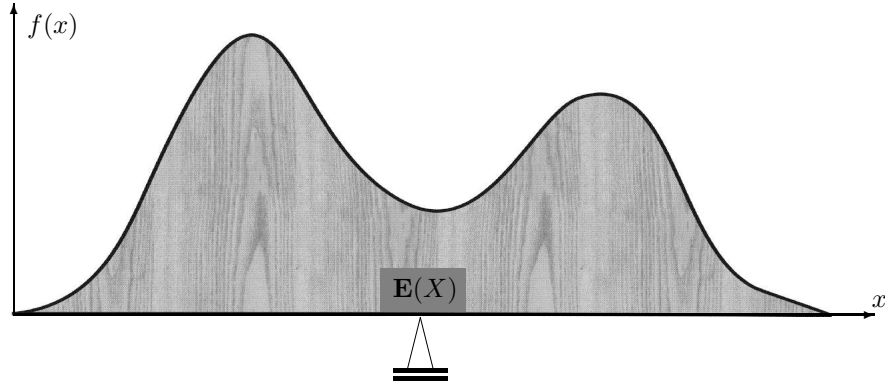


FIGURE 2.2: Expectation of a continuous variable as a center of gravity.

DEFINITION 2.18

For a vector of random variables, the **joint cumulative distribution function** is defined as

$$F_{(X,Y)}(x, y) = \mathbf{P}\{X \leq x \cap Y \leq y\}.$$

The **joint density** is the *mixed derivative* of the joint cdf,

$$f_{(X,Y)}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{(X,Y)}(x, y).$$

The most commonly used distribution families are summarized in Section 13.1.2. Let us only review how to compute Normal probabilities using Table A1 – it comes handy in a number of problems.

DEFINITION 2.19

Normal distribution with “standard parameters” $\mu = 0$ and $\sigma = 1$ is called **Standard Normal distribution**.

Distribution	Discrete	Continuous
Marginal distributions	$P(x) = \sum_y P(x, y)$ $P(y) = \sum_x P(x, y)$	$f(x) = \int f(x, y) dy$ $f(y) = \int f(x, y) dx$
Independence	$P(x, y) = P(x)P(y)$	$f(x, y) = f(x)f(y)$
Computing probabilities	$\mathbf{P}\{(X, Y) \in A\}$ $= \sum_{(x,y) \in A} P(x, y)$	$\mathbf{P}\{(X, Y) \in A\}$ $= \iint_{(x,y) \in A} f(x, y) dx dy$

TABLE 2.2: Joint and marginal distributions in discrete and continuous cases.

$$\begin{array}{l} \text{NOTATION} \left\{ \begin{array}{l} Z = \text{Standard Normal random variable} \\ \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \text{ Standard Normal pdf} \\ \Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz, \text{ Standard Normal cdf} \end{array} \right. \end{array}$$

A Standard Normal variable, usually denoted by Z , can be obtained from a non-standard Normal(μ, σ) random variable X by **standardizing**, that is, subtracting the mean and dividing by the standard deviation,

$$Z = \frac{X - \mu}{\sigma}. \quad (2.15)$$

Unstandardizing Z , we can reconstruct the initial variable X ,

$$X = \mu + \sigma Z. \quad (2.16)$$

To compute probabilities about an arbitrary Normal random variable X , we have to standardize it first, as in (2.15), then use Table A1.

Example 2.1 (COMPUTING NON-STANDARD NORMAL PROBABILITIES). Suppose that the average household income in some country is 900 coins, and the standard deviation is 200 coins. Assuming the Normal distribution of incomes, compute the proportion of “the middle class,” whose income is between 600 and 1200 coins.

Solution. Standardize and use Table A1. For a Normal($\mu = 900, \sigma = 200$) variable X ,

$$\begin{aligned} P\{600 < X < 1200\} &= P\left\{\frac{600 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{1200 - \mu}{\sigma}\right\} \\ &= P\left\{\frac{600 - 900}{200} < Z < \frac{1200 - 900}{200}\right\} = P\{-1.5 < Z < 1.5\} \\ &= \Phi(1.5) - \Phi(-1.5) = 0.9332 - 0.0668 = \underline{0.8664}. \end{aligned}$$

Discrete	Continuous
$\mathbf{E}(X) = \sum_x xP(x)$ $\text{Var}(X) = \mathbf{E}(X - \mu)^2$ $= \sum_x (x - \mu)^2 P(x)$ $= \sum_x x^2 P(x) - \mu^2$ $\text{Cov}(X, Y) = \mathbf{E}(X - \mu_X)(Y - \mu_Y)$ $= \sum_x \sum_y (x - \mu_X)(y - \mu_Y) P(x, y)$ $= \sum_x \sum_y (xy) P(x, y) - \mu_x \mu_y$	$\mathbf{E}(X) = \int x f(x) dx$ $\text{Var}(X) = \mathbf{E}(X - \mu)^2$ $= \int (x - \mu)^2 f(x) dx$ $= \int x^2 f(x) dx - \mu^2$ $\text{Cov}(X, Y) = \mathbf{E}(X - \mu_X)(Y - \mu_Y)$ $= \iint (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$ $= \iint (xy) f(x, y) dx dy - \mu_x \mu_y$

TABLE 2.3: Moments for discrete and continuous distributions.

◇

Theorem 1 (CENTRAL LIMIT THEOREM) *Let $X_1, X_2, \dots$ be independent random variables with the same expectation $\mu = \mathbf{E}(X_i)$ and the same standard deviation $\sigma = \text{Std}(X_i)$, and let*

$$S_n = \sum_{i=1}^n X_i = X_1 + \dots + X_n.$$

As $n \rightarrow \infty$, the standardized sum

$$Z_n = \frac{S_n - \mathbf{E}(S_n)}{\text{Std}(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to a Standard Normal random variable, that is,

$$F_{Z_n}(z) = \mathbf{P} \left\{ \frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z \right\} \rightarrow \Phi(z) \quad (2.17)$$

for all z .

In particular, Binomial variable is a sum of i.i.d. Bernoulli variables, therefore,

$$\text{Binomial}(n, p) \approx \text{Normal} \left(\mu = np, \sigma = \sqrt{np(1-p)} \right) \quad (2.18)$$

These approximations are widely used for large-sample statistics.

Example 2.2. A new computer virus attacks a folder consisting of 200 files. Each file gets damaged with probability 0.2 independently of other files. What is the probability that fewer than 50 files get damaged?

Solution. The number X of damaged files has Binomial distribution with $n = 200$, $p = 0.2$, $\mu = np = 40$, and $\sigma = \sqrt{np(1-p)} = 5.657$. Applying the Central Limit Theorem with the continuity correction,

$$\begin{aligned} \mathbf{P}\{X < 50\} &= \mathbf{P}\{X < 49.5\} = \mathbf{P}\left\{ \frac{X - 40}{5.657} < \frac{49.5 - 40}{5.657} \right\} \\ &= \Phi(1.68) = \underline{0.9535}. \end{aligned}$$

Notice that the properly applied continuity correction replaces 50 with 49.5, not 50.5. Indeed, we are interested in the event that X is *strictly* less than 50. This includes all values up to 49 and corresponds to the interval $[0, 49]$ that we *expand* to $[0, 49.5]$. In other words, events $\{X < 50\}$ and $\{X < 49.5\}$ are the same; they include the same possible values of X . Events $\{X < 50\}$ and $\{X < 50.5\}$ are different because the former includes $X = 50$, and the latter does not. Replacing $\{X < 50\}$ with $\{X < 50.5\}$ would have changed its probability and would have given a wrong answer. ◇



Chapter 3

Review of Basic Statistics

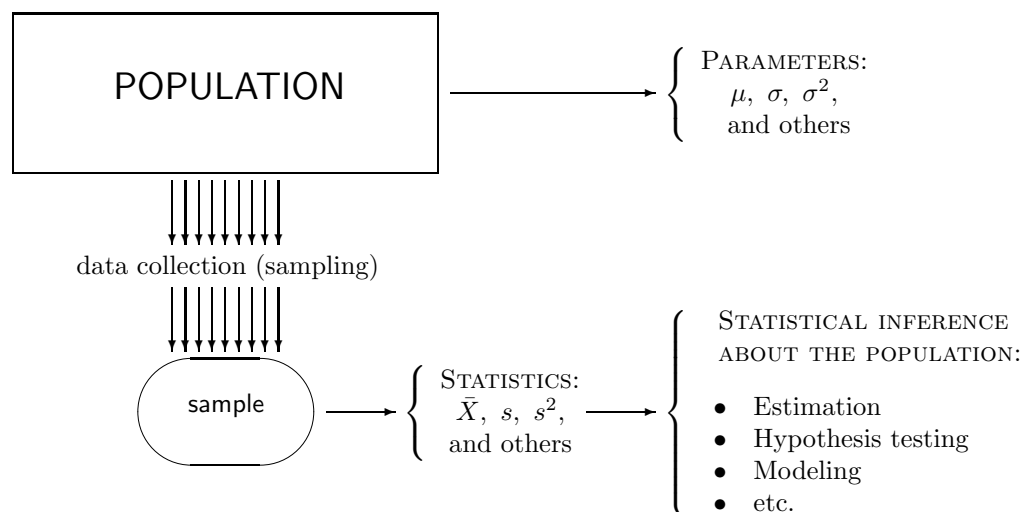
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3.1 Parameters and their estimates

DEFINITION 3.1

A **population** consists of all units of interest. Any numerical characteristic of a population is a **parameter**. A **sample** consists of observed units collected from the population. It is used to make statements about the population. Any function of a sample is called **statistic**.

NOTATION $\left\| \begin{array}{l} \theta = \text{population parameter} \\ \hat{\theta} = \text{its estimator, obtained from a sample} \end{array} \right\|$



h!

FIGURE 3.1: Population parameters and sample statistics.

3.1.1 Mean

DEFINITION 3.2

Sample mean $\bar{X}$ is the arithmetic average,

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

DEFINITION 3.3

An estimator $\hat{\theta}$ is **unbiased** for a parameter θ if its expectation equals the parameter,

$$\mathbf{E}(\hat{\theta}) = \theta$$

for all possible values of θ .

Bias of $\hat{\theta}$ is defined as $\text{Bias}(\hat{\theta}) = \mathbf{E}(\hat{\theta} - \theta)$.

Sample mean $\bar{X}$ estimates the population mean $\mu = \mathbf{E}(X)$ unbiasedly.

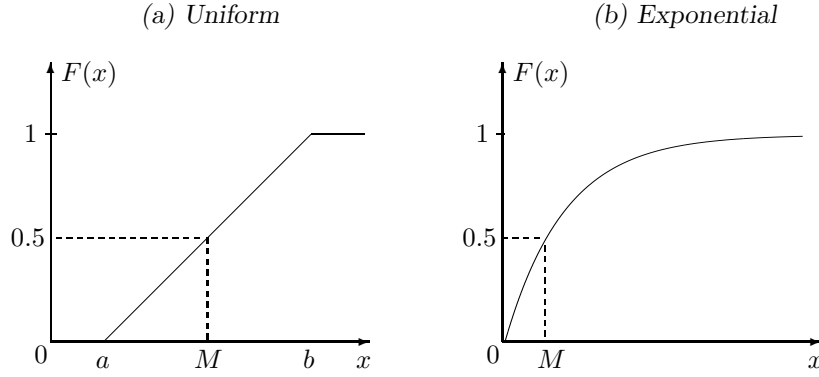


FIGURE 3.2: Computing medians of continuous distributions.

DEFINITION 3.4

An estimator $\hat{\theta}$ is **consistent** for a parameter θ if the probability of its sampling error of any magnitude converges to 0 as the sample size increases to infinity. Stating it rigorously,

$$\mathbf{P} \left\{ |\hat{\theta} - \theta| > \varepsilon \right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

for any $\varepsilon > 0$.

3.1.2 Median**DEFINITION 3.5**

Median means a “central” value.

Sample median $\hat{M}$ is a number that is exceeded by at most a half of observations and is preceded by at most a half of observations.

Population median M is a number that is exceeded with probability no greater than 0.5 and is preceded with probability no greater than 0.5. That is, M is such that

$$\begin{cases} \mathbf{P} \{X > M\} & \leq 0.5 \\ \mathbf{P} \{X < M\} & \leq 0.5 \end{cases}$$

For continuous distributions, computing a **population median** reduces to solving one equation:

$$\begin{cases} \mathbf{P} \{X > M\} & = 1 - F(M) \leq 0.5 \\ \mathbf{P} \{X < M\} & = F(M) \leq 0.5 \end{cases} \Rightarrow F(M) = 0.5.$$

For discrete distributions, equation $F(x) = 0.5$ has either a whole interval of roots or no roots at all (see Figure 3.3).

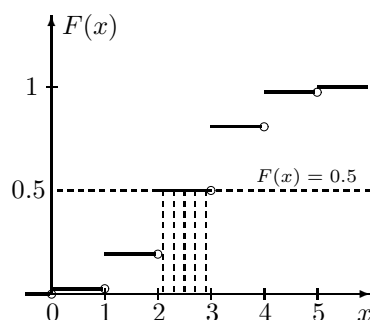
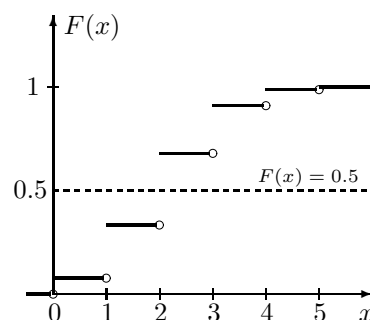
(a) Binomial ($n=5, p=0.5$)
many roots(b) Binomial ($n=5, p=0.4$)
no roots

FIGURE 3.3: Computing medians of discrete distributions.

In the first case, any number in this interval, excluding the ends, is a median. Notice that the median in this case is not unique (Figure 3.3a). Often the middle of this interval is reported as the median.

In the second case, the smallest x with $F(x) \geq 0.5$ is the median. It is the value of x where the cdf jumps over 0.5 (Figure 3.3b).

Sample median

If n is odd, the $\left(\frac{n+1}{2}\right)$ -th smallest observation is a median.
 If n is even, any number between the $\left(\frac{n}{2}\right)$ -th smallest and the $\left(\frac{n+2}{2}\right)$ -th smallest observations is a median.

3.1.3 Quantiles

DEFINITION 3.6

A **p -quantile** of a population is such a number x that solves equations

$$\begin{cases} P\{X < x\} & \leq p \\ P\{X > x\} & \leq 1 - p \end{cases}$$

A **sample p -quantile** is any number that exceeds at most $100p\%$ of the sample, and is exceeded by at most $100(1 - p)\%$ of the sample.

A **γ -percentile** is (0.01γ) -quantile.

First, second, and third **quantiles** are the 25th, 50th, and 75th percentiles. They split a population or a sample into four equal parts.

A **median** is at the same time a 0.5-quantile, 50th percentile, and 2nd quartile.

Computing quantiles is very similar to computing medians.

Example 3.1 (SAMPLE QUANTILES). Let us compute the 1st and 3rd quartiles of CPU times. The ordered sample is below.

9	15	19	22	24	25	30	34	35	35
36	36	37	38	42	43	46	48	54	55
56	56	59	62	69	70	82	82	89	139

First quartile $\hat{Q}_1$. For $p = 0.25$, we find that 25% of the sample equals $np = 7.5$, and 75% of the sample is $n(1 - p) = 22.5$ observations. From the ordered sample, we see that only the 8th element, 34, has no more than 7.5 observations to the left and no more than 22.5 observations to the right of it. Hence, $\hat{Q}_1 = 34$.

Third quartile $\hat{Q}_3$. Similarly, the third sample quartile is the 23rd smallest element, $\hat{Q}_3 = 59$.

◇

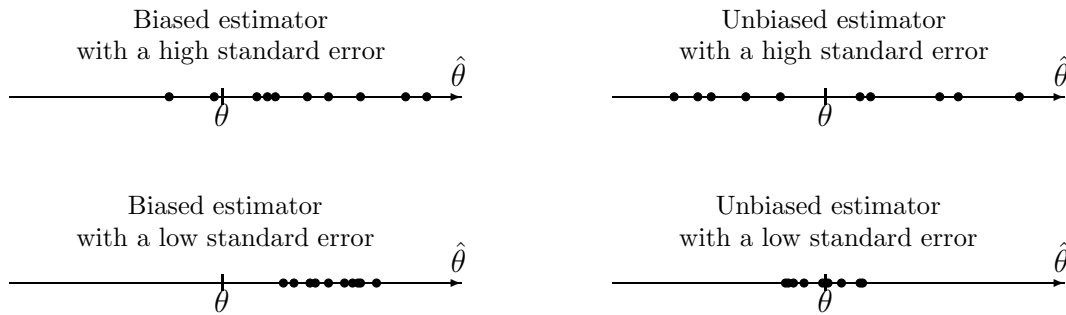


FIGURE 3.4: Bias and standard error of an estimator. In each case, the dots represent parameter estimators $\hat{\theta}$ obtained from 10 different random samples.

3.1.4 Variance and standard deviation

DEFINITION 3.7

For a sample $(X_1, X_2, \dots, X_n)$, a **sample variance** is defined as

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2. \quad (3.1)$$

It measures variability among observations and estimates the population variance $\sigma^2 = \text{Var}(X)$.

Sample standard deviation is a square root of a sample variance, $s = \sqrt{s^2}$. It measures variability in the same units as X and estimates the population standard deviation $\sigma = \text{Std}(X)$.

DEFINITION 3.8

Standard error of an estimator $\hat{\theta}$ is its standard deviation, $\sigma(\hat{\theta}) = \text{Std}(\hat{\theta})$.

<u>NOTATION</u>	$\left\ \begin{array}{ll} \sigma(\hat{\theta}) &= \text{standard error of estimator } \hat{\theta} \text{ of parameter } \theta \\ s(\hat{\theta}) &= \text{estimated standard error} = \hat{\sigma}(\hat{\theta}) \end{array} \right\ $
-----------------	---

3.1.5 Method of moments

DEFINITION 3.9

The k -th **population moment** is defined as

$$\mu_k = \mathbf{E}(X^k).$$

The k -th **sample moment**

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k$$

estimates μ_k from a sample $(X_1, \dots, X_n)$.

The first sample moment is the sample mean $\bar{X}$.

Central moments are computed similarly, after centralizing the data, that is, subtracting the mean.

To estimate k parameters by the **method of moments**, equate the first k population and sample moments,

$$\begin{cases} \mu_1 &= m_1 \\ \dots &\dots \dots \\ \mu_k &= m_k \end{cases}$$

The left-hand sides of these equations depend on the distribution parameters. The right-hand sides can be computed from data. The **method of moments estimator** is the solution of this system of equations.

Example 3.2 (METHOD OF MOMENTS FOR THE GAMMA DISTRIBUTION). Apparently, CPU times in Example 3.1 on p. 33 have Gamma distribution with some parameters α and λ . To estimate these parameters, we need two equations. From the given data, we compute

$$m_1 = \bar{X} = 48.2333 \quad \text{and} \quad m'_2 = S^2 = 679.7122.$$

and write two equations,

$$\begin{cases} \mu_1 &= \mathbf{E}(X) &= \alpha/\lambda &= m_1 \\ \mu'_2 &= \text{Var}(X) &= \alpha/\lambda^2 &= m'_2. \end{cases}$$

It is convenient to use the second *central* moment here because we already know the expression for the variance $m'_2 = \text{Var}(X)$ of a Gamma variable.

Solving this system in terms of α and λ , we get the method of moment estimates

$$\begin{cases} \hat{\alpha} &= m_1^2/m'_2 &= 3.4227 \\ \hat{\lambda} &= m_1/m'_2 &= 0.0710. \end{cases}$$

◇

3.1.6 Method of maximum likelihood

DEFINITION 3.10

Maximum likelihood estimator is the parameter value that maximizes the likelihood of the observed sample. For a discrete distribution, we maximize the joint pmf of data $P(X_1, \dots, X_n)$. For a continuous distribution, we maximize the joint density $f(X_1, \dots, X_n)$.

Manual application method manually requires knowledge of Calculus I.

For a discrete distribution, the probability of a given sample is the joint pmf of data,

$$P\{\mathbf{X} = (X_1, \dots, X_n)\} = P(\mathbf{X}) = P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i),$$

because in a simple random sample, all observed X_i are independent. For a continuous distribution, the joint density of the data $f(\mathbf{X}) = \prod_{i=1}^n f(X_i)$ is the likelihood.

To maximize this likelihood, we consider the critical points by taking derivatives with respect to all unknown parameters and equating them to 0. The maximum can only be attained at such parameter values θ where the derivative $\frac{\partial}{\partial \theta} P(\mathbf{X})$ equals 0, where it does not exist, or at the boundary of the set of possible values of θ .

A nice computational shortcut is to take logarithms first. Differentiating the sum

$$\ln \prod_{i=1}^n P(X_i) = \sum_{i=1}^n \ln P(X_i)$$

is easier than differentiating the product $\prod P(X_i)$. Besides, logarithm is an increasing function, so the **likelihood** $P(\mathbf{X})$ and the **log-likelihood** $\ln P(\mathbf{X})$ are maximized by exactly the same parameters.

Example 3.3 (POISSON). The pmf of Poisson distribution is

$$P(x) = e^{-\lambda} \frac{\lambda^x}{x!},$$

and its logarithm is

$$\ln P(x) = -\lambda + x \ln \lambda - \ln(x!).$$

Thus, we need to maximize

$$\ln P(\mathbf{X}) = \sum_{i=1}^n (-\lambda + X_i \ln \lambda) + C = -n\lambda + \ln \lambda \sum_{i=1}^n X_i + C,$$

where $C = -\sum \ln(x!)$ is a constant that does not contain the unknown parameter λ .

Find the critical point(s) of this log-likelihood. Differentiating it and equating its derivative to 0, we get

$$\frac{\partial}{\partial \lambda} \ln P(\mathbf{X}) = -n + \frac{1}{\lambda} \sum_{i=1}^n X_i = 0.$$

This equation has only one solution

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

Since this is the only critical point, and since the likelihood vanishes (converges to 0) as $\lambda \downarrow 0$ or $\lambda \uparrow \infty$, we conclude that $\hat{\lambda}$ is the maximizer. Therefore, it is the maximum likelihood estimator of λ . $\diamond$

3.2 Confidence intervals

DEFINITION 3.11

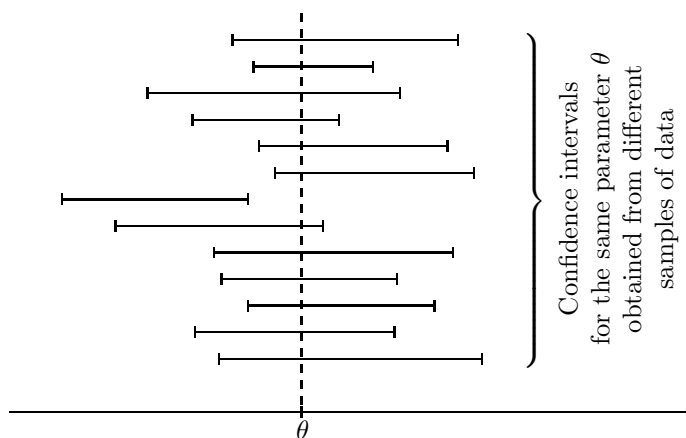
An interval $[a, b]$ is a $(1 - \alpha)100\%$ **confidence interval** for the parameter θ if it contains the parameter with probability $(1 - \alpha)$,

$$P\{a \leq \theta \leq b\} = 1 - \alpha.$$

The **coverage probability** $(1 - \alpha)$ is also called a **confidence level**.

This **confidence level** is illustrated in Figure 3.5. Suppose that we collect many random samples and produce a confidence interval from each of them. If these are $(1 - \alpha)100\%$ confidence intervals, then we expect $(1 - \alpha)100\%$ of them to cover θ and $100\alpha\%$ of them to miss it. This is the classical frequentist “long-run” measure of performance.

Therefore, without a prior distribution, it is *wrong* to say, “*I computed a 90% confidence interval, it is $[3, 6]$. Parameter belongs to this interval with probability 90%.*” For a frequentist, the parameter is constant; it either belongs to the interval $[3, 6]$ (with probability 1) or does not. In this case, 90% refers to the proportion of confidence intervals that contain the unknown parameter in a long run.

FIGURE 3.5: Confidence intervals and coverage of parameter θ .

**Confidence
interval,
Normal
distribution**

If parameter θ has an unbiased, Normally distributed estimator $\hat{\theta}$, then

$$\hat{\theta} \pm z_{\alpha/2} \cdot \sigma(\hat{\theta}) = \left[\hat{\theta} - z_{\alpha/2} \cdot \sigma(\hat{\theta}), \hat{\theta} + z_{\alpha/2} \cdot \sigma(\hat{\theta}) \right]$$

is a $(1 - \alpha)100\%$ **confidence interval** for θ .

(3.2)

If the distribution of $\hat{\theta}$ is *approximately* Normal, this is an *approximately* $(1 - \alpha)100\%$ confidence interval.

**Confidence interval
for the mean;
 σ is known**

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

(3.3)

**Confidence
interval
for the mean;
 σ is unknown**

$$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2}$ is a critical value from T-distribution with $n - 1$ degrees of freedom

(3.4)

Example 3.4. Construct a 95% confidence interval for the population mean based on a sample of measurements

2.5, 7.4, 8.0, 4.5, 7.4, 9.2

if measurement errors have Normal distribution, and the measurement device guarantees a standard deviation of $\sigma = 2.2$.

Solution. This sample has size $n = 6$ and sample mean $\bar{X} = 6.50$. To attain a confidence level of

$$1 - \alpha = 0.95,$$

we need $\alpha = 0.05$ and $\alpha/2 = 0.025$. Hence, we are looking for quantiles

$$q_{0.025} = -z_{0.025} \quad \text{and} \quad q_{0.975} = z_{0.025}.$$

From Table A1, we find that $q_{0.975} = 1.960$. Substituting these values into (3.3), we obtain a 95% confidence interval for μ ,

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 6.50 \pm (1.960) \frac{2.2}{\sqrt{6}} = \underline{6.50 \pm 1.76} \text{ or } \underline{[4.74, 8.26]}.$$

◇

3.3 Testing Hypotheses

To begin, we need to state exactly what we are testing. These are *hypothesis* and *alternative*.

$$\begin{array}{l} \text{NOTATION} \quad \left\| \begin{array}{ll} H_0 & = \text{hypothesis (the null hypothesis)} \\ H_A & = \text{alternative (the alternative hypothesis)} \end{array} \right\| \end{array}$$

H_0 and H_A are simply two mutually exclusive statements. Each test results either in **acceptance** of H_0 or its **rejection** in favor of H_A .

DEFINITION 3.12

Alternative of the type $H_A : \mu \neq \mu_0$ covering regions on both sides of the hypothesis ($H_0 : \mu = \mu_0$) is a **two-sided alternative**.

Alternative $H_A : \mu < \mu_0$ covering the region to the left of H_0 is **one-sided, left-tail**.

Alternative $H_A : \mu > \mu_0$ covering the region to the right of H_0 is **one-sided, right-tail**.

DEFINITION 3.13

A **type I error** occurs when we reject the true null hypothesis.

A **type II error** occurs when we accept the false null hypothesis.

DEFINITION 3.14

Probability of a type I error is the **significance level** of a test,

$$\alpha = P\{\text{reject } H_0 \mid H_0 \text{ is true}\}.$$

Probability of rejecting a false hypothesis is the **power of the test**,

$$p(\theta) = P\{\text{reject } H_0 \mid \theta; H_A \text{ is true}\}.$$

It is usually a function of the parameter θ because the alternative hypothesis includes a set of parameter values. Also, the **power** is the probability to avoid a Type II error.

3.3.1 Level α tests

A standard algorithm for a level α test of a hypothesis H_0 against an alternative H_A consists of 3 steps.

Step 1. Test statistic. Testing hypothesis is based on a **test statistic** T , a quantity computed from the data that has some known, tabulated distribution F_0 if the hypothesis H_0 is true. Test statistics are used to discriminate between the hypothesis and the alternative.

Step 2. Acceptance region and rejection region. Next, we consider the **null distribution** F_0 . This is the distribution of test statistic T when the hypothesis H_0 is true. If it has a density f_0 , then the whole area under the density curve is 1, and we can always find a portion of it whose area is α , as shown in Figure 3.6. It is called **rejection region** (R). The remaining part, the complement of the rejection region, is called **acceptance region** ($A = \bar{R}$). By the complement rule, its area is $(1 - \alpha)$.

These regions are selected in such a way that the values of test statistic T in the rejection region provide a stronger support of H_A than the values $T \in A$. For example, suppose that T is expected to be large if H_A is true. Then the rejection region corresponds to the right tail of the null distribution F_0 (Figure 3.6). Also, see Figure 3.7 for the right-tail, left-tail, and two-sided Z-test.

Step 3: Result and its interpretation. Accept the hypothesis H_0 if the test statistic T belongs to the acceptance region. Reject H_0 in favor of the alternative H_A if T belongs to the rejection region.

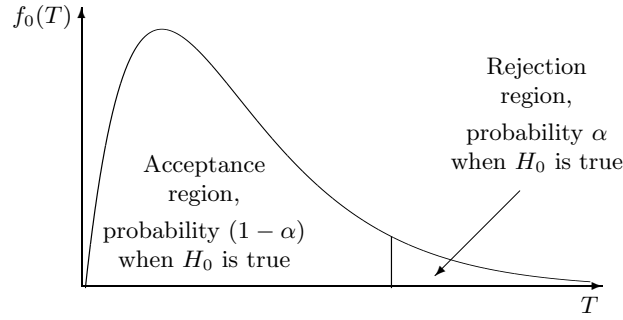


FIGURE 3.6: Acceptance and rejection regions.

Notice the [frequentist interpretation of the significance level](#) – in a long run of level α tests, the proportion of rejected null hypotheses is α , provided that the null hypotheses were actually true.

Even after the test, we cannot tell whether the null hypothesis is true or false. We need to see the entire population to tell that. A test only determines whether the data provided sufficient evidence against H_0 in favor of H_A .

3.3.2 Z-tests

Consider testing a hypothesis about a population parameter θ . Suppose that its estimator $\hat{\theta}$ has Normal distribution, at least approximately, and we know $\mathbf{E}(\hat{\theta})$ and $\text{Var}(\hat{\theta})$ if the hypothesis is true.

Then the test statistic

$$Z = \frac{\hat{\theta} - \mathbf{E}(\hat{\theta})}{\sqrt{\text{Var}(\hat{\theta})}} \quad (3.5)$$

has Standard Normal distribution, and we can construct acceptance and rejection regions for a level α test from Table A1. The most popular Z-tests are summarized in Table 3.1.

Example 3.5 (TWO-SAMPLE Z-TEST OF PROPORTIONS). A quality inspector finds 10 defective parts in a sample of 500 parts received from manufacturer A. Out of 400 parts from manufacturer B, she finds 12 defective ones. A computer-making company uses these parts in their computers and claims that the quality of parts produced by A and B is the same. At the 5% level of significance, do we have enough evidence to disprove this claim?

Solution. We test $H_0 : p_A = p_B$, or $H_0 : p_A - p_B = 0$, against $H_A : p_A \neq p_B$. This is a two-sided test because no direction of the alternative has been indicated. We only need to verify whether or not the proportions of defective parts are equal for manufacturers A and B.

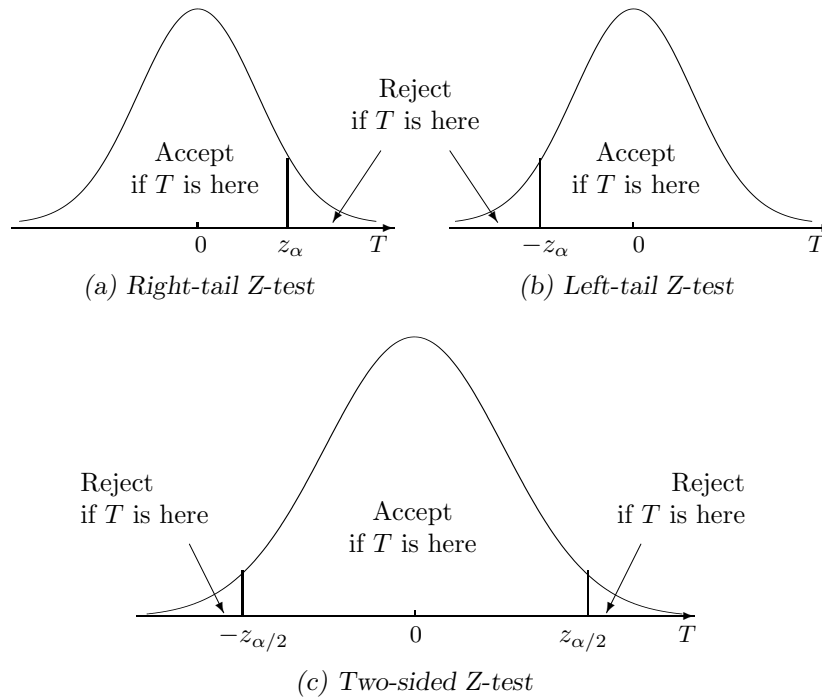


FIGURE 3.7: Acceptance and rejection regions for a Z-test with (a) a one-sided right-tail alternative; (b) a one-sided left-tail alternative; (c) a two-sided alternative.

Step 1: Test statistic. We are given: $\hat{p}_A = 10/500 = 0.02$ from a sample of size $n = 500$; $\hat{p}_B = 12/400 = 0.03$ from a sample of size $m = 400$. The tested value is $D = 0$.

As we know, for these Bernoulli data, the variance depends on the unknown parameters p_A and p_B which are estimated by the sample proportions $\hat{p}_A$ and $\hat{p}_B$.

The test statistic then equals

$$Z = \frac{\hat{p}_A - \hat{p}_B - D}{\sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n} + \frac{\hat{p}_B(1 - \hat{p}_B)}{m}}} = \frac{0.02 - 0.03}{\sqrt{\frac{(0.02)(0.98)}{500} + \frac{(0.03)(0.97)}{400}}} = -0.945.$$

Step 2: Acceptance and rejection regions. This is a two-sided test; thus we divide α by 2, find $z_{0.05/2} = z_{0.025} = 1.96$, and

$$\begin{cases} \text{reject } H_0 & \text{if } |Z| \geq 1.96; \\ \text{accept } H_0 & \text{if } |Z| < 1.96. \end{cases}$$

Step 3: Result. The evidence against H_0 is insufficient because $|Z| < 1.96$. Although *sample proportions* of defective parts are unequal, the difference between them appears too small to claim that *population proportions* are different. $\diamond$

Null hypothesis H_0	Parameter, estimator $\theta, \hat{\theta}$	If H_0 is true:		Test statistic $Z = \frac{\hat{\theta} - \theta_0}{\sqrt{\text{Var}(\hat{\theta})}}$
		$\mathbf{E}(\hat{\theta})$	$\text{Var}(\hat{\theta})$	
One-sample Z-tests for means and proportions, based on a sample of size n				
$\mu = \mu_0$	$\mu, \bar{X}$	μ_0	$\frac{\sigma^2}{n}$	$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
$p = p_0$	$p, \hat{p}$	p_0	$\frac{p_0(1-p_0)}{n}$	$\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$
Two-sample Z-tests comparing means and proportions of two populations, based on independent samples of size n and m				
$\mu_X - \mu_Y = D$	$\mu_X - \mu_Y, \bar{X} - \bar{Y}$	D	$\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$	$\frac{\bar{X} - \bar{Y} - D}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$
$p_1 - p_2 = D$	$p_1 - p_2, \hat{p}_1 - \hat{p}_2$	D	$\frac{p_1(1-p_1)}{n} + \frac{p_2(1-p_2)}{m}$	$\frac{\hat{p}_1 - \hat{p}_2 - D}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}}$
$p_1 = p_2$	$p_1 - p_2, \hat{p}_1 - \hat{p}_2$	0	$p(1-p)\left(\frac{1}{n} + \frac{1}{m}\right),$ where $p = p_1 = p_2$	$\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n} + \frac{1}{m}\right)}}$ where $\hat{p} = \frac{n\hat{p}_1 + m\hat{p}_2}{n+m}$

TABLE 3.1: Summary of Z-tests.

Example 3.6 (EXAMPLE 3.5, CONTINUED). When testing equality of two proportions, we can also use the **pooled sample proportion** that estimates both p_A and p_B that are equal under H_0 . Here the pooled proportion equals

$$\hat{p} = \frac{10 + 12}{500 + 400} = 0.0244,$$

so that

$$Z = \frac{0.02 - 0.03}{\sqrt{(0.0244)(0.9756) \left(\frac{1}{500} + \frac{1}{400} \right)}} = -0.966.$$

This does not affect our result. We obtained a different value of Z-statistic, but it also belongs to the acceptance region. We still don't have a significant evidence against the equality of two population proportions. $\diamond$

Hypothesis H_0	Conditions	Test statistic t	Degrees of freedom
$\mu = \mu_0$	Sample size n ; unknown σ	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	$n - 1$
$\mu_X - \mu_Y = D$	Sample sizes n, m ; unknown but equal standard deviations, $\sigma_X = \sigma_Y$	$t = \frac{\bar{X} - \bar{Y} - D}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$	$n + m - 2$
$\mu_X - \mu_Y = D$	Sample sizes n, m ; unknown, unequal standard deviations, $\sigma_X \neq \sigma_Y$	$t = \frac{\bar{X} - \bar{Y} - D}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}$	Satterthwaite approximation

TABLE 3.2: Summary of T-tests.

3.3.3 T-tests

When we don't know the population standard deviation, we estimate it. The resulting *T-statistic* has the form

$$t = \frac{\hat{\theta} - \mathbf{E}(\hat{\theta})}{s(\hat{\theta})} = \frac{\hat{\theta} - \mathbf{E}(\hat{\theta})}{\sqrt{\widehat{\text{Var}}(\hat{\theta})}}.$$

In the case when the distribution of $\hat{\theta}$ is Normal, the test is based on *Student's T-distribution* with acceptance and rejection regions according to the direction of H_A , see Table 3.2.

3.3.4 Duality: two-sided tests and two-sided confidence intervals

A level α Z-test of $H_0 : \theta = \theta_0$ vs $H_A : \theta \neq \theta_0$

accepts the null hypothesis if and only if a symmetric $(1 - \alpha)100\%$ confidence Z-interval for θ contains θ_0 (see Fig. 3.8).

3.3.5 P-value

So far, we were testing hypotheses by means of acceptance and rejection regions. *Significance level* α is needed for this; results of our test depend on it.

How do we choose α , the probability of making type I sampling error, rejecting the true hypothesis? Of course, when it seems too dangerous to reject true H_0 , we choose a low significance level. How low? Should we choose $\alpha = 0.01$? Perhaps, 0.001? Or even 0.0001?

Also, if our observed test statistic $Z = Z_{\text{obs}}$ belongs to a rejection region but it is “too

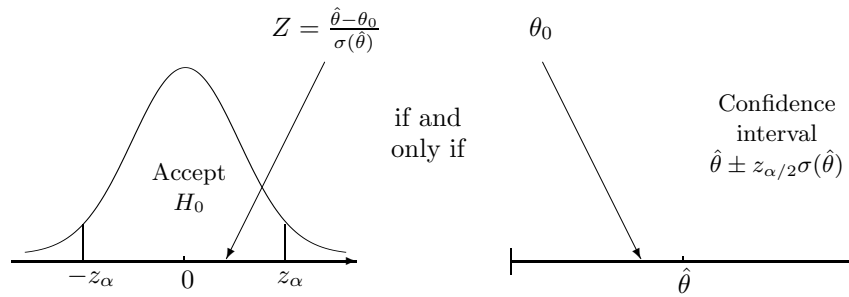


FIGURE 3.8: Duality of tests and confidence intervals.

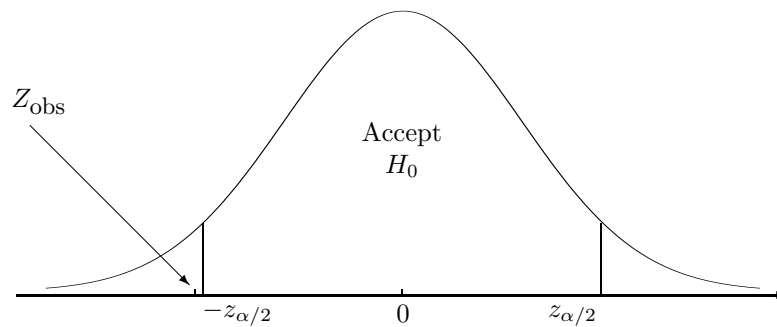


FIGURE 3.9: This test is “too close to call”: formally we reject the null hypothesis although the Z-statistic is almost at the boundary.

close to call” (see, for example, Figure 3.9), then how do we report the result? Formally, we should reject the null hypothesis, but practically, we realize that a slightly different significance level α could have expanded the acceptance region just enough to cover Z_{obs} and force us to accept H_0 .

Suppose that the result of our test is crucially important. For example, the choice of a business strategy for the next ten years depends on it. In this case, can we rely so heavily on the choice of α ? And if we rejected the true hypothesis just because we chose $\alpha = 0.05$ instead of $\alpha = 0.04$, then how do we explain to the chief executive officer that the situation was marginal? What is the statistical term for “too close to call”?

Using a P-value approach, we try not to rely on the level of significance. In fact, let us try to test a hypothesis using *all levels of significance*! Considering all $\alpha \in (0, 1)$ (because it is the probability of Type I error), we notice:

Case 1. If a level of significance is *very low*, we *accept* the null hypothesis (see Figure 3.10a).

A low value of

$$\alpha = \mathbf{P} \{ \text{reject the null hypothesis when it is true} \}$$

makes it very unlikely to reject the hypothesis because it yields a very small **rejection region**. The right-tail area above the rejection region equals α .

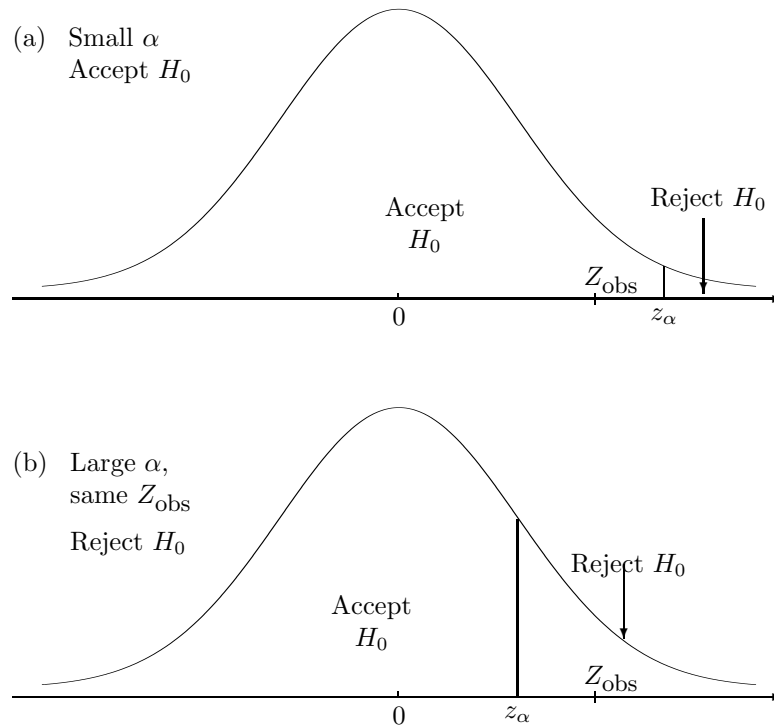


FIGURE 3.10: (a) Under a low level of significance α , we accept the null hypothesis. (b) Under a high level of significance, we reject it.

Case 2. On the other extreme end, a *high significance level* α makes it likely to reject the null hypothesis and corresponds to a large rejection region. A sufficiently large α will produce such a large **rejection region** that will cover our test statistic, forcing us to *reject* H_0 (see Figure 3.10b).

Conclusion: there exists a boundary value between α -to-accept (case 1) and α -to-reject (case 2). This number is the **P-value** (Figure 3.11).

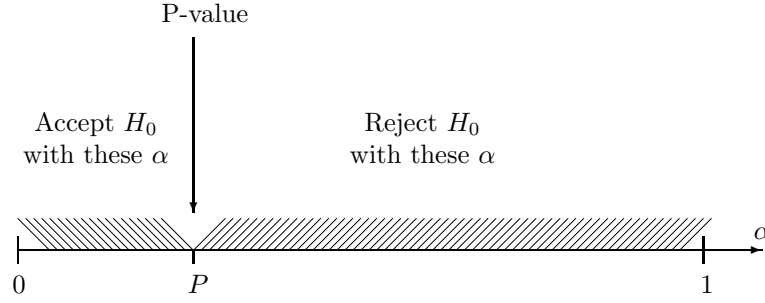
DEFINITION 3.15

P-value is the lowest significance level α that forces rejection of the null hypothesis.

P-value is also the highest significance level α that forces acceptance of the null hypothesis.

Testing hypotheses with a P-value

Once we know a P-value, we can test hypotheses at *all* significance levels. Figure 3.11 shows that for all $\alpha < P$ we accept the null hypothesis, and for all $\alpha > P$, we reject it.

FIGURE 3.11: *P-value separates α -to-accept and α -to-reject.*

Computing P-values

Let us look at Figure 3.10 again. Start from Figure 3.10a, gradually increase α , and keep your eye at the vertical bar separating the acceptance and rejection region. It will move to the left until it hits the observed test statistic Z_{obs} . At this point, our decision changes, and we switch from case 1 (Figure 3.10a) to case 2 (Figure 3.10b). Increasing α further, we pass the Z-statistic and start accepting the null hypothesis.

What happens at the border of α -to-accept and α -to-reject? Definition 3.15 says that this borderline α is the **P-value**, $P = \alpha$. Also, at this border our observed Z-statistic coincides with the critical value z_α , $Z_{\text{obs}} = z_\alpha$, and thus,

$$P = \alpha = \mathbf{P}\{Z \geq z_\alpha\} = \mathbf{P}\{Z \geq Z_{\text{obs}}\}.$$

In this formula, Z is any Standard Normal random variable, and Z_{obs} is our observed test statistic, which is a concrete number, computed from data. First, we compute Z_{obs} , then use Table A1 to calculate

$$\mathbf{P}\{Z \geq Z_{\text{obs}}\} = 1 - \Phi(Z_{\text{obs}}).$$

P-values for the left-tail and for the two-sided alternatives are computed similarly, as given in Table 3.3.

Hypothesis H_0	Alternative H_A	P-value	Computation
$\theta = \theta_0$	right-tail $\theta > \theta_0$	$\mathbf{P}\{Z \geq Z_{\text{obs}}\}$	$1 - \Phi(Z_{\text{obs}})$
	left-tail $\theta < \theta_0$	$\mathbf{P}\{Z \leq Z_{\text{obs}}\}$	$\Phi(Z_{\text{obs}})$
	two-sided $\theta \neq \theta_0$	$\mathbf{P}\{ Z \geq Z_{\text{obs}} \}$	$2(1 - \Phi(Z_{\text{obs}}))$

TABLE 3.3: *P-values for Z-tests.*

Understanding P-values

Looking at Table 3.3, we see that *P-value* is the probability of observing a test statistic *at least as extreme as* Z_{obs} or t_{obs} . Being “extreme” is determined by the alternative. For a right-tail alternative, large numbers are extreme; for a left-tail alternative, small numbers are extreme; and for a two-sided alternative, both large and small numbers are extreme. In general, the more extreme test statistic we observe, the stronger support of the alternative it provides.

This creates another interesting definition of a P-value.

DEFINITION 3.16

P-value is the probability of observing a test statistic that is as extreme as or more extreme than the test statistic computed from a given sample.

The following philosophy can be used when we test hypotheses by means of a P-value.

We are deciding between the null hypothesis H_0 and the alternative H_A . Observed is a test statistic Z_{obs} . If H_0 were true, how likely would it be to observe such a statistic? In other words, are the observed data consistent with H_0 ?

A high P-value tells that this or even more extreme value of Z_{obs} is quite possible under H_0 , and therefore, we see no contradiction with H_0 . The null hypothesis is not rejected.

Conversely, a low P-value signals that such an extreme test statistic is unlikely if H_0 is true. However, we really observed it. Then, our data are not consistent with the hypothesis, and we should reject H_0 .

For example, if $P = 0.0001$, there is only 1 chance in 10,000 to observe what we really observed. The evidence supporting the alternative is highly significant in this case.

Example 3.7 (QUALITY INSPECTION). In Example 3.5, we compared the quality of parts produced by two manufacturers by a two-sided test. We obtained a test statistic

$$Z_{\text{obs}} = -0.94.$$

The P-value for this test equals

$$P = \mathbf{P}\{|Z| \geq |-0.94|\} = 2(1 - \Phi(0.94)) = 2(1 - 0.8264) = 0.3472.$$

This is a rather high P-value (greater than 0.1), and the null hypothesis is not rejected. Given H_0 , there is a 34% chance of observing what we really observed. No contradiction with H_0 , and therefore, no evidence that the quality of parts is not the same. $\diamond$

Chapter 4

Choice of a Prior Distribution

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Bayes decision rules are based on three components, the data, the prior, and the loss,

$$\delta_{\text{Bayes}} = \delta(\mathbf{X}, \pi, L).$$

So, for *Bayesian decision making*, we need

- To collect data $\mathbf{X} = (X_1, \dots, X_n)$;
- To choose a prior distribution of unknown parameters $\pi(\theta)$;
- To choose a loss function $L(\theta, \delta)$.

Data collection is pretty well understood and well covered in Sampling courses. The prior and the loss function are to be chosen, and surely, our final decision will depend on that choice. We discuss the choice of a prior distribution in this chapter and the choice of a loss function in the next one.

There are perhaps four general ways to choose a prior distribution:

- **Subjectively:** quantify your personal beliefs, express your uncertainty about the parameter θ in a form of a distribution;
- **Empirically:** let the data suggest the prior distribution. People often use historical data or data on similar cases;
- **Conveniently:** take a convenient form of the prior $\pi(\theta)$ in order to get a mathematically tractable posterior distribution $\pi(\theta | \mathbf{X})$.
- **Non-informatively:** in the absence of any information about the parameter prior to the experiment, which prior distribution would most fairly reflect this situation?

4.1 Subjective choice of a prior distribution

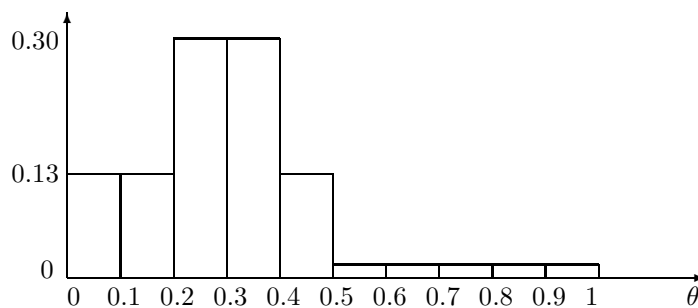
Subjectively determined prior does not have a direct mathematical formula. It is just an attempt to express your original beliefs about the unknown parameter and your uncertainty about it in a usable mathematical form. Something that we did in Example 1.9 on p. 6.

A few suggestions. Often we can determine a few related probabilities and fit a distribution to them. Sometimes we can compare probabilities of different values of θ or probabilities of intervals. Usually, it is easy to compare the chances of events and their complements like $P\{\theta \in [a, b]\}$ and $P\{\theta \notin [a, b]\}$ - which one is more likely? Sometimes one can determine some percentiles, say, with probability 25%, parameter θ does not exceed what?

Example 4.1 (SUBJECTIVE PRIOR). Before introducing a smoke-free campus rule, we would like to explore the proportion of smokers among AU students. That's an unknown parameter $\theta \in [0, 1]$. How likely do you think that $0.2 \leq \theta \leq 0.4$? I think it is about 60% likely, $\pi([0.2, 0.4]) \approx 0.6$. Also, I am almost sure that $\theta < 0.5$, i.e., majority do not smoke. So, let $\pi((0.5, 1]) \approx 0.01$. It remains to compare $\pi([0, 0.2])$ with $\pi(0.4, 0.5)$. I think $\theta \in [0, 0.2)$ is twice as likely as $\theta \in [0.4, 0.5)$. That means, $\pi([0, 0.2)) \approx 0.26$ and $\pi([0.4, 0.5)) \approx 0.13$, in order to make

$$\pi([0, 1]) = \pi([0, 0.2)) + \pi([0.2, 0.4)) + \pi(0.4, 0.5) + \pi((0.5, 1]) \approx 0.26 + 0.6 + 0.13 + 0.01 = 1.$$

Our constructed prior distribution of θ is given as a following histogram,



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4.2 Empirical Bayes solutions

The general idea of empirical Bayes analysis is to estimate the prior distribution from the data. This can be done in several ways:

- **Parametric empirical Bayes.** A family of prior distributions $\pi(\theta|\lambda)$ is chosen, but its parameter(s) λ is unknown. This λ will then be estimated from the marginal distribution $m(x) = m(x|\lambda)$.
- **Nonparametric empirical Bayes.** No family of prior distributions is assumed. Thus, there is no form of the posterior as well. Instead, the form of a Bayes decision rule is obtained directly, bypassing the posterior.
- **Hierarchical Bayes.** We can also take a “completely Bayes” approach, assume a family of prior distributions $\pi(\theta|\lambda)$ and estimate unknown λ in the Bayesian way. That is, we put a prior distribution $\rho(\lambda)$ on λ and estimate it, before estimating the parameter of interest θ . This second-level prior is called a *hyperprior*, and parameter λ is a *hyperparameter*. Sometimes this hierarchy of priors and their parameters has more than two levels.

We’ll study the *empirical Bayes* approach later in this course.

4.3 Conjugate priors

Let us focus on the mathematically convenient families of prior distributions.

A suitably chosen prior distribution of θ may lead to a very tractable form of the posterior.

DEFINITION 4.1

A family of prior distributions π is **conjugate** to the model $f(\mathbf{x}|\theta)$ if the posterior distribution belongs to the same family.

Three classical examples of conjugate families are given below.

4.3.1 Gamma family is conjugate to the Poisson model

Let $(X_1, \dots, X_n)$ be a sample from $\text{Poisson}(\theta)$ distribution with a $\text{Gamma}(\alpha, \lambda)$ prior distribution of θ .

Then

$$f(\mathbf{x}|\theta) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!} \sim e^{-n\theta} \theta^{\sum x_i}. \quad (4.1)$$

Remark about dropping constant coefficients. In the end of (4.1), we dropped $(x_i!)$ and wrote that the result is “proportional” ($\sim$) to $e^{-n\theta} \theta^{\sum x_i}$. Dropping terms that don’t contain θ often

simplifies the computation. The form of the posterior distribution can be obtained without the constant term. If needed, we can eventually evaluate the normalizing constant in the end, making $\pi(\theta|\mathbf{x})$ a fine distribution with the total probability 1. In particular, the marginal distribution $m(\mathbf{x})$ can be dropped because it is θ -free. But for the computations, keep in mind that in this case we obtain the posterior distribution “up to a constant coefficient.”

The Gamma prior distribution of θ has density

$$\pi(\theta) \sim \theta^{\alpha-1} e^{-\lambda\theta}.$$

As a function of θ , this prior density has the same form as the model $f(\mathbf{x}|\theta)$ – a power of θ multiplied by an exponential function. This is the general idea behind conjugate families.

Then, the posterior distribution of θ given $\mathbf{X} = \mathbf{x}$ is

$$\pi(\theta|\mathbf{x}) \sim f(\mathbf{x}|\theta)\pi(\theta) \sim \left(e^{-n\theta}\theta^{\sum x_i}\right) (\theta^{\alpha-1}e^{-\lambda\theta}) \sim \theta^{\alpha+\sum x_i-1}e^{-(\lambda+n)\theta}.$$

Comparing with the general form of a Gamma density (sec. 13.1.2), we see that $\pi(\theta|\mathbf{x})$ is the Gamma distribution with new parameters,

$$\alpha_x = \alpha + \sum_{i=1}^n x_i \text{ and } \lambda_x = \lambda + n.$$

We can conclude that:

1. Gamma family of prior distributions is conjugate to Poisson models.
2. Having observed a Poisson sample $\mathbf{X} = \mathbf{x}$, we update the $\text{Gamma}(\alpha, \lambda)$ prior distribution of θ to the $\text{Gamma}(\alpha + \sum x_i, \lambda + n)$ posterior.

Gamma distribution family is rather rich; it has two parameters. There is often a good chance to find a member of this large family that suitably reflects our knowledge about θ .

Example 4.2 (NETWORK BLACKOUTS). The number of network blackouts each week has $\text{Poisson}(\theta)$ distribution. The weekly rate of blackouts θ is not known exactly, but according to the past experience with similar networks, it averages 4 blackouts with a standard deviation of 2.

There exists a Gamma distribution with the given mean $\mu = \alpha/\lambda = 4$ and standard deviation $\sigma = \sqrt{\alpha}/\lambda = 2$. Its parameters α and λ can be obtained by solving the system,

$$\begin{cases} \alpha/\lambda &= 4 \\ \sqrt{\alpha}/\lambda &= 2 \end{cases} \Rightarrow \begin{cases} \alpha &= (4/2)^2 = 4 \\ \lambda &= 2^2/4 = 1 \end{cases}$$

Hence, we can assume the $\text{Gamma}(\alpha = 4, \lambda = 1)$ prior distribution θ . It is convenient to have a conjugate prior because the posterior will then belong to the Gamma family too.

Suppose there were $X_1 = 2$ blackouts this week. Given that, the posterior distribution of θ is Gamma with parameters

$$\alpha_x = \alpha + 2 = 6, \quad \lambda_x = \lambda + 1 = 2.$$

If no blackouts occur during the next week, the updated posterior parameters become

$$\alpha_x = \alpha + 2 + 0 = 6, \quad \lambda_x = \lambda + 2 = 3.$$

This posterior distribution has the average weekly rate of $6/3 = 2$ blackouts per week. Two weeks with very few blackouts reduced our estimate of the average rate from 4 to 2. $\diamond$

4.3.2 Beta family is conjugate to the Binomial model

A sample from Binomial(k, θ) distribution (assume k is known) has the probability mass function

$$f(\mathbf{x} | \theta) = \prod_{i=1}^n \binom{k}{x_i} \theta^{x_i} (1 - \theta)^{k - x_i} \sim \theta^{\sum x_i} (1 - \theta)^{nk - \sum x_i}.$$

Density of Beta(α, β) prior distribution has the same form, as a function of θ ,

$$\pi(\theta) \sim \theta^{\alpha-1} (1 - \theta)^{\beta-1} \quad \text{for } 0 < \theta < 1$$

(see Section 13.1.2 in the Appendix). Then, the posterior density of θ is

$$\pi(\theta | \mathbf{x}) \sim f(\mathbf{x} | \theta) \pi(\theta) \sim \theta^{\alpha + \sum_{i=1}^n x_i - 1} (1 - \theta)^{\beta + nk - \sum_{i=1}^n x_i - 1},$$

and we recognize the Beta density with new parameters

$$\alpha_x = \alpha + \sum_{i=1}^n x_i \quad \text{and} \quad \beta_x = \beta + nk - \sum_{i=1}^n x_i.$$

Hence,

1. Beta family of prior distributions is conjugate to the Binomial model.
2. Posterior parameters are $\alpha_x = \alpha + \sum x_i$ and $\beta_x = \beta + nk - \sum x_i$.

4.3.3 Normal family is conjugate to the Normal model

Consider now a sample from Normal distribution with an unknown mean θ and a known variance σ^2 :

$$\begin{aligned} f(\mathbf{x} | \theta) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\} \sim \exp \left\{ -\sum_{i=1}^n \frac{(x_i - \theta)^2}{2\sigma^2} \right\} \\ &\sim \exp \left\{ \theta \frac{\sum x_i}{\sigma^2} - \theta^2 \frac{n}{2\sigma^2} \right\} = \exp \left\{ \left(\theta \bar{X} - \frac{\theta^2}{2} \right) \frac{n}{\sigma^2} \right\}. \end{aligned}$$

If the prior distribution of θ is also Normal, with prior mean μ and prior variance τ^2 , then

$$\pi(\theta) \sim \exp \left\{ -\frac{(\theta - \mu)^2}{2\tau^2} \right\} \sim \exp \left\{ \left(\theta\mu - \frac{\theta^2}{2} \right) \frac{1}{\tau^2} \right\},$$

and again, it has a similar form as $f(\mathbf{x}|\theta)$.

The posterior density of θ equals

$$\pi(\theta | \mathbf{x}) \sim f(\mathbf{x} | \theta)\pi(\theta) \sim \exp \left\{ \theta \left(\frac{n\bar{X}}{\sigma^2} + \frac{\mu}{\tau^2} \right) - \frac{\theta^2}{2} \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2} \right) \right\} \sim \exp \left\{ -\frac{(\theta - \mu_x)^2}{2\tau_x^2} \right\},$$

where

$$\mu_x = \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} \text{ and } \tau_x^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}. \quad (4.2)$$

This posterior distribution is certainly Normal with parameters μ_x and τ_x .

We can conclude that:

1. Normal family of prior distributions is conjugate to the Normal model with unknown mean;
2. Posterior parameters are given by (4.2).

Our results are summarized in Table 4.1. We'll see more examples of conjugate prior distributions in quizzes and exercises.

Model $f(\mathbf{x} \theta)$	Prior $\pi(\theta)$	Posterior $\pi(\theta \mathbf{x})$
Poisson(θ)	Gamma(α, λ)	Gamma($\alpha + n\bar{X}, \lambda + n$)
Binomial(k, θ)	Beta(α, β)	Beta($\alpha + n\bar{X}, \beta + n(k - \bar{X})$)
Normal(θ, σ)	Normal(μ, τ)	Normal $\left(\frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2}, \frac{1}{\sqrt{n/\sigma^2 + 1/\tau^2}} \right)$

TABLE 4.1: Three classical conjugate families.

4.3.4 Combining the data and the prior

Let us look again at the last example of a Normal model and a Normal prior for the population mean. We see in (4.2) that the posterior mean

$$\mu_x = \mathbf{E}(\theta | \mathbf{X}) = \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} = \left(\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \right) \bar{X} + \left(\frac{1/\tau^2}{n/\sigma^2 + 1/\tau^2} \right) \mu$$

is a weighted average of the prior mean μ and the sample mean $\bar{X}$. This is how the prior information and the observed data are combined in case of Normal distributions.

How will this posterior mean behave when it is computed from a large sample? As the sample

size n increases, we get more information from the data, and as a result, the frequentist estimator will dominate. According to (4.2), the posterior mean converges to the sample mean $\bar{X}$ as $n \rightarrow \infty$,

$$\mu_x = \left(\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \right) \bar{X} + \left(\frac{1/\tau^2}{n/\sigma^2 + 1/\tau^2} \right) \mu \longrightarrow \bar{X} \text{ as } n \rightarrow \infty.$$

Posterior mean will also converge to $\bar{X}$ when $\tau \rightarrow \infty$. Large τ means a lot of uncertainty in the prior distribution of θ ; thus, naturally, we should rather use observed data as a more reliable source of information in this case.

On the other hand, large σ indicates a lot of uncertainty in the observed sample. If that is the case, the prior distribution is more reliable, and as we see in (4.2), $\mu_x \approx \mu$ for large σ ,

$$\mu_x = \left(\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2} \right) \bar{X} + \left(\frac{1/\tau^2}{n/\sigma^2 + 1/\tau^2} \right) \mu \longrightarrow \mu \text{ as } \sigma \rightarrow \infty.$$

We get the same result when $\tau \rightarrow 0$ which implies that the prior mean μ is a very accurate estimator of θ , and there is practically no uncertainty about it.

4.4 Non-informative prior distributions and generalized Bayes rules

One of the main arguments of non-Bayesians against Bayesians was the *subjective choice of the prior distribution*. Indeed, it is not always trivial to come up with a realistic distribution that truly reflects our prior knowledge and uncertainty about the unknown parameter.

As an extreme case, what can we do if we have *no prior knowledge* whatsoever? No information about the parameter until we see the actual data... However, we still would like to use Bayesian methods because of their nice properties that we'll see in further chapters. Is there a “fair” prior distribution that reflects our absence of knowledge? Such a distribution would be called a **non-informative prior distribution**.

Example 4.3 (CAUTION: THERE ARE TRANSFORMATIONS!). Consider estimation of parameter θ of Binomial(k, θ) distribution. We know that $\theta \in [0, 1]$, and suppose that nothing else is known about θ . Then, we should choose a prior distribution that gives equal weights to all values of θ , making them all “equally likely”. So, let $\pi(\theta) \sim \text{Uniform}(0, 1)$?

Indeed, this seems to be the most natural non-informative choice. However... any non-linear transformation of θ , its reparameterization, appears non-Uniform!

For example, consider another parameter $\eta = \theta^2$. It also belongs to $[0, 1]$. Its cumulative distribution function

$$F_\eta(t) = \mathbf{P}\{\eta \leq t\} = \mathbf{P}\{\theta^2 \leq t\} = \mathbf{P}\{\eta \leq \sqrt{t}\} = \sqrt{t},$$

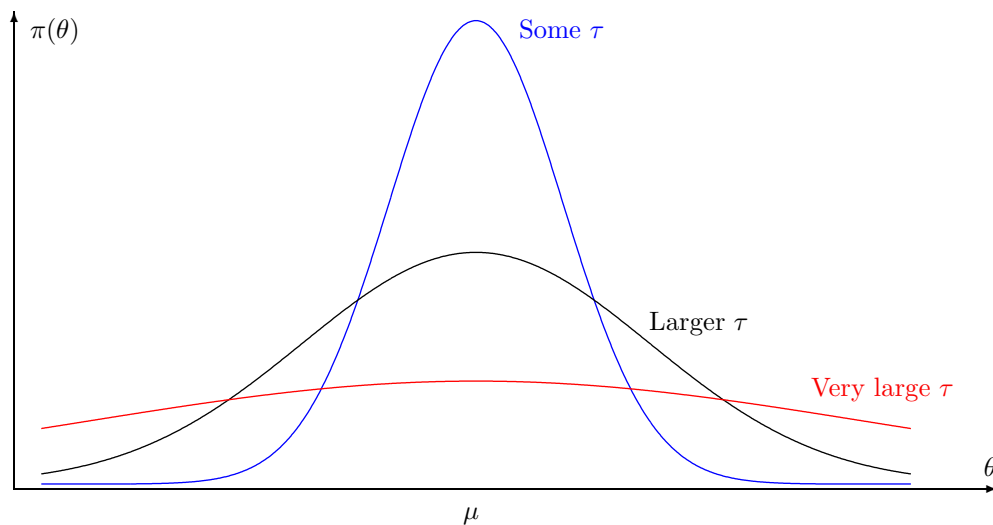


FIGURE 4.1: Normal prior densities with increasing prior variance $\text{Var}(\theta) = \tau^2$.

which leads to a density

$$f_{\eta}(t) = F'_{\eta}(t) = \frac{1}{2\sqrt{t}}.$$

This density is no longer Uniform! How comes that we had no information about parameter θ , but somehow we do possess some information about $\eta = \theta^2$??? $\diamond$

So, a simple approach did not work. What seemed to be a non-informative prior distribution of θ appears informative for θ^2 . Let us try a different way.

Looking again at the case of a Normal model with a Normal prior in Sec. 4.3.4, we noticed that a *high prior variance* $\tau^2 = \text{Var}(\theta)$ means a lot of uncertainty about the unknown parameter, and therefore, little information about it. Taking it to the extreme, when $\tau^2 \rightarrow \infty$, we are infinitely uncertain about θ . As $\tau^2 \rightarrow \infty$, the $\text{Normal}(\mu, \tau)$ prior density becomes more and more flat, converging to a constant (Fig. 4.1).

One problem though... There is no constant density on $\mathbf{R} = (-\infty, \infty)$. There is a constant measure, Lebesgue measure, with $\pi(\theta) \equiv 1$ for all $\theta \in \mathbf{R}$, but

$$\int_{-\infty}^{\infty} \pi(\theta) d\theta = \int_{-\infty}^{\infty} d\theta = \infty,$$

so this is not a probability measure...

Nevertheless, Lebesgue measure gives us a fine posterior distribution

$$\pi(\theta | \mathbf{x}) \sim f(\mathbf{x} | \theta) \pi(\theta) \sim \exp \left\{ \left(\theta \bar{X} - \frac{\theta^2}{2} \right) \frac{n}{\sigma^2} \right\} \sim \exp \left\{ -\frac{(\theta - \bar{X})^2}{2\sigma^2/n} \right\} \sim \text{Normal}(\bar{X}, \sigma^2/n).$$

Relations between the sample mean $\bar{X}$ and the parameter pretty symmetric in this case:

$$\text{Given } \theta, \quad \bar{X} \sim \text{Normal}(\theta, \sigma^2/n)$$

$$\text{Given } \mathbf{X}, \quad \theta \sim \text{Normal}(\bar{X}, \sigma^2/n)$$

That is, without any prior information, after seeing the data, we have exactly as much uncertainty about θ as the data contain. This is quite reasonable. So, such a prior worth consideration, even though it is not, strictly speaking, a distribution.

DEFINITION 4.2

An **improper prior distribution** is an infinite measure on the parameter space Θ (that is, $\int_{\Theta} d\pi(\theta) = \infty$) which produces a proper posterior distribution.

Decision that minimizes the posterior risk under an improper prior is called a **generalized Bayes rule**.

Notice from (4.2) that $\mu_x \neq \bar{X}$ for any proper Normal prior. However, $\mu_x \rightarrow \bar{X}$ under proper priors, as $\tau \rightarrow \infty$, and in the limiting case, $\mu_x = \bar{X}$ under the improper constant prior.

Generalized Bayes rules are limits of proper Bayes rules.

4.4.1 Invariant non-informative priors

So, which priors are “truly” non-informative in different situations? Let us try to generalize what we have achieved. Our solution will depend on a model, a family of distributions of $\mathbf{X}$.

Location family

Parameter θ is a **location parameter** if the distribution of data has the form

$$F_{\theta}(x) = F(x - \theta).$$

Then, different θ cause shifts in the distribution of $\mathbf{X}$ without changing its shape, variation, and other properties.

In this case, a **non-informative prior distribution** should equally treat θ and its shift $\eta = \theta + c$. That is, for any Borel (that is, measurable, a set that has probability) set $A \subset \mathbf{R}$, we would like to have equality

$$\mathbf{P}\{\theta \in A\} = \mathbf{P}\{\eta \in A\} = \mathbf{P}\{\theta + c \in A\} = \mathbf{P}\{\theta \in A - c\}.$$

Which measure π can satisfy this condition for all (Borel) sets A ? Computing these probabilities by integrating the prior density and making a substitution, we get

$$\int_A \pi(\theta) d\theta = \int_{A-c} \pi(t) dt = \int_A \pi(\theta + c) d\theta$$

for any A , and therefore, $\pi(\theta) = \pi(\theta + c)$ for $\forall c \in \mathbf{R}$.

This means, $\pi(\theta) \equiv \text{const.}$ For example, $\pi(\theta) \equiv 1$, a that we considered for the Normal model. Turns out, it suits any location family.

Scale family

Parameter $\theta > 0$ is a **scale parameter** if the distribution of data has the form

$$F_\theta(x) = F\left(\frac{x}{\theta}\right).$$

Seeking invariance under *rescaling*, let's look for a **non-informative prior distribution** that equally treats θ and its multiple $\eta = c\theta$. Then

$$\mathbf{P}\{\theta \in A\} = \mathbf{P}\{\eta \in A\} = \mathbf{P}\{c\theta \in A\} = \mathbf{P}\{\theta \in A/c\},$$

and (substituting $\theta = ct$ in the integral),

$$\int_A \pi(\theta) d\theta = \int_{A/c} \pi(t) dt = \int_A \pi\left(\frac{\theta}{c}\right) \frac{d\theta}{c}$$

for any A , and therefore, $\pi(\theta) = \frac{1}{c} \pi\left(\frac{\theta}{c}\right)$ for $\forall c > 0$.

We can easily solve this! Put $c = \theta$ and obtain $\pi(\theta) = \frac{1}{\theta} \pi(1) \sim \frac{1}{\theta}$. Thus, we derived a non-informative prior distribution for the scale family of models,

$$\pi(\theta) = \frac{1}{\theta}, \quad \theta > 0.$$

Since $\int_0^\infty \pi(\theta) d\theta = \int_0^\infty \frac{1}{\theta} d\theta = \infty$, this is also an improper prior.

Example 4.4 (IT MAKES SENSE! 4.14 OF BERGER). Assume $\mathbf{X}$ is a sample from $\text{Poisson}(\theta)$ distribution. Use the improper non-informative prior $\pi(\theta) = \frac{1}{\theta}$, $\theta > 0$.

- (a) Find the posterior density of θ given $\mathbf{X} = \mathbf{x}$ for $\mathbf{x} \neq (0, \dots, 0)$.
- (b) If $\mathbf{x} = (0, \dots, 0)$, the posterior distribution does not exist... and it makes sense... why? What would be a standard non-Bayesian estimator in this case, and does it make sense?

... ..

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Chapter 5

Loss Functions, Utility Theory, and Rao-Blackwellization

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Similarly to the prior distributions, loss functions $L(\theta, \delta)$ can be chosen either in a [subjective](#) way or in a [convenient and standard](#) way.

5.1 Utility Theory and Subjectively Chosen Loss Functions

Loss functions reflect our anticipation of possible results of our decisions, gains and penalties. These are not trivial to predict, especially when randomness is involved. For example, when a student chooses a major or an investor chooses a portfolio, results of their decisions are not immediately clear at all! Did you choose your major successfully? Success of many ventures depends on a number of factors and perhaps, on a random chance as well.

Utility theory explains how one can quantify and order these random rewards and use this ordering in decision making. It is understood that:

1. Every statistical decision results in a **random reward** r , which has some *distribution of rewards* $\mathcal{P}(r)$.
2. **Utility function** of rewards $U(r)$ shows how valuable reward r is for you. You should prefer rewards with higher $U(r)$.

3. Let $\mathcal{P}$ be a distribution of rewards. Its value is given as the expectation $\mathbf{E}^{\mathcal{P}(r)}U(r)$. That is, distribution of rewards $\mathcal{P}_2$ is **preferred** to distribution $\mathcal{P}_1$ if

$$\mathbf{E}^{\mathcal{P}_2(r)}U(r) > \mathbf{E}^{\mathcal{P}_1(r)}U(r).$$

NOTATION			
	$\mathcal{P}_1 \prec \mathcal{P}_2$	means	distribution of rewards $\mathcal{P}_2$ is preferred to $\mathcal{P}_1$
	$\mathcal{P}_1 \approx \mathcal{P}_2$	means	$\mathcal{P}_1$ and $\mathcal{P}_2$ are equivalent rewards , i.e., none of them is preferred to the other
	$w\langle r_1 \rangle + (1-w)\langle r_2 \rangle$	means	a random reward that equals r_1 with probability w and equals r_2 with probability $(1-w)$

This is how we can compare different random rewards and choose statistical decisions that maximize the value of these rewards for us.

It only remains to construct the utility function $U(r)$ that truly reflects our preference among rewards. Then, the loss function will be understood as the negative utility,

$$L(\theta, \delta) = -U(r(\theta, \delta)).$$

In general, we choose decisions to maximize the utility or to minimize the loss.

5.1.1 Construction of a utility function

Step # 1. Two initial rewards. Choose two non-random rewards $r_1 \prec r_2$ and assign some arbitrary utilities, $U(r_1) < U(r_2)$. For example, simply take $U(r_1) = 0$ and $U(r_2) = 1$. If there exist the smallest and the largest rewards, then it is convenient to name them r_1 and r_2 , but it is not necessary.

Well, indeed, in order to compare our decisions, we only need to determine utilities and losses up to a constant shift and a constant coefficient. That is, a loss $L(\theta, \delta)$ is just as good to use as $a + bL(\theta, \delta)$ for any a and any $b > 0$.

So, we have two degrees of freedom, and we can choose $U(r_1)$ and $U(r_2)$ arbitrarily.

Step # 2. Intermediate rewards. Consider a **distribution of rewards** $\mathcal{P}_3 = \frac{1}{2}\langle r_1 \rangle + \frac{1}{2}\langle r_2 \rangle$, which is a gamble, a lottery ticket that wins either r_1 or r_2 with a 50-50 chance. Which non-random reward r_3 is equivalent to $\mathcal{P}_3$? If we are talking about monetary rewards, how much are you willing to pay for this lottery ticket? Let a fair price be r_3 . It reflects your personal preferences, how much you like to risk or to avoid risk (uncertainty, randomness).

When r_3 is determined, compute its **utility** as

$$U(r_3) = \mathbf{E}^{\mathcal{P}_3}U(r) = \frac{1}{2}U(r_1) + \frac{1}{2}U(r_2),$$

which is necessary for rewards $\langle r_3 \rangle$ and $\mathcal{P}_3 = \frac{1}{2}\langle r_1 \rangle + \frac{1}{2}\langle r_2 \rangle$ to be equivalent.

If $U(r_1) = 0$ and $U(r_2) = 1$, then $U(r_3) = 0.5$. So, now we have determined utilities for three rewards r_1 , r_2 , and r_3 .

More steps # 2. To build a **utility function** $U(r)$, we need more points. Pick two rewards r_i and r_j whose utilities are already determined and find a non-random reward r_k such that

$$\langle r_k \rangle \approx \frac{1}{2}\langle r_i \rangle + \frac{1}{2}\langle r_j \rangle.$$

Then set its utility

$$U(r_k) = \frac{1}{2}U(r_i) + \frac{1}{2}U(r_j).$$

Variations of step # 2.

2a. Asymmetric gambles. Instead of a 50-50 gamble $\frac{1}{2}\langle r_i \rangle + \frac{1}{2}\langle r_j \rangle$, we can consider a distribution

$$\mathcal{P} = w\langle r_i \rangle + (1 - w)\langle r_j \rangle,$$

a lottery ticket that gives reward r_i with probability w and reward r_j with probability $(1 - w)$.

Find a non-random reward $r_k \approx \mathcal{P}$ and set its utility as

$$U(r_k) = wU(r_i) + (1 - w)U(r_j).$$

2b. Determine probabilities. Instead of a fair price for lottery ticket $\mathcal{P}$, we can set some non-random reward r and then determine the probabilities w and $(1 - w)$ that make $r \approx \mathcal{P}$. (This is similar to a Progressive Insurance “Name-your-price tool”, where you can choose what you want to pay for the insurance coverage, and they will find a coverage for that price. Again, set

$$U(r) = wU(r_i) + (1 - w)U(r_j).$$

2c. Rewards below r_1 . So far, we played with rewards between the initially chosen r_1 and r_2 . However, if there are rewards $r \prec r_1$, then it is worse (less valuable for us) than any reward considered so far.

Apply Step 2b to this case by finding the probability w that make r_1 a fair price for a gamble between r and r_2 (or another award),

$$\langle r_1 \rangle \approx w\langle r \rangle + (1 - w)\langle r_2 \rangle.$$

This implies $U(r_1) = wU(r) + (1 - w)U(r_2)$, and we can find utility $U(r)$ from here,

$$U(r) = \frac{U(r_1) - (1 - w)U(r_2)}{w}.$$

2d. Rewards above r_2 . Similarly, for rewards $r \succ r_2$ that are more valuable than r_2 , find probability w that makes

$$\langle r_2 \rangle \approx w \langle r_1 \rangle + (1 - w) \langle r \rangle.$$

This implies $U(r_2) = wU(r_1) + (1 - w)U(r)$, hence,

$$U(r) = \frac{U(r_2) - wU(r_1)}{1 - w}.$$

Repeat Step 2 and its variations to obtain a sufficient number of points that you can connect and draw a utility function $U(r)$.

Step # 3. Cross-validation. When $U(r)$ is sketched, verify a few random combinations of rewards. Choose some pairs r_m and r_n that have not been used so far and determine either a reward r or a probability w , applying some version of Step 2,

$$\langle r \rangle \approx w \langle r_m \rangle + (1 - w) \langle r_n \rangle.$$

This should imply $U(r) = wU(r_m) + (1 - w)U(r_n)$. Compare this $U(r)$ with the utility of reward r obtained from the sketched function U . Does the point $(r, U(r))$ lie on the graph of our utility function? If not, even approximately, then revise some of the previous steps.

Example 5.1 (INVEST OR NOT INVEST?). Let's decide whether we should invest in today's economy or not, based on our personal **utility function**. For simplicity, let the action space consist of two decisions

$$\mathcal{A} = \{\delta_1, \delta_2\} = \{\text{invest, don't invest}\}$$

and the parameter space consist of two possible states of economy,

$$\Theta = \{\theta_1, \theta_2\} = \{\text{market goes up, market goes down}\}.$$

Then there are 4 possible non-random rewards,

	Invest	Don't invest
Market up	r_1	r_2
Market down	r_3	r_4

Here r_1 is a real gain, r_2 is a lost opportunity, r_3 is a real loss, and r_4 a correct decision although it is not a monetary gain. Therefore, these rewards are ordered as

$$r_3 \prec r_2 \prec r_4 \prec r_1.$$

Let us determine all the utilities and then decide whether or not we should invest today, given a priori that the market will go up with probability $\pi(\theta_1) = 0.52$.

Solution.

Step 1. Let $U(r_3) = 0$ and $U(r_1) = 1$.

Step 2. Rewards are already given, so apply the version 2b.

I'd say, r_2 (lost opportunity) is closer to r_3 (loss) than to r_1 (gain), and so, it is about equivalent to a gamble $0.55\langle r_3 \rangle + 0.45\langle r_1 \rangle$. Therefore,

$$U(r_2) = 0.55U(r_3) + 0.45U(r_1) = 0.45.$$

Also, r_4 (right choice for no gain) is a lot closer to r_2 (lost opportunity but also no dollar gain or loss) than to r_1 (real gain). Therefore, I'd estimate that

$$\langle r_4 \rangle \approx 0.9\langle r_2 \rangle + 0.1\langle r_1 \rangle,$$

so that

$$U(r_4) = (0.9)(0.45) + (0.1)(1) = 0.505.$$

We have the utility function $U(r_3) = 0$, $U(r_2) = 0.45$, $U(r_4) = 0.505$, and $U(r_1) = 1$.

Step 3. Let's use some unused combinations for cross-validation. For me, r_2 would be about equivalent to an 80-20 gamble between r_4 and r_3 . This is pretty subjective, isn't it, but I know that two rewards r_2 and r_4 , that carry no monetary gain or loss, are much closer to each other than to a real gain or loss.

So, I claim

$$\langle r_2 \rangle \approx 0.8\langle r_4 \rangle + 0.2\langle r_3 \rangle,$$

from where

$$U(r_2) = (0.8)U(r_4) + (0.2)U(r_3) = (0.8)(0.505) + (0.2)(0) = 0.404.$$

That's close to our previously determined $U(r_2) = 0.45$, so I would not revise our utility function.

Having determined all the utilities, **should we invest?** With the given prior distribution, let us find the **Bayes decision**.

Compute the expected utilities of both possible decisions,

$$\begin{aligned} \mathbf{E}_{\delta_1}^{\pi(\theta)} U(r) &= \pi(\theta_1)U(r_1) + \pi(\theta_2)U(r_3) = (0.52)(1) + (0.48)(0) = 0.52 \\ \mathbf{E}_{\delta_2}^{\pi(\theta)} U(r) &= \pi(\theta_1)U(r_2) + \pi(\theta_2)U(r_4) = (0.52)(0.45) + (0.48)(0.505) = 0.4764 \end{aligned}$$

Translating these into our old terms, the Bayes risk of δ_1 is -0.52 , and the Bayes risk of δ_2 is -0.4764 .

We go with the higher expected utility and the lower Bayes risk, and therefore, our best (Bayes) decision is δ_1 , **to invest**.

Notice that with a more pessimistic opinion about the state of economy, say, $\pi(\theta_1) = 0.4$ and $\pi(\theta_2) = 0.6$, the expected utilities would have been $\mathbf{E}_{\delta_1}^{\pi(\theta)} U(r) = 0.4$ and $\mathbf{E}_{\delta_2}^{\pi(\theta)} U(r) = 0.483$, preventing us from investing. $\diamond$

Example 5.2 (PERSONAL UTILITY FUNCTION FOR MONEY). Let me briefly try to apply steps 1-3 to construct my own personal utility function for money.

Two meaningful rewards for me are \$0 and \$1,000, so let $U(0) = 0$ and $U(1000) = 1$.

Then,

$$400 \approx \frac{1}{2}\langle 0 \rangle + \frac{1}{2}\langle 1000 \rangle \Rightarrow U(400) = 0.5U(0) + 0.5U(1) = 0.5$$

(that is, I'd agree to pay \$400 for a 50-50 gamble between \$0 and \$1000, but I'd refuse to pay \$500 for such a gamble; I'd prefer \$500 in cash)

$$600 \approx \frac{1}{2}\langle 400 \rangle + \frac{1}{2}\langle 1000 \rangle \Rightarrow U(600) = 0.5U(400) + 0.5U(1000) = 0.75$$

$$750 \approx \frac{1}{2}\langle 600 \rangle + \frac{1}{2}\langle 1000 \rangle \Rightarrow U(750) = 0.5U(600) + 0.5U(1000) = 0.875$$

$$480 \approx \frac{1}{2}\langle 400 \rangle + \frac{1}{2}\langle 600 \rangle \Rightarrow U(480) = 0.5U(400) + 0.5U(600) = 0.625$$

$$140 \approx \frac{1}{2}\langle 0 \rangle + \frac{1}{2}\langle 400 \rangle \Rightarrow U(140) = 0.5U(0) + 0.5U(400) = 0.25$$

$$240 \approx \frac{1}{2}\langle 140 \rangle + \frac{1}{2}\langle 400 \rangle \Rightarrow U(240) = 0.5U(140) + 0.5U(400) = 0.375$$

Check: is $400 \approx \frac{1}{2}\langle 240 \rangle + \frac{1}{2}\langle 480 \rangle$? No, I would not pay \$400 for this gamble. It is probably \$250 worth for me.

Then, let us redefine $U(350) = 0.5$ and $U(580) = 0.75$, because $580 \approx \frac{1}{2}\langle 350 \rangle + \frac{1}{2}\langle 1000 \rangle$.

Also, include some negative rewards, or money losses:

$$0 \approx \frac{2}{5}\langle -500 \rangle + \frac{3}{5}\langle 500 \rangle \Rightarrow U(-500) = -1.5U(500) \approx (-1.5)(0.7) = -1.05$$

$$-100 \approx \frac{1}{2}\langle -500 \rangle + \frac{1}{2}\langle 500 \rangle \Rightarrow U(-100) = 0.5U(-500) + 0.5U(500) = -0.175$$

$$-150 \approx \frac{1}{2}\langle -500 \rangle + \frac{1}{2}\langle 0 \rangle \Rightarrow U(-150) = 0.5U(-500) + 0.5U(0) = -0.0525$$

$$-500 \approx \frac{1}{4}\langle -1000 \rangle + \frac{3}{4}\langle 0 \rangle \Rightarrow U(-1000) = 4U(-500) = -4.2$$

(Some more checks would be useful for the final tuning.) Then, we have obtained the following utility function for money between $r = -1000$ and $r = 1000$.

r	-1000	-500	-150	-100	0	140	240	350	480	500	580	750	1000
$U(r)$	-4.2	-1.05	-0.525	-0.175	0	0.25	0.375	0.5	0.625	0.7	0.75	0.875	1

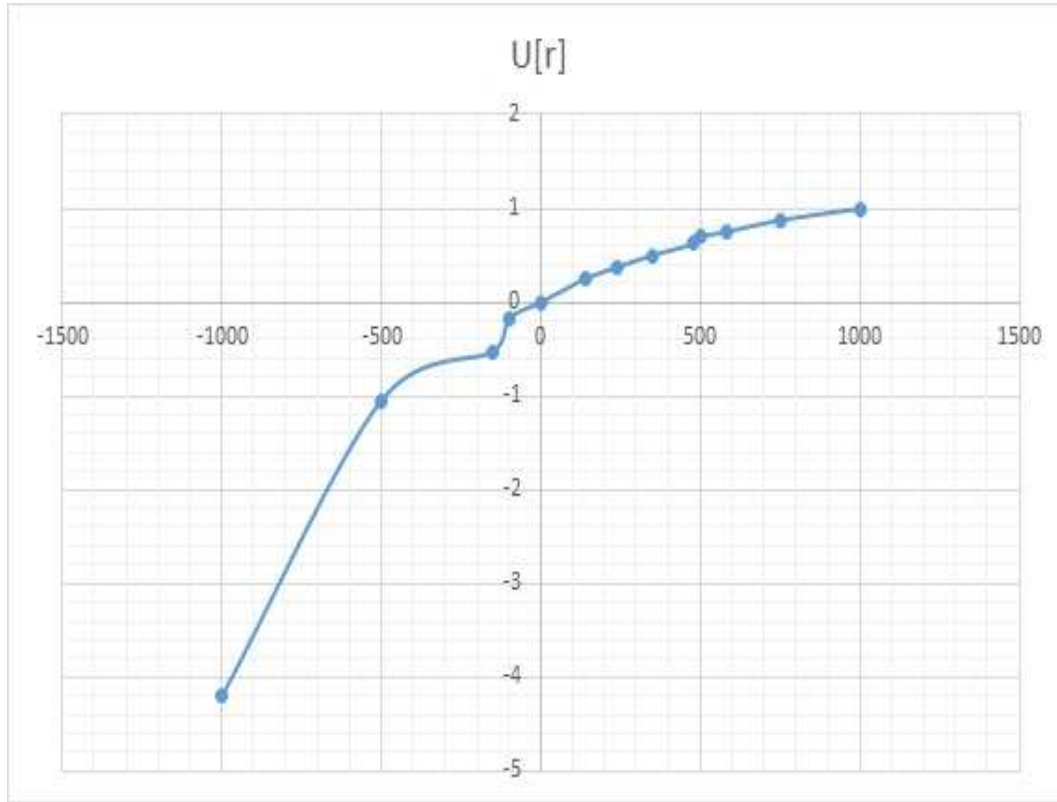
Its graph is on Figure 5.1. ◇

Notice several important and common features of $U(r)$, the personal utility function for money:

1. $U(r)$ is **concave** for $r > 0$. This means *risk aversion*. That is, I would not pay for the additional risk (uncertainty), and I would agree to pay for the risk only if its expected reward is higher than my payment,

$$wU(r_1) + (1-w)U(r_2) = \mathbf{E}\{U(w\langle r_1 \rangle + (1-w)\langle r_2 \rangle)\} < U(wr_1 + (1-w)r_2),$$

i.e., a hard cash reward of $wr_1 + (1-w)r_2$ is more valuable for me than a gamble $w\langle r_1 \rangle + (1-w)\langle r_2 \rangle$ with a similar expected value.

FIGURE 5.1: *Constructed utility function for money.*

- (Notice that $wU(r_1) + (1 - w)U(r_2) < U(wr_1 + (1 - w)r_2)$ is one of the definitions of convexity.)
2. $U(r)$ is practically flat and most likely, bounded for large r . It means that really large rewards r make little difference for an individual. While I can feel the difference between \$10 and \$20, the difference between \$10,000 and \$10,010 is very unclear to me. Further, since I have no way to spend one billion dollars, a gift of one billion means the same to me as a gift of one trillion.
 3. $U(r)$ is almost linear around $r \approx 0$. For small amounts, I do not mind to gamble.

Since the utility function is mostly **concave**, the loss $L = -U$ is mostly **convex**. Almost all commonly used loss functions are convex.

Example 5.3 (WE HAVE NO REASON TO FIGHT! BERGER 4.9 FROM DEGROOT). Assume that Mr. A and Mr. B have the same utility function for money, given by $U(x) = x^{1/3} = \sqrt[3]{x}$. Suppose now that one of the men receives, as a gift, a lottery ticket which yields either a reward of $r > 0$ dollars or a reward of 0 dollars, with probability $\frac{1}{2}$ each. Show that there exists a number $b > 0$ having the following property: regardless of which man receives the

lottery ticket, he can sell it to the other man for b dollars and the sale will be advantageous to both men.

... ..

◇

Example 5.4 (FOOD FOR THOUGHT). The statement of the previous example remains correct even if the ticket's winning probability is some $p \in (0, 1)$ instead of $\frac{1}{2}$. ◇

Example 5.5 (ANOTHER EXERCISE). All right, Mr. A and Mr. B calculated the good price b , and then Mr. A sold it to Mr. B for $\$b$. Now, perhaps Mr. A would like to buy it back? Show that now there is no such price that can make this deal possible. ◇

5.2 Standard loss functions and corresponding Bayes decision rules

In this section, we list several loss functions that are most frequently used in statistical decision making. Notice that one decision will remain better than another in Bayes or minimax sense if the entire loss function is increased by a constant or multiplied by a positive coefficient. Therefore,

- We can drop constant coefficients and shifts;
- We can consider only non-negative losses. As we observed in the previous section, utility functions are usually bounded from above. Then, losses are bounded from below. Without loss of generality, add this (constant) lower bound to the loss function, making it non-negative, $L(\theta, \delta) \geq 0$.

5.2.1 Squared-error loss

The **squared-error loss function**

$$L(\theta, \delta) = (\theta - \delta)^2.$$

is used for parameter estimation problems. The parameter is θ , and its estimator is δ .

The corresponding squared-error **posterior risk** is therefore

$$\rho(\pi, \delta | \mathbf{X}) = \mathbf{E}^\theta \{(\theta - \delta)^2 | \mathbf{X}\} = \mathbf{E}^\theta \{(\theta - \mu_X)^2 | \mathbf{X}\} + (\mu_X - \delta(\mathbf{X}))^2,$$

which can be interpreted as the **posterior variance** of θ plus **posterior bias** squared. Here $\mu_X = \mathbf{E}\{\theta | \mathbf{X}\}$ is the **posterior mean** of θ .

What is the **Bayes decision** with respect to the squared-error loss? It is the one that minimizes the posterior risk which we'll write as

$$\rho(\pi, \delta | \mathbf{X}) = \mathbf{E}^\theta \{(\theta - \delta)^2 | \mathbf{X}\} = \delta^2 - 2\delta \mathbf{E}\{\theta | \mathbf{X}\} + \mathbf{E}\{\theta^2 | \mathbf{X}\}.$$

The minimum of this Bayes risk is attained at

$$\delta_{\text{Bayes}} = -\frac{-2 \mathbf{E}\{\theta | \mathbf{X}\}}{2} = \mathbf{E}\{\theta | \mathbf{X}\} = \mu_X.$$

Conclusion: **posterior mean** of θ is the Bayes decision with respect to the squared-error loss.

5.2.2 Absolute-error loss

The **absolute-error loss function** is

$$L(\theta, \delta) = |\theta - \delta|.$$

It does not penalize as much as the squared-error loss for large deviations of the estimator δ from the parameter θ .

How should the Bayes decision look like, under this loss? Apparently, it's the **posterior median**.

PROOF: Consider M , the median of $\pi(\theta | \mathbf{X})$, and δ , some decision, and compare their losses.

Case 1, let $\delta < M$. The difference of losses is

$$L(\theta, \delta) - L(\theta, M) = |\theta - \delta| - |\theta - M| = \begin{cases} M - \delta & \text{for } \theta \geq M \\ \text{a linear continuous function} \\ \text{between } \delta \text{ and } M \\ -(M - \delta) & \text{for } \theta \leq \delta \end{cases}$$

This function of θ is on Figure 5.2. So, $L(\theta, \delta) - L(\theta, M) \geq -(M - \delta)$ for $\theta < M$, and it equals $(M - \delta)$ for $\theta \geq M$.

Taking expected values with respect to the posterior distribution $\pi(\theta | \mathbf{X})$, we obtain the difference of posterior risks,

$$\begin{aligned} \rho(\pi, \delta | \mathbf{X}) - \rho(\pi, M | \mathbf{X}) &\geq -(M - \delta) \mathbf{P}\{\theta < M | \mathbf{X}\} + (M - \delta) \mathbf{P}\{\theta \geq M | \mathbf{X}\} \\ &= (M - \delta) (\mathbf{P}\{\theta \geq M | \mathbf{X}\} - \mathbf{P}\{\theta < M | \mathbf{X}\}). \end{aligned}$$

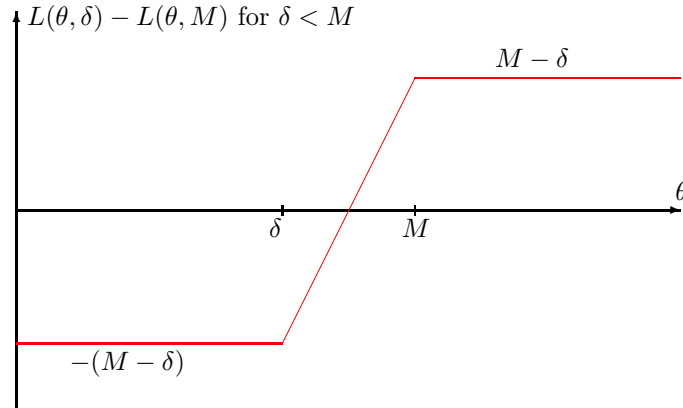


FIGURE 5.2: Difference of loss functions for the posterior median M and another decision $\delta < M$.

It's time to recall that M is the median of $\pi(\theta | \mathbf{X})$, and therefore,

$$\mathbf{P}\{\theta \geq M | \mathbf{X}\} \geq \mathbf{P}\{\theta < M\}.$$

(For a continuous posterior, $\mathbf{P}\{\theta < M | \mathbf{X}\} = \mathbf{P}\{\theta \geq M | \mathbf{X}\} = 1/2$.)

Conclusion? $\rho(\pi, \delta | \mathbf{X}) - \rho(\pi, M | \mathbf{X}) \geq 0$.

Case 2, $\delta > M$, is very similar. In both cases, we get $\rho(\pi, \delta | \mathbf{X}) \geq \rho(\pi, M | \mathbf{X})$. This precisely means that **M is the Bayes rule**. No other decision δ can have a lower posterior risk. $\square$

Conclusion: **posterior median** of θ is the Bayes decision with respect to the absolute-error loss.

5.2.3 Zero-one loss

Generally, the **zero-one loss function** gives a penalty of 1 for any wrong decision and no penalty for the correct decision. This makes sense in hypothesis testing - Type I and Type II errors are penalized while correct acceptance and correct rejections aren't.

In estimation, a 0-1 loss can be defined as

$$L(\theta, \delta) = I\{\theta \neq \delta\} = \begin{cases} 1 & \text{if } \theta \neq \delta \\ 0 & \text{if } \theta = \delta \end{cases}$$

However, this only makes sense in discrete cases, when $\theta = \delta$ with a non-zero probability. Then, what is the **Bayes decision** under this loss function?

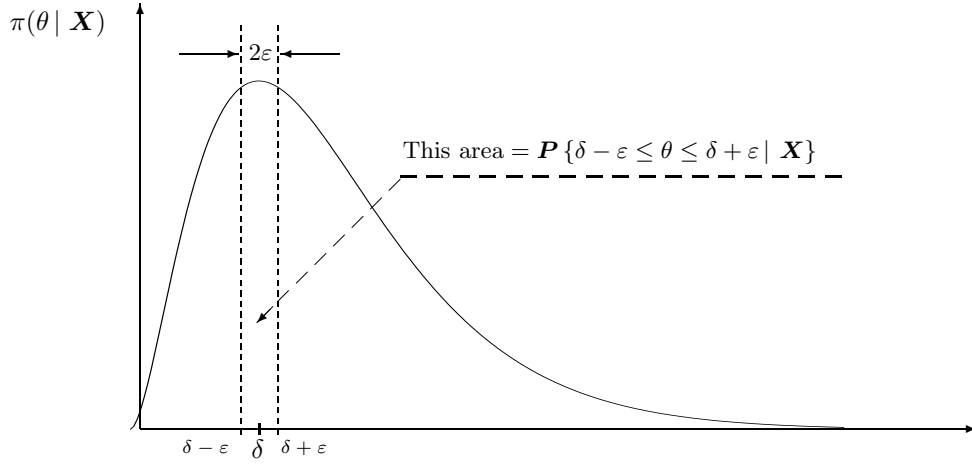


FIGURE 5.3: Bayes estimator δ of a continuous parameter θ under the 0-1 loss function has to maximize the posterior probability of the interval $\delta \pm \varepsilon$.

Compute the posterior risk,

$$\rho(\pi, \delta | \mathbf{X}) = \mathbf{E}^\theta \{I\{\theta \neq \delta\} | \mathbf{X}\} = \mathbf{P}\{\theta \neq \delta | \mathbf{X}\} = 1 - \mathbf{P}\{\theta = \delta | \mathbf{X}\} = 1 - \pi(\delta | \mathbf{X}).$$

Now, the Bayes rule should minimize this posterior risk. To do that, it maximizes $\pi(\delta | \mathbf{X})$. The Bayes rule is the point of maximum of the posterior distribution (probability mass function) $\pi(\theta | \mathbf{X})$! How do we call it? It is the **posterior mode**.

Conclusion: For a discrete parameter θ , **posterior mode** is the Bayes decision with respect to the zero-one loss.

In continuous cases, the probability of $\theta = \delta$ is 0, so there is nothing to maximize. For this reason, the 0-1 loss is often defined as

$$L(\theta, \delta) = I\{|\theta - \delta| > \varepsilon\} = \begin{cases} 1 & \text{if } |\theta - \delta| > \varepsilon \\ 0 & \text{if } |\theta - \delta| \leq \varepsilon \end{cases}$$

allowing the estimator δ to differ from the parameter θ by at most a small ε . The Bayes decision δ in this case maximizes the probability $\mathbf{P}\{\delta - \varepsilon \leq \theta \leq \delta + \varepsilon | \mathbf{X}\}$, and it converges to the **posterior mode** as $\varepsilon \rightarrow 0$, as on Figure 5.3.

DEFINITION 5.1

Generalized maximum likelihood estimator of the parameter θ is the posterior mode, the value of θ that maximizes the posterior density or posterior probability mass function

Do you see a connection with the method of maximum likelihood? Maximum likelihood estimators (**MLE**) maximize the joint density or probability mass function $f(\mathbf{X} \mid \theta)$, “the model” while generalized maximum likelihood estimators (**GMLE**) maximize the posterior, which is proportional to $f(\mathbf{X} \mid \theta)\pi(\theta)$. Thus, MLE is a special case of GMLE when the prior distribution π is *Uniform*.

We’ve just obtained an interesting result: *maximum likelihood estimators are Bayes decisions, under the zero-one loss function and Uniform prior distribution*. If θ is continuous, its maximum likelihood estimator is asymptotically Bayes, under the zero-one loss function $L(\theta, \delta) = I\{|\theta - \delta| > \varepsilon\}$ with $\varepsilon \downarrow 0$.

5.2.4 Other loss functions

A few words about some other commonly used loss functions.

Weighted loss functions allow larger errors when the parameter is large. For example, a weighted squared-error loss function is

$$L_w(\theta, \delta) = w(\theta)(\theta - \delta)^2,$$

where the weight $w(\theta)$ decreases as $|\theta|$ gets large. It just means that a 1-inch error on a 1-mile parameter θ causes less penalty than a 1-inch error on a 1-foot θ .

A **LinEx loss function** (linear-exponential)

$$L(\theta, \delta) = e^{-\alpha(\delta - \theta)} - 1 + \alpha(\delta - \theta)$$

reflects different penalties for underestimation and overestimation. For example, underestimating the value of a contract by \$1000, we probably lose just \$1000. Overestimating the same contract by the same amount can lead to a loss of a customer and result in no deal. The squared-error, absolute-error, and zero-one losses considered so far are symmetric.

Asymptotically, the LinEx loss function is exponentially growing on one side and linear on the other. As shown on Figure 5.4, it strongly penalizes for underestimation (exponentially strong) if $\alpha > 0$, and it strongly penalizes for overestimation if $\alpha < 0$.

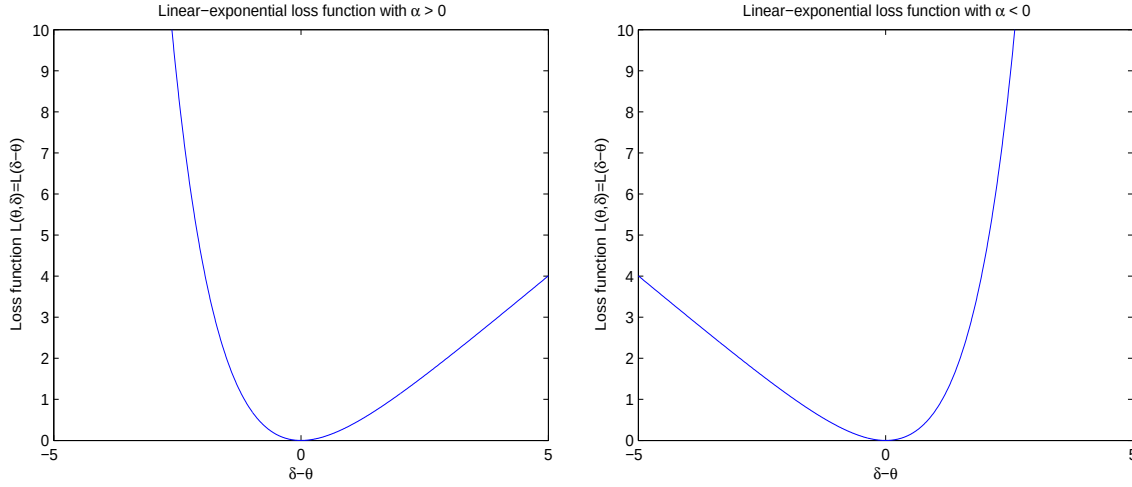
Other **asymmetric loss functions** are also used, and some of them give comparable penalties on both sides. For example, a **linear loss function**

$$L(\theta, \delta) = \begin{cases} K_1(\theta - \delta) & \text{if } \theta \geq \delta \\ K_2(\delta - \theta) & \text{if } \theta \leq \delta \end{cases}$$

is an asymmetric analogue of the absolute-error loss.

For confidence sets, a loss function

$$L(\theta, S) = \lambda|S| + I\{\theta \notin S\}$$

FIGURE 5.4: Linear-exponential loss function (*LinEx*) with $\alpha > 0$ and $\alpha < 0$.

is often used, where S is a set for parameter θ , $|S|$ is the size of this set such as its length, volume, or number of elements, and λ is a constant that shows how much we can neglect the size of the confidence set in favor of the coverage probability. The risk corresponding to this loss is a combination of the expected size of the set and its probability of non-coverage, missing θ .

In *sequential experiments*, when observations are collected one by one, and a statistician decides when to stop data collection, the cost of observations is usually added to the loss function,

$$L(\theta, \delta(X_1, \dots, X_n)) + Cn.$$

The optimal decisions (stopping rules) will then find an optimal balance between the sample size n and the accuracy of reported results.

5.3 Convexity and the Rao-Blackwell Theorem

As we noticed in Section 5.1, the constructed loss function for money appears to be [convex](#), reflecting our risk-averting nature. The commonly used squared-error, absolute-error, and LinEx loss functions are also convex, as functions of θ and δ . In this section, we'll use this property to obtain better statistical decisions.

5.3.1 Convex functions

Let us start with two equivalent definitions of convexity.

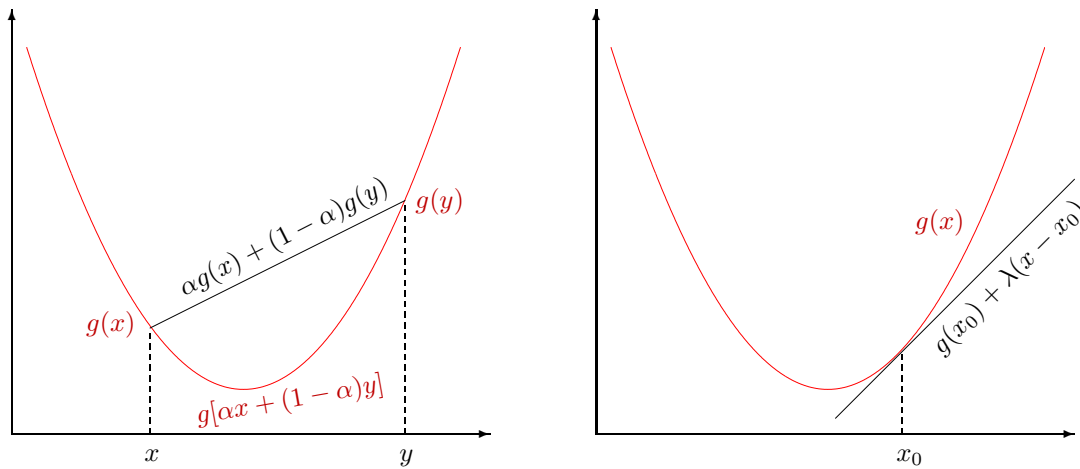


FIGURE 5.5: Two definitions of a convex function.

DEFINITION 5.2

Function $g(x)$ is **convex** on an interval $[a, b]$ if for any points $x, y \in [a, b]$ and any coefficient $\alpha \in [0, 1]$,

$$g[\alpha x + (1 - \alpha)y] \leq \alpha g(x) + (1 - \alpha)g(y).$$

Function $g(x)$ is **convex** on an interval $[a, b]$ if for any point $x_0 \in [a, b]$ there exists a slope $\lambda \in [-\infty, \infty]$, such that for all $x \in [a, b]$

$$g(x_0) + \lambda(x - x_0) \leq g(x).$$

The first definition says that a convex function stays below any secant line connecting two points on its graph, as on Figure 5.5 (left). By the second definition, the function is convex if a tangent line drawn at any arbitrary point of its graph stays below the function itself on its entire domain, as on Figure 5.5 (right).

Next, we'll need a classical result about the expected value of a convex function, called Jensen's Inequality.

5.3.2 Jensen's Inequality

For a convex function $g(x)$ and a random variable X with finite expectation $\mathbf{E}(X) < \infty$,

$$g(\mathbf{E} X) \leq \mathbf{E} g(X)$$

PROOF: Since g is convex, for any x_0 , there exists a slope λ , for which

$$g(x_0) + \lambda(x - x_0) \leq g(x)$$

holds for any x . Substitute $x = X$, $x_0 = \mathbf{E} X$,

$$g(\mathbf{E} X) + \lambda(X - \mathbf{E} X) \leq g(X).$$

Finally, take expectations of both sides,

$$g(\mathbf{E} X) \leq \mathbf{E} g(X).$$

□

5.3.3 Sufficient statistics

Next, let's recall (from Mathematical Statistics or Statistical Inference courses) the definition of a **sufficient statistic**.

DEFINITION 5.3

Statistic $T(\mathbf{X})$ computed from a sample $\mathbf{X}$ is a **sufficient statistic** for parameter θ if $f(\mathbf{X} | T(\mathbf{X}))$, the conditional distribution of $\mathbf{X}$ given $T(\mathbf{X})$, is independent of θ .

It means that in order to do inference about θ , it is *sufficient* to use just the sufficient statistic T instead of the whole sample $\mathbf{X}$. Given T , the distribution of data is the same for all θ , so no more information beyond $T(\mathbf{X})$ can be of any use for the inference about θ .

A working tool for finding sufficient statistics is the Neyman-Fisher **Factorization Theorem**.

Factorization Theorem

Statistic $T(\mathbf{X})$ is **sufficient** for parameter θ
if and only if
the joint density (or pmf) of $\mathbf{X}$ factorizes as

$$f(\mathbf{X} | \theta) = g_\theta(T(\mathbf{X}))h(\mathbf{X}),$$

where $g_\theta(T(\mathbf{X}))$ depends on data $\mathbf{X}$ only through the statistic T ,
and $h(\mathbf{X})$ does not contain θ

Example 5.6. Use the Factorization Theorem to find sufficient statistics for samples from

- Poisson(θ) distribution,
- Normal(μ, σ) distribution with known σ ,
- Normal(μ, σ) distribution with both μ and σ being unknown,
- Cauchy(θ) distribution that has a density

$$f(x | \theta) = \frac{\theta}{\pi(\theta^2 + x^2)}, \quad x \in (-\infty, \infty)$$

... ..

◇

We are now ready for the main result of this section - Rao-Blackwell Theorem in its general form, for any convex loss functions.

5.3.4 Rao-Blackwell Theorem

Let

- $L(\theta, \delta)$ be a loss function that is convex in δ for any θ ,
- T be a sufficient statistic for θ ,
- δ_0 be some decision.

Define a new decision $\delta_1 := \mathbf{E}\{\delta_0 \mid T\}$ as a conditional expected value of δ , given $T(\mathbf{X})$.

Then the frequentist risk of δ_1 never exceeds the frequentist risk of δ_0 ,

$$R(\theta, \delta_1) \leq R(\theta, \delta_0) \text{ for } \forall \theta \in \Theta.$$

PROOF: By the Jensen's inequality,

$$L(\theta, \delta_1) = L(\theta, \mathbf{E}\{\delta_0 \mid T\}) \leq \mathbf{E}\{L(\theta, \delta_0) \mid T\}.$$

Take expectations,

$$\begin{aligned} R(\theta, \delta_1) &= \mathbf{E} L(\theta, \delta_1(T)) \\ &= \mathbf{E} L(\theta, \delta_1(T)) \\ &\leq \mathbf{E} \mathbf{E}\{L(\theta, \delta_0) \mid T\} \\ &= \mathbf{E} L(\theta, \delta_0) \\ &= R(\theta, \delta_0). \end{aligned}$$

□

Question: so, where did we use sufficiency in this proof? Does it work for statistics that are not sufficient?

Example 5.7. Consider a sample from Uniform(0, θ) distribution. To estimate θ , someone uses a statistic $\delta_0 = 2\bar{X}$. It is unbiased! Can you propose a better estimator, under the squared-error loss or **any** convex loss function?

... ..

◇

Chapter 6

Intro to WinBUGS

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WinBUGS (**B**ayesian inference **U**sing **G**ibbs **S**ampling) for Windows is a software created specifically for Bayesian analysis of complex statistical models using Markov chain Monte Carlo (MCMC) techniques.

6.1 Installation

With the creation of a new version of BUGS called OpenBUGS, WinBUGS became free of charge; it can be downloaded and installed from the Medical Research Council site

<http://www.mrc-bsu.cam.ac.uk/software/bugs/the-bugs-project-winbugs/>

This is a version for Windows. For Mac users, a different software package is recommended called **JAGS** (**J**ust **A**nother **G**ibbs **S**ampler). It is also free, available for download on <http://mcmc-jags.sourceforge.net/>.

Installation was explained in detail in a separate handout. The main steps are:

1. Find out whether you are running a 32-bit or a 64-bit machine.
2. For a 32-bit machine, download and install WinBUGS14.exe from the site above. For a 64-bit machine, download a zipped version and unzip it.
3. Find and execute the file WinBUGS14.exe. Create a shortcut of WinBUGS.exe to desktop for your convenience.
4. Register for eternal free use. From the same site, download a patch under the link “Download and install the patch for 1.4.3”, right-click on it, and save as txt file. Open this file in WinBUGS with **File** → **Open** → [this patch file], then decode it with **Tools** → **Decode** → **Decode all**.

5. Install the “immortality key” with similar steps. Find the link under “Get the free key for unrestricted use by clicking here”. Again, save as txt file, open it in WinBUGS, and decode.

WinBUGS is now ready for our use!

6.2 Basic Program: Model, Data, and Initialization

Three basic parts of a WinBUGS program are the model specification, the data set(s), and the initial values for the Markov chain Monte Carlo (we’ll discuss MCMC a little later). The model block is necessary, the other two blocks are not. In no-data problems, the data block can be omitted. Also, instead of initial values, we can ask WinBUGS to generate them.

6.2.1 Model specification

The model block contains description of all the distributions in the problem, including the distribution of data, the prior distribution of unknown parameters, and possible hyper-prior distributions of the parameters of the prior.

Each distribution name starts with letter “d” followed by an abbreviated distribution name. The list of distributions supported by WinBUGS in Table 6.1. A few remarks on them:

- `dcat(p[])` is an arbitrary discrete distribution given by probabilities $p[i]$ specified by us.
- Negative Binomial variable here represents the number of failures before the r -th success, so that $x = 0, 1, 2, \dots$
- Exponential distribution parameter is the frequency λ , so the density is $f(x) = \lambda e^{-\lambda x}$.
- The Double Exponential distribution has the density $f(x) = \frac{\tau}{2} e^{-\tau|x-\mu|}$ for $x \in \mathbf{R}$.
- For $\text{Pareto}(\alpha, c)$, c is the scale parameter, so the density is $f(x) = \alpha c^\alpha x^{-(\alpha+1)}$.
- A general version of a *non-central* Student’s t-distribution is used, with k being degrees of freedom, and also, μ and τ being the location and scale parameters.
- Parameters of multivariate distributions are vectors and matrices that have to be specified in the model or data blocks.

Discrete distributions	
Bernoulli	<code>dbern(p)</code>
Binomial	<code>dbin(p,n)</code>
Categorical	<code>dcat(p[])</code>
Negative Binomial	<code>dnegbin(p,r)</code>
Poisson	<code>dpois(lambda)</code>

Continuous distributions	
Beta	<code>dbeta(a,b)</code>
Chi-squared	<code>dchisqr(k)</code>
Double Exponential	<code>ddexp(mu,tau)</code>
Exponential	<code>dexp(lambda)</code>
Gamma	<code>dgamma(r,lambda)</code>
Log-normal	<code>dlnorm(mu,tau)</code>
Logistic	<code>dlogis(mu,tau)</code>
Normal	<code>dnorm(mu,tau)</code>
Pareto	<code>dpar(alpha,c)</code>
Student's t	<code>dt(mu,tau,k)</code>
Uniform	<code>dunif(a,b)</code>
Weibull	<code>dweib(v,lambda)</code>

Discrete multivariate distributions	
Multinomial	<code>dmulti(p[],N)</code>

Continuous multivariate distributions	
Dirichlet	<code>ddirch(alpha[])</code>
Multivariate Normal	<code>dmnorm(mu[],T[,])</code>
Multivariate Student's t	<code>dmt(mu[],T[,],k)</code>
Wishart	<code>dwish(R[,],k)</code>

TABLE 6.1: List of distributions supported by WinBUGS.

For a simple example, consider the following problem,

Model: $X \sim \text{Normal}(\theta, \sigma)$
Prior: $\theta \sim \text{Normal}(0, 1)$
where $\sigma = 10$

A WinBUGS code for this model is:

```
model {  
  X ~ dnorm(theta,sigma)  
  theta ~ dnorm(0,1)  
  sigma <- 10  
}
```

To execute this code in WinBUGS, we follow several steps:

1. Write. Open a new project (*File* $\rightarrow$ *New* or Ctrl-N) and write this code.
2. Verify. Go to *Model* $\rightarrow$ *Specification* to open the specification tool box. Highlight the word "model" in your code and click "check model" in the specification tool. Watch for the response in the lower left corner. A correct response is "model is synthetically correct", otherwise WinBUGS will point to the first error.
3. Compile. Press "compile" in the specification tool. Watch for the response "model compiled" or find errors.
4. Initialize. Since we did not write any initial values, ask WinBUGS to generate its own ones by pressing "Gen inits" in the specification tool. The correct response is "initial values generated, model initialized".
5. Specify the monitored variables. Go to *Inference* $\rightarrow$ *Samples* to open the Sample Monitor Tool and enter variables of interest in the first box "node", one at a time. These are the variables that we want to monitor, get their estimated parameters, plot their densities, etc. After entering the name of each monitored variable, press "set". For our example, let us enter "X", press "set", then enter "theta", and press "set" again. We can also specify the percentiles that we would like to monitor (for example, highlight "median" to find the Bayes estimator of θ under the absolute-error loss).
6. Execute the code. Go to *Model* $\rightarrow$ *Update* to open the Update tool, then press "update". Right before this, we can also specify the desired number of Monte Carlo runs in the box "updates". The default number is 1000.
7. Read results. In the Sample Monitor Tool, enter a star "*" into the box "node", then press the option we'd like to use. Press "stats" to see the main statistics such as the mean, median, standard error, as well as the *Monte Carlo standard error* that shows

node	mean	sd	MC error	2.5%	median	97.5%	start	sample
X	0.04486	1.029	0.03788	-2.002	0.03069	2.003	1	1000
theta	0.04249	1.005	0.03748	-1.973	0.02815	2.008	1	1000

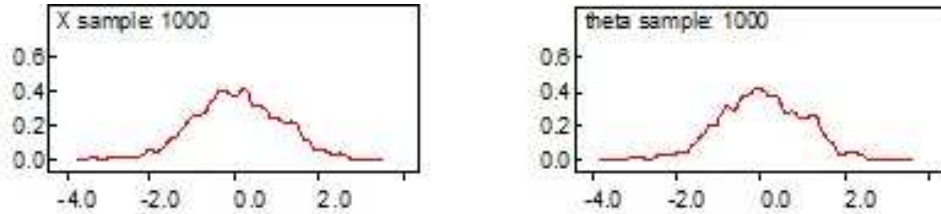


FIGURE 6.1: Statistics and densities generated by WinBUGS over 1000 Monte Carlo runs.

node	mean	sd	MC error	2.5%	median	97.5%	start	sample
X	-0.002583	1.044	0.006926	-2.051	-0.004549	2.06	1	20000
theta	-5.412E-4	0.9968	0.006703	-1.968	-0.00746	1.972	1	20000

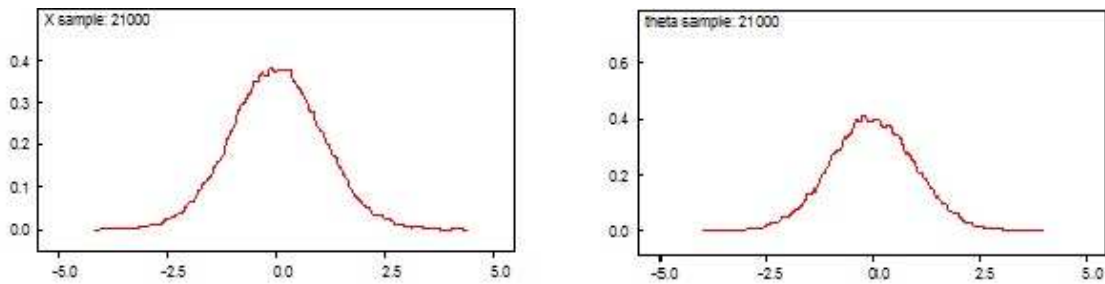


FIGURE 6.2: Statistics and densities generated by WinBUGS over 21,000 MC runs.

the accuracy of Monte Carlo generated estimators. Press “density” to see graphs of densities and probability mass functions of all the monitored variables. Press “history” to see the plot of all the simulated values of these variables, and press “coda” if you’d like to get a list of all these values.

In our example, the statistics and densities will look like on Figure 6.1. They are generated by the algorithm and averaged over 1000 Monte Carlo runs. Choosing a larger number of runs will lead to more accurate statistics and smoother density estimates, under a smaller MC error. For example, we used 21,000 MC runs to get results on Figure 6.2, for the same problem.

6.3 Entering Data

The data $\mathbf{X} = (X_1, X_2, \dots, X_n)$ as well as some constant values of participating parameters can be entered separately from the model statement. They will be listed in the “list” block.

For example, the following WinBUGS code can be used to solve one of our future homework exercises:

“Six measurements $X_1, \dots, X_6$ from $\text{Normal}(\theta, 1)$ distribution are made. The first five are 3, 4, 5, 6, and 7, but X_6 got lost. Parameter θ is unknown; it has the $\text{Normal}(4, 1)$ prior distribution. Derive the predictive distribution of X_6 and a 90% predictive interval for it.”

```
# MODEL model{      for (i in 1:5) {X[i] ~ dnorm(theta,sigma)}
  theta ~ dnorm(mu,tau)
  sigma <- 1
  tau <- 1
  mu <- 4
  Z ~ dnorm(theta,1)
}
# DATA
  list(X=c(3,4,5,6,7))
```

In the first line of the model block, we indicated a sample of size 5 from Normal distribution. Then, we specified the prior distribution of θ and the constant “known” parameters. Also, we introduced the variable Z that we need to predict in this problem.

The actual data are written as a list `list(X=c(3,4,5,6,7))`. To let the WinBUGS use these data, we should enter them **after checking the model but before compiling it**. That is, between Step 2 “Verify” and Step 3 “Compile”, highlight the word “list” and click on “load data” in the Specification Tool. The correct response is “data loaded”.

Constant parameters can also be entered in the same list, as part of the data. So, here is an equivalent code:

```
# MODEL model{      for (i in 1:5) {X[i] ~ dnorm(theta,sigma)}
  theta ~ dnorm(mu,tau)
  Z ~ dnorm(theta,1)
}
# DATA
  list(X=c(3,4,5,6,7), sigma=1, tau=1, mu=4)
```

The lines that start from `#` are just comments for us.

In this problem, we should monitor variable Z which we need to predict. The shape of its density suggests the Normal distribution, and its estimated parameters are given in “stats”. After 10,000 MC runs, I obtained $\widehat{\mathbf{E}}(Z|\mathbf{X}) = 4.836$ and $\widehat{\text{Std}}(Z) = 1.067$.

6.3.1 Initialization

The Markov chain Monte Carlo (MCMC) underlying the computations in WinBUGS is an *iterative algorithm*. It starts with some initial values of all non-constant random variables and proceed through a sequence of iterations until it is assumed “converged”. Convergence is postulated at the moment when the generated results get very close to each other.

However, especially in multi-layer hierarchical models involving a number of parameters, convergence may not be trivial. In particular, the final result may depend on where the algorithm started, the initial values. One way to verify the stability of the algorithm and reliability of its results is to start from totally different initial values and see if the algorithm nevertheless converges to roughly the same conclusion.

Initial values are also written as “list”, just like the data. We can specify more than one list, and in this case we have to tell WinBUGS how many of them we’ll like to use.

Before compiling the model, enter the desired number of initializations under “num of chains” in the Specification Tool. Then, press “compile”, and instead of generating initial values with “gen inits”, enter our prepared lists.

Highlight the first word “list” and press **“load inits”**. WinBUGS will accept our initial values for chain 1 but will let us know in its response statement that the other chains are still uninitialized. Then highlight the second word “list” and press **“load inits”** again. We’ll repeat this until all the initial values are entered. At this time, the correct response will be “model is initialized”.

For example, the following code can be used to solve the next exercise, #2 of Homework #7, where we modified the previous example by letting the prior parameter μ have its own prior distribution:

“ Consider the previous exercise again, with one correction. Replace the Normal(4,1) prior distribution of θ with the Normal(μ ,1) distribution, where the hyper-parameter μ is unknown, and it has its own prior distribution, which is Normal(4,1).”

```
# MODEL
  model{ for (i in 1:5) {X[i] ~ dnorm(theta,1)}
    theta ~ dnorm(mu,1)
    mu ~ dnorm(4,1)
    Z ~ dnorm(theta,1)
  }

# DATA
  list(X=c(3,4,5,6,7))

# INITIAL VALUES
  list(theta=-10,Z=0,mu=1)
  list(theta=0,Z=0,mu=1)
  list(theta=10,Z=0,mu=1)
```

The last three lines specify the initial values. All the random quantities have to be initialized. Each line generates a separate sequence of Monte Carlo runs that starts from the

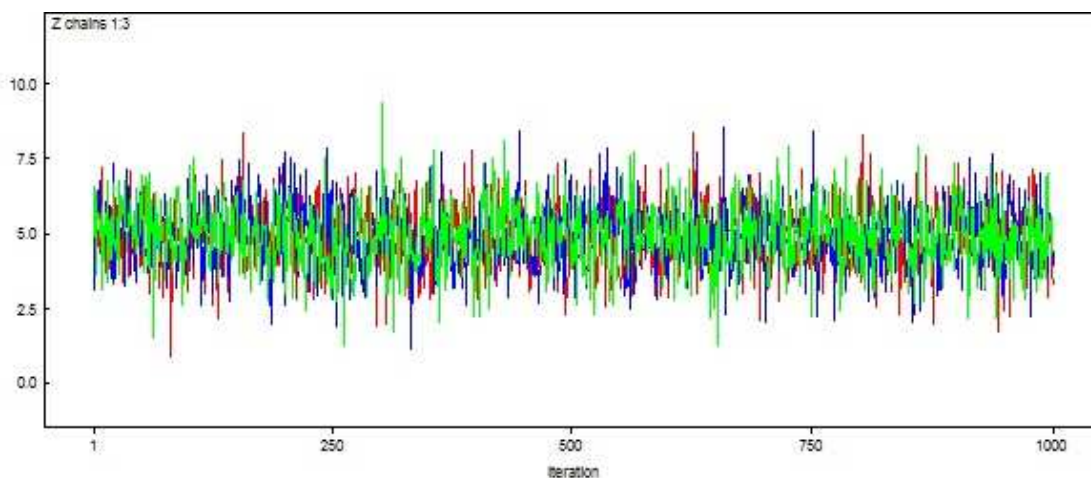


FIGURE 6.3: *Statistics and densities generated by WinBUGS over 21,000 MC runs.*

given values. (Seeking brevity, I wrote the non-random parameters as part of the specified distributions.)

In this problem, we trace variable Z , so we indicate in the Sample Monitor Tool node. After executing the code, enter “*” for this node, to request results. Then press “history” to see the actually generated values of Z . Here is what I obtained, Figure 6.3.

As it says in the top right corner, these are simulations of 3 Markov chains Monte Carlo, each chain starting from its own initial values that we requested. Over 1000 Monte Carlo runs, there were 3000 generated values of variable Z . The three chains are depicted with different colors. As we can see from this plot, they seemingly converged to the same result, no color makes a graph too different from the others.

This is one way to measure stability of our Monte Carlo algorithm. Regardless of the initial values, the results are roughly the same, only subject to the random variation. This variation cannot be avoided due to the use random simulations but it can be reduced by choosing a larger number of Monte Carlo runs.

Chapter 7

Bayesian Inference: Estimation, Hypothesis Testing, Prediction

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By now, we have obtained all the three components needed for Bayesian decision making. We collected data, and we determined the prior distribution and the loss function. Then, combining the data and the prior, we obtained the **posterior distribution**. All the knowledge about the unknown parameter is now included in the posterior, and that is what we'll use for further statistical analysis (Figure 7.1).

7.1 Bayesian estimation

In this section, we use Bayesian techniques to compute a point estimator of the unknown parameter θ and to evaluate its performance (in a Bayesian way!).

As we saw in chap. 5, the Bayes estimator make take different forms, depending on the loss function. For example, under the squared-error, the absolute-error, and the zero-one loss functions, the Bayes estimators are respectively, the mean, the median, and the mode of the posterior distribution.

The most commonly used estimator is the **posterior mean**,

$$\hat{\theta}_{\text{Bayes}} = \mathbf{E}\{\theta|\mathbf{X}\} = \begin{cases} \sum_{\theta} \theta \pi(\theta|\mathbf{X}) & = \frac{\sum \theta f(\mathbf{X}|\theta) \pi(\theta)}{\sum f(\mathbf{X}|\theta) \pi(\theta)} & \text{if } \theta \text{ is discrete} \\ \int_{\theta} \theta \pi(\theta|\mathbf{X}) d\theta & = \frac{\int \theta f(\mathbf{X}|\theta) \pi(\theta) d\theta}{\int f(\mathbf{X}|\theta) \pi(\theta) d\theta} & \text{if } \theta \text{ is continuous} \end{cases}$$

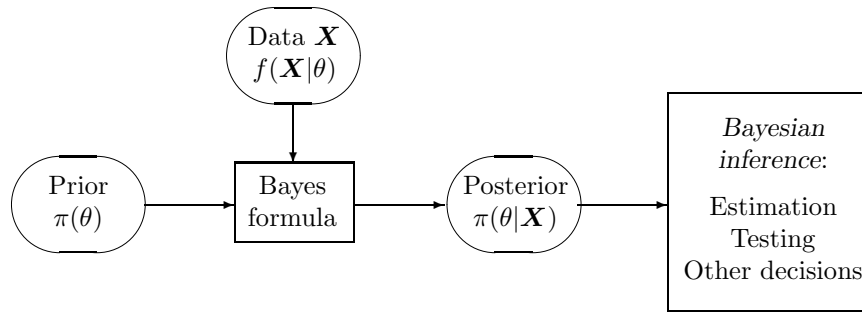


FIGURE 7.1: Posterior distribution is the basis for Bayesian inference.

Posterior mean is the conditional expectation of θ given data $\mathbf{X}$. In abstract terms, the **Bayes estimator** $\hat{\theta}_{\text{Bayes}}$ is what we “expect” θ to be, after we observed a sample.

How accurate is this estimator? Among all estimators, $\hat{\theta}_{\text{Bayes}} = \mathbf{E}\{\theta|\mathbf{X}\}$ has the lowest squared-error **posterior risk**

$$\mathbf{E}\{(\hat{\theta} - \theta)^2 | \mathbf{X}\}$$

and also, the lowest **Bayes risk**

$$\mathbf{E}\mathbf{E}(\hat{\theta} - \theta)^2,$$

where this double expectation is taken over the joint distribution of $\mathbf{X}$ and θ .

For the Bayes estimator $\hat{\theta}_{\text{Bayes}}$, posterior risk equals **posterior variance**, which shows variability of θ around $\hat{\theta}_{\text{Bayes}}$.

Example 7.1 (NORMAL CASE). The Bayes estimator of the mean θ of $\text{Normal}(\theta, \sigma)$ distribution with a $\text{Normal}(\mu, \tau)$ prior is

$$\hat{\theta}_{\text{Bayes}} = \mu_x = \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2},$$

and its posterior risk is

$$\tau_x^2 = \frac{1}{n/\sigma^2 + 1/\tau^2},$$

(Table 4.1). As we expect, this risk decreases to 0 as the sample size grows to infinity. $\diamond$

Example 7.2 (NETWORK BLACKOUTS, CONTINUED). After two weeks of data, the weekly rate of network blackouts, according to Example 4.2 on p. 52, has Gamma posterior distribution with parameters $\alpha_x = 6$ and $\lambda_x = 3$.

The Bayes estimator of the weekly rate θ is

$$\hat{\theta}_{\text{Bayes}} = \mathbf{E}\{\theta|\mathbf{X}\} = \frac{\alpha_x}{\lambda_x} = 2 \text{ (blackouts per week)}$$

with a posterior risk of

$$\text{Var} \{ \theta | \mathbf{X} \} = \frac{\alpha_x}{\lambda_x^2} = 0.6667.$$

◇

Although conjugate priors simplify our statistics, Bayesian inference can certainly be done for other priors too.

Example 7.3 (QUALITY INSPECTION, CONTINUED). In Example 1.10 on p. 8, we computed posterior distribution of the proportion of defective parts θ . This was a discrete distribution,

$$\pi(0.05 | \mathbf{X}) = 0.2387; \quad \pi(0.10 | \mathbf{X}) = 0.7613.$$

Now, the Bayes estimator of θ is

$$\hat{\theta}_{\text{Bayes}} = \sum_{\theta} \theta \pi(\theta | \mathbf{X}) = (0.05)(0.2387) + (0.10)(0.7613) = 0.0881.$$

It does not agree with the manufacturer (who claims $\theta = 0.05$) or with the quality inspector (who feels that $\theta = 0.10$) but its value is much closer to the inspector's estimate.

The posterior risk of $\hat{\theta}_{\text{Bayes}}$ is

$$\begin{aligned} \text{Var} \{ \theta | \mathbf{X} \} &= \mathbf{E} \{ \theta^2 | \mathbf{X} \} - \mathbf{E}^2 \{ \theta | \mathbf{X} \} \\ &= (0.05)^2(0.2387) + (0.10)^2(0.7613) - (0.0881)^2 = 0.0004, \end{aligned}$$

which means a rather low posterior standard deviation of 0.02.

◇

7.1.1 Precision evaluation

A parameter θ is estimated by an estimator $\delta = \hat{\theta}$. How accurate is this decision?

A **frequentist** measure of precision is the **mean-squared error**

$$\text{MSE}(\delta) = \mathbf{E}_{\theta}^{\mathbf{X}} (\delta - \theta)^2 = \mathbf{E}_{\theta} (\delta - \mathbf{E}_{\theta} \delta)^2 + (\mathbf{E}_{\theta} \delta - \theta)^2 = \text{Var}(\delta) + \{\text{Bias}(\delta)\}^2.$$

Here all the expectations are taken in the frequentist way, i.e., they integrate over all samples of $\mathbf{X}$, given a fixed parameter θ .

Following **Bayesian approach**, the same expectations are taken with respect to the posterior distribution of θ , given a fixed, already observed sample $\mathbf{X}$. This is called **posterior variance** of estimator δ ,

$$\begin{aligned} V_{\mathbf{X}}(\delta) &= \mathbf{E}_{\mathbf{X}}^{\pi(\theta|\mathbf{X})} (\delta - \theta)^2 \\ &= \mathbf{E} \{ (\theta - \mu_x)^2 | \mathbf{X} \} + (\delta - \mu_x)^2 \\ &= \tau_x^2 + (\delta - \mu_x)^2, \end{aligned}$$

where $\mu_x = \mathbf{E}\{\theta|\mathbf{X}\}$ is the mean and $\tau_x^2 = \text{Var}\{\theta|\mathbf{X}\}$ is the variance of the posterior distribution $\pi(\theta|\mathbf{X})$.

There are two variance components in the posterior variance of δ . One is the posterior variance of the parameter θ , and the other is the squared deviation of estimator δ from the posterior mean μ_x . So, the total variability of our estimator consists of variability of θ around its mean and the distance between that posterior mean and our estimator.

**Posterior
variance**

$\text{Var}(\theta \mathbf{X})$ or τ_x^2	$= \mathbf{E}\{(\theta - \mu_x)^2 \mathbf{X}\}$ posterior variance of parameter θ , variance of the posterior distribution of θ
$V_{\mathbf{X}}(\delta)$	$= \tau_x^2 + (\delta - \mu_x)^2$, posterior variance of estimator δ

Example 7.4 (NORMAL-NORMAL). For data $\mathbf{X} \sim \text{Normal}(\theta, \sigma^2)$ and prior $\pi(\theta) \sim \text{Normal}(\mu, \tau^2)$, we have derived the posterior variance in sec. 4.3.3

$$\tau_x^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}.$$

If we decide to estimate the population mean θ by the sample mean $\delta = \bar{X}$, it will be a *biased* estimator with respect to the posterior distribution, with the bias

$$\delta - \mu_x = \bar{X} - \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} = \frac{\bar{X} - \mu}{n\tau^2/\sigma^2 + 1}.$$

Then, the sample mean has the posterior variance

$$V_{\mathbf{X}}(\bar{X}) = \frac{1}{n/\sigma^2 + 1/\tau^2} + \left(\frac{\bar{X} - \mu}{n\tau^2/\sigma^2 + 1} \right)^2.$$

Depending on how well the data agree with the prior distribution, how close the sample mean and the prior mean are, this posterior variance of $\bar{X}$ may exceed its mean squared error

$$\text{MSE}(\bar{X}) = \frac{\sigma^2}{n}.$$

◇

Example 7.5 (NONINFORMATIVE PRIOR). If in the previous example, we choose a non-informative improper prior $\pi(\theta) \equiv 1$, it cannot disagree with the data - because it carries no

disagreeing information. As a result, we see that $\bar{X}$ is unbiased not only in the frequentist sense but also with respect to the posterior distribution

$$\pi(\theta | \mathbf{X}) \sim \text{Normal}(\bar{X}, \sigma^2/n).$$

In this case,

$$V_{\mathbf{X}}(\bar{X}) = \frac{\sigma^2}{n} = \text{MSE}(\bar{X}),$$

precision of $\delta = \bar{X}$ is the same in the Bayesian and in the frequentist sense. $\diamond$

Example 7.6 (GAMMA POSTERIOR. 4.14, 4.19G, 4.20I OF BERGER). In Example 4.4 on 58, we considered a $\text{Poisson}(\theta)$ sample $\mathbf{X}$ and the improper non-informative prior $\pi(\theta) = \frac{1}{\theta}$, $\theta > 0$. We figured that in this case, the posterior distribution is

$$\pi(\theta | \mathbf{X}) \sim \text{Gamma}(\sum X_i, 1/n),$$

as long as $\mathbf{X} \neq (0, \dots, 0)$.

- (a) Find the posterior mean and the posterior variance.
- (b) Find the generalized maximum likelihood estimator of θ and its posterior variance.

... ..

$\diamond$

7.2 Bayesian credible sets

Bayesian and frequentist approaches to confidence estimation are quite different. In Bayesian analysis, having a posterior distribution of θ , we no longer have to explain the confidence level $(1 - \alpha)$ in terms of a long run of samples. Instead, we can give an interval $[a, b]$ or a set C that has a posterior probability $(1 - \alpha)$ and state that *the parameter θ belongs to this set with probability $(1 - \alpha)$* . Such a statement was impossible before we considered prior and posterior distributions. This set is called a $(1 - \alpha)100\%$ *credible set*.

DEFINITION 7.1

Set C is a $(1 - \alpha)100\%$ **credible set** for the parameter θ if the posterior probability for θ to belong to C equals $(1 - \alpha)$. That is,

$$\mathbf{P} \{ \theta \in C | \mathbf{X} \} = \int_C \pi(\theta | \mathbf{X}) d\theta = 1 - \alpha.$$

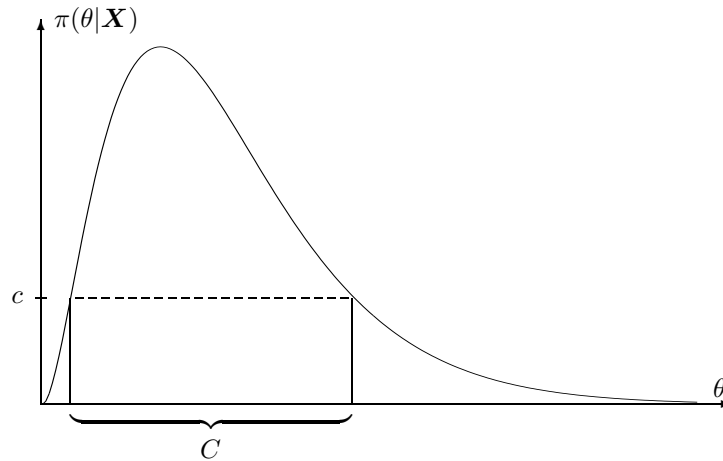


FIGURE 7.2: The $(1 - \alpha)100\%$ highest posterior density credible set.

Such a set is not unique. Recall that for two-sided, left-tail, and right-tail hypothesis testing, we took different portions of the area under the Normal curve, all equal $(1 - \alpha)$.

Minimizing the length of set C among all $(1 - \alpha)100\%$ credible sets, we just have to include all the points θ with a high posterior density $\pi(\theta|\mathbf{X})$,

$$C = \{\theta : \pi(\theta|\mathbf{X}) \geq c\}$$

(see Figure 7.2). Such a set is called the **highest posterior density credible set**, or just the **HPD set**.

For the $\text{Normal}(\mu_x, \tau_x)$ posterior distribution of θ , the $(1 - \alpha)100\%$ HPD set is

$$\mu_x \pm z_{\alpha/2}\tau_x = [\mu_x - z_{\alpha/2}\tau_x, \mu_x + z_{\alpha/2}\tau_x].$$

Example 7.7 (SALARIES). In Example 1.9 on p. 6, we “decided” that the most likely range for the mean starting salary θ of AU graduates is between \$40,000 and \$70,000. Expressing this in a form of a prior distribution, we let the prior mean be $\mu = (40,000 + 70,000)/2 = 55,000$. Further, if we feel that the range $[40,000; 70,000]$ is 95% likely, and we accept a Normal prior distribution for θ , then this range should be equal

$$[40,000; 70,000] = \mu \pm z_{0.025/2}\tau = \mu \pm 1.96\tau,$$

where τ is the prior standard deviation (Figure 7.3). It can be computed from the information given,

$$\tau = \frac{70,000 - 40,000}{2(1.96)} = 7,653.$$

This is the advantage of using a rich (two-parameter) family of prior distributions: we are likely to find a member of this family that reflects our prior beliefs adequately.

Then, *prior to collecting any data*, the 95% HPD credible set of the mean starting salary θ is

$$\mu \pm z_{0.025}\tau = [40,000; 70,000].$$

Suppose a random sample of 100 graduates has the mean starting salary of $\bar{X} = 48,000$ with a sample standard deviation $s = 12,000$. Then we compute the *posterior* mean and standard deviation,

$$\begin{aligned}\mu_x &= \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} = \frac{(100)(48,000)/(12,000)^2 + (55,000)/(7,653)^2}{100/(12,000)^2 + 1/(7653)^2} \\ &= 48,168; \\ \tau_x &= \frac{1}{\sqrt{n/\sigma^2 + 1/\tau^2}} = \frac{1}{\sqrt{100/(12,000)^2 + 1/(7653)^2}} = 11,855.\end{aligned}$$

We used the sample standard deviation s in place of the population standard deviation σ assuming that a sample of size 100 estimates the latter rather accurately. Alternatively, we could put a prior distribution on unknown σ too and estimate it by Bayesian methods. Since the observed sample mean is smaller than our prior mean, the resulting posterior distribution is shifted to the left of the prior (Figure 7.4).

Conclusion. After seeing the data, the Bayes estimator for the mean starting salary of CS graduates is

$$\hat{\theta}_{\text{Bayes}} = \mu_x = 48,168 \text{ dollars,}$$

and the 95% HPD credible set for this mean salary is

$$\mu_x \pm z_{0.025}\tau_x = 48,168 \pm (1.96)(11,855) = 48,168 \pm 23,236 = [24,932; 71,404]$$

Lower than a priori predicted starting salaries extended the lower end of our credible interval. $\diamond$

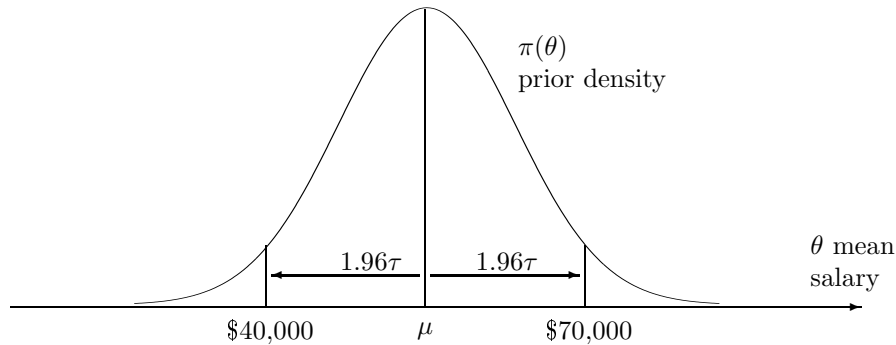


FIGURE 7.3: Normal prior distribution and the 95% HPD credible set for the mean starting salary of Computer Science graduates (Example 7.7).

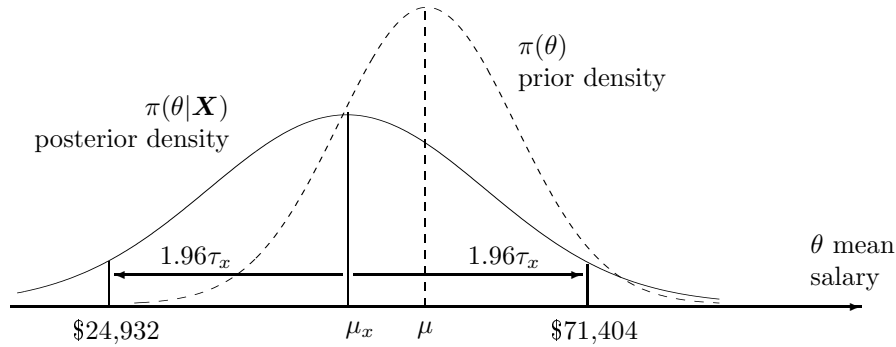


FIGURE 7.4: Normal prior and posterior distributions for the mean starting salary (Example 7.7).

Example 7.8 (TELEPHONE COMPANY). A new telephone company predicts to handle an average of 1000 calls per hour. During 10 randomly selected hours of operation, it handled 7265 calls.

How should it update the initial estimate of the frequency of telephone calls? Construct a 95% HPD credible set. Telephone calls are placed according to a Poisson process. Exponential prior distribution of the hourly rate of calls is applicable.

Solution. We need a Bayesian estimator of the frequency θ of telephone calls. The number of calls during 1 hour has $\text{Poisson}(\theta)$ distribution, where θ is unknown, with

$$\text{Exponential}(\lambda) = \text{Gamma}(1, \lambda)$$

prior distribution that has an expectation of

$$\mathbf{E}(\theta) = \frac{1}{\lambda} = 1000 \text{ calls.}$$

Hence, $\lambda = 0.001$. We observe a sample of size $n = 10$, totaling

$$\sum_{i=1}^n X_i = n\bar{X} = 7265 \text{ calls.}$$

As we know (see Table 4.1 on p. 54), the posterior distribution in this case is $\text{Gamma}(\alpha_x, \lambda_x)$ with

$$\begin{aligned} \alpha_x &= \alpha + n\bar{X} = 7266, \\ \lambda_x &= \lambda + n = 10.001. \end{aligned}$$

This distribution has mean

$$\mu_x = \alpha_x / \lambda_x = 726.53$$

and standard deviation

$$\tau_x = \sqrt{\alpha_x} / \lambda_x = 26.95.$$

The Bayes estimator of θ is

$$\mathbf{E}(\theta|\mathbf{X}) = \mu_x = \underline{726.53 \text{ calls per hour}}.$$

It almost coincides with the sample mean $\bar{X}$ showing the sample was informative enough to dominate over the prior information.

For the credible set, we notice that α_x is sufficiently large to make the Gamma posterior distribution approximately equal the Normal distribution with parameters μ_x and τ_x . The 95% HPD credible set is then

$$\mu_x \pm z_{0.05/2}\tau_x = 726.53 \pm (1.96)(26.95) = 726.53 \pm 16.71 = \underline{[709.82, 743.24]}$$

◇

All the HPD credible sets are Bayes decisions under the loss function

$$L(\theta, C) = \lambda|C| + I_{\theta \notin C},$$

where $|C|$ is the size, usually the length of set C , and λ is a coefficient. This loss requires a reasonably narrow set, but it penalizes if the set does not contain the actual parameter. Instead of satisfying the given confidence level α , the Bayes set finds the optimal balance between the expected size of a set $|C|$ and the probability of missing the parameter, $\mathbf{P}\{\theta \notin C\}$.

Example 7.9 (BETA POSTERIOR). Two independent Bernoulli(θ) variables are observed, $X_1 = 0$ and $X_2 = 1$. Under the Uniform(0,1) prior distribution,

- (a) compute the 80% HPD credible set for parameter θ ;
- (b) obtain the *Bayes confidence set* under the loss function $L(\theta, C) = 0.2|C| + I_{(\theta \notin C)}$.

... ..

◇

7.3 Bayesian hypothesis testing

Bayesian hypothesis testing is very easy to interpret. We can compute prior and posterior probabilities for the hypothesis H_0 and alternative H_A to be true and decide from there which one to accept or to reject.

Computing such probabilities was not possible without prior and posterior distributions of the parameter θ . In non-Bayesian statistics, θ was not random, thus H_0 and H_A were either true (with probability 1) or false (with probability 1).

For Bayesian tests, in order for H_0 and H_A to have meaningful, non-zero probabilities, they often represent disjoint sets of parameter values,

$$H_0 : \theta \in \Theta_0 \text{ vs } H_A : \theta \in \Theta_1$$

(which makes sense because exact equality $\theta = \theta_0$ is unlikely to hold anyway, and in practice it is understood as $\theta \approx \theta_0$).

Comparing posterior probabilities of H_0 and H_A ,

$$P\{\Theta_0 | \mathbf{X}\} \text{ and } P\{\Theta_1 | \mathbf{X}\},$$

we decide whether $P\{\Theta_1 | \mathbf{X}\}$ is large enough to present significant evidence and to reject the null hypothesis. One can again compare it with $(1 - \alpha)$ such as 0.90, 0.95, 0.99, or state the result in terms of likelihood, “the null hypothesis is this much likely to be true.”

Example 7.10 (TELEPHONE COMPANY, CONTINUED). Let us test whether the telephone company in Example 7.8 can actually face a call rate of 1000 calls *or more* per hour. We are testing

$$H_0 : \theta \geq 1000 \text{ vs } H_A : \theta < 1000,$$

where θ is the hourly rate of telephone calls.

According to the $\text{Gamma}(\alpha_x, \lambda_x)$ posterior distribution of θ and its Normal ($\mu_x = 726.53, \tau_x = 72.65$) approximation,

$$P\{H_0 | \mathbf{X}\} = P\left\{\frac{\theta - \mu_x}{\tau_x} \geq \frac{1000 - \mu_x}{\tau_x}\right\} = 1 - \Phi(3.76) = 0.0001.$$

By the complement rule, $P\{H_A | \mathbf{X}\} = 0.9999$, and this presents a sufficient evidence against H_0 .

We conclude that it's extremely unlikely for this company to face a frequency of 1000+ calls per hour. $\diamond$

7.3.1 Zero- w_i loss function

Often one can anticipate the consequences of Type I and Type II errors in hypothesis testing and assign a **loss** $L(\theta, a)$ associated with each possible error. Here θ is the parameter, and a is our action, the decision on whether we accept or reject the null hypothesis.

Each decision then has its **posterior risk** $\rho(a)$, defined as the expected loss computed under the posterior distribution. The action with the lower posterior risk is our **Bayes decision**.

Suppose that the Type I error causes the loss

$$w_0 = \text{Loss}(\text{Type I error}) = L(\theta, \text{reject } H_0), \text{ for } \theta \in \Theta_0$$

and the Type II error causes the loss

$$w_1 = \text{Loss}(\text{Type II error}) = L(\theta, \text{accept } H_0), \text{ for } \theta \in \Theta_1$$

This is a **zero- w_i loss function**, and it generalizes the zero-one loss.

Posterior risks of each possible action are then computed as

$$\begin{aligned} \rho(\pi, \text{reject } H_0 \mid \mathbf{X}) &= w_0 \pi(\Theta_0 \mid \mathbf{X}), \\ \rho(\pi, \text{accept } H_0 \mid \mathbf{X}) &= w_1 \pi(\Theta_1 \mid \mathbf{X}). \end{aligned}$$

Now we can determine the Bayes action. If $w_0 \pi(\Theta_0 \mid \mathbf{X}) \leq w_1 \pi(\Theta_1 \mid \mathbf{X})$, the Bayes action is to accept H_0 . If $w_0 \pi(\Theta_0 \mid \mathbf{X}) \geq w_1 \pi(\Theta_1 \mid \mathbf{X})$, the Bayes action is to reject H_0 .

Example 7.11 (QUALITY INSPECTION, CONTINUED). In Example 1.10 on p. 8, we are testing

$$H_0 : \theta = 0.05 \text{ vs } H_A : \theta = 0.10$$

for the proportion θ of defective parts. Suppose that the Type I error is three times as costly here as the Type II error. What is the Bayes action in this case?

Example 7.3 gives posterior probabilities

$$\pi(\Theta_0 \mid \mathbf{X}) = 0.2387 \text{ and } \pi(\Theta_1 \mid \mathbf{X}) = 0.7613.$$

Since $w_0 = 3w_1$, the posterior risks are

$$\begin{aligned} \rho(\text{reject } H_0) &= w_0 \pi(\Theta_0 \mid \mathbf{X}) = 3w_1(0.2387) = 0.7161w_1, \\ \rho(\text{accept } H_0) &= w_1 \pi(\Theta_1 \mid \mathbf{X}) = 0.7613w_1. \end{aligned}$$

Thus, rejecting H_0 has a lower posterior risk, and therefore, it is the Bayes action. Reject H_0 . $\diamond$

7.3.2 Bayesian classification

Bayesian approach to hypothesis testing naturally generalizes to the case of more than two hypotheses. Instead of classifying the unknown parameter into either Θ_0 (accept H_0) or Θ_1 (reject H_0), it can be classified into one of disjoint subsets of Θ ,

$$H_1 : \theta \in \Theta_1, \quad H_2 : \theta \in \Theta_2, \quad \dots, \quad H_k : \theta \in \Theta_k.$$

A loss function will then include $L(\theta, \delta) = w_{ij}$, a penalty for accepting hypothesis H_j whereas hypothesis H_i is true.

Similarly, we can include an option of making **no decision** and concluding that there is not enough information for or against either hypothesis. This also carries a pre-determined penalty w_{0j} , and sometimes “no decision” may be optimal comparing with the penalty of accepting a wrong hypothesis.

7.3.3 Bayes factors

A popular tool for the Bayesian hypothesis testing is Bayes factors.

DEFINITION 7.2

Bayes factor is defined as

$$B = \frac{\pi(\Theta_0 | \mathbf{X}) / \pi(\Theta_1 | \mathbf{X})}{\pi(\Theta_0) / \pi(\Theta_1)},$$

the posterior odds ratio divided by the prior odds ratio.

Often Bayes factors are quite stable or insensitive to the choice of prior probabilities $\pi(\Theta_0)$ and $\pi(\Theta_1)$.

How do we use the Bayes factors for hypothesis testing? Given the Bayes factor B , anyone multiplies it by her or his prior odds ratio $\pi(\Theta_0) / \pi(\Theta_1)$ and obtains the posterior odds ratio

$$\pi(\Theta_0 | \mathbf{X}) / \pi(\Theta_1 | \mathbf{X}) = \{\pi(\Theta_0) / \pi(\Theta_1)\} B$$

and uses it to decide between H_0 and H_1 .

Example 7.12 (SIMPLE VS SIMPLE). Suppose each $\Theta_0 = \{\theta_0\}$ and $\Theta_1 = \{\theta_1\}$ each consist of just one parameter value. In this case, the posterior probabilities are

$$\pi(\theta_0 | \mathbf{X}) = \frac{f(\mathbf{X} | \theta_0)\pi(\theta_0)}{f(\mathbf{X} | \theta_0)\pi(\theta_0) + f(\mathbf{X} | \theta_1)\pi(\theta_1)}, \quad \pi(\theta_1 | \mathbf{X}) = \frac{f(\mathbf{X} | \theta_1)\pi(\theta_1)}{f(\mathbf{X} | \theta_0)\pi(\theta_0) + f(\mathbf{X} | \theta_1)\pi(\theta_1)},$$

and the Bayes factor is

$$B = \frac{f(\mathbf{X} | \theta_0)\pi(\theta_0) / f(\mathbf{X} | \theta_1)\pi(\theta_1)}{\pi(\theta_0) / \pi(\theta_1)} = \frac{f(\mathbf{X} | \theta_0)}{f(\mathbf{X} | \theta_1)}.$$

The Bayes factor is simply the likelihood ratio! And yes, it is perfectly insensitive to the prior probabilities in this case. $\diamond$

7.4 Bayesian prediction

In this section, we are going to predict a random variable Z that is somehow related to the unknown parameter θ . Here is how the problem is formulated.

**Bayesian
prediction
problem**

A sample $\mathbf{X} = (X_1, \dots, X_n)$ is observed from $f(\mathbf{X} | \theta)$.
 Unknown parameter θ has a prior distribution $\pi(\theta)$.
 Random variable Z has distribution $g(z | \theta)$.
 Find the **predictive density** of Z , which is $p(z | \mathbf{X})$,
 the distribution of Z given the observed $\mathbf{X}$

Example 7.13 (ELECTRICITY PRICES). This is a practical example. On Fig. 7.5, you can see predictive densities of spot (instantaneous) electricity prices for each day of the next year and 5 years ahead¹. Such predictions are widely used for *option valuation*, estimating the value and risk of various financial instruments involving commodity prices. $\diamond$

As always in Bayesian analysis, the solution to the Bayesian prediction problem is based on the posterior distribution of θ .

¹From M. Rosenberg, J. Bryngelson, M. Baron, and A. Papalexopoulos. *Transmission Valuation Analysis Based on Real Options with Price Spikes*, in *Handbook of Power Systems*, Springer, 2010, pp. 101-125

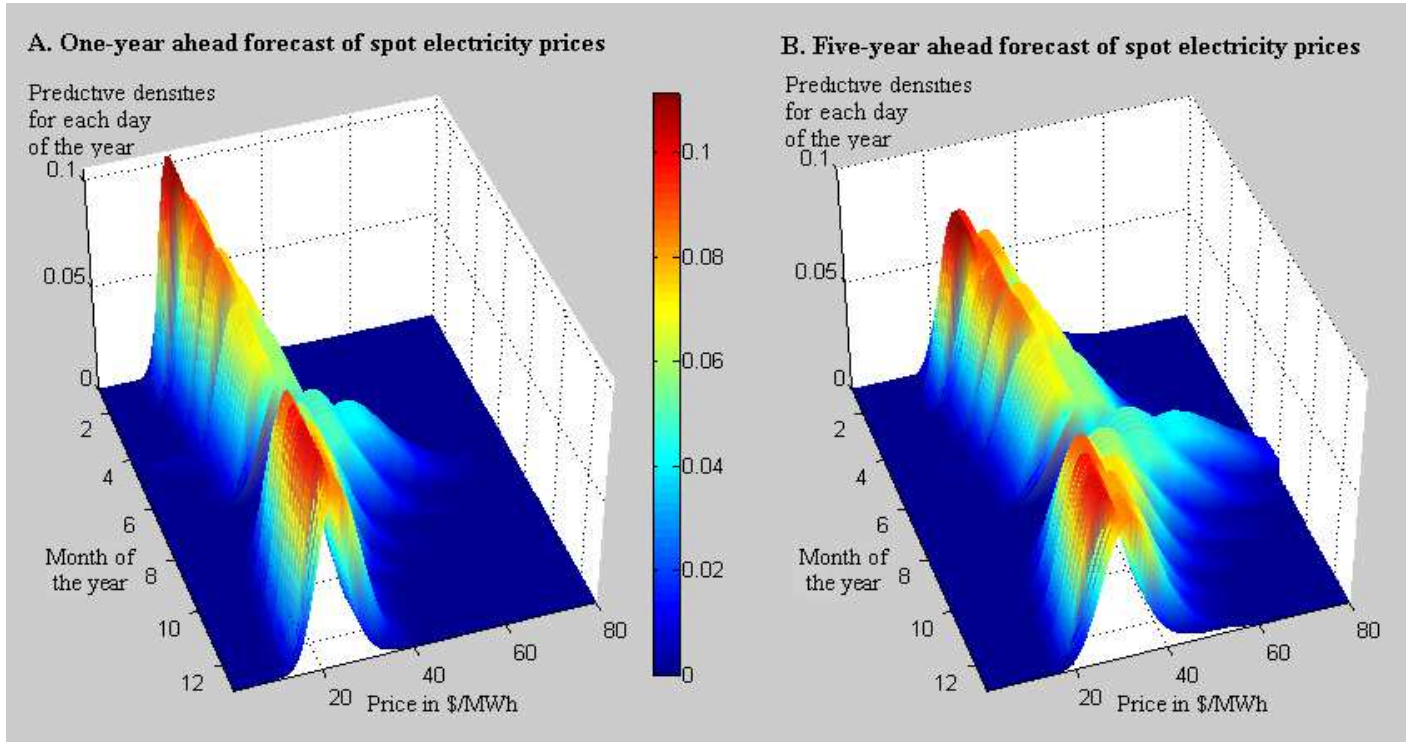


FIGURE 7.5: Predictive densities of electricity prices.

First, combine the posterior of θ with the distribution of Z given θ to obtain the joint distribution of Z and θ ,

$$g(z, \theta | \mathbf{X}) = g(z | \theta) \pi(\theta | \mathbf{X}).$$

Then, integrate the joint density over θ , to obtain the marginal distribution of Z ,

$$p(z | \mathbf{X}) = \int g(z, \theta | \mathbf{X}) d\theta = \int g(z | \theta) \pi(\theta | \mathbf{X}) d\theta.$$

This is the prediction density of Z . This integral also represents the expectation $\mathbf{E}\{g(z | \theta) | \mathbf{X}\}$, taken with respect to the posterior distribution of θ .

**Predictive
density**

$$p(z | \mathbf{X}) = \int g(z | \theta) \pi(\theta | \mathbf{X}) d\theta = \mathbf{E}\{g(z | \theta) | \mathbf{X}\}$$

If $\mathbf{X}$ and Z are not independent, given θ , the density of Z is $g(z | \theta, \mathbf{X})$.

Example 7.14 (POISSON + GAMMA = NEGATIVE BINOMIAL). Let's predict the number of accidents Z on a new segment of a highway during one year. It has a $\text{Poisson}(\theta)$ distribution, but we do not know parameter θ . On the other hand, we observed the accident counts $X_1, \dots, X_n$ on n existing similar segments of this highway, and presumably, they are also $\text{Poisson}(\theta)$. The prior distribution of θ is chosen to belong to the conjugate $\text{Gamma}(\alpha, \lambda)$ family. What is the predictive distribution of Z ?

We already know, from sec. 4.3.1, that the posterior distribution for this situation is $\text{Gamma}(\alpha_x, \beta_x)$,

$$\pi(\theta | \mathbf{X}) \sim \theta^{\alpha_x - 1} e^{-\lambda_x \theta}, \quad \text{where } \alpha_x = \alpha + \sum_{i=1}^n X_i \text{ and } \lambda_x = \lambda + n.$$

Variable Z is discrete, so we are looking for its probability mass function $p(z | \mathbf{X}) = \mathbf{P}\{Z = z | \mathbf{X}\}$. Given θ , Z is $\text{Poisson}(\theta)$, i.e.,

$$g(z | \theta) = e^{-\theta} \theta^z / z!$$

(but we cannot use this formula for predictions because θ is unknown). Notice that I am not dropping a “constant” $z!$ because we are looking for the distribution of Z , so we need to keep the terms containing not only θ but also z .

Then, the predictive distribution is

$$\begin{aligned} p(z | \mathbf{X}) &= \int g(z | \theta) \pi(\theta | \mathbf{X}) d\theta \\ &\sim \int_0^\infty e^{-\theta} \theta^z / z! \cdot \theta^{\alpha_x - 1} e^{-\lambda_x \theta} d\theta \\ &= \frac{1}{z!} \int_0^\infty e^{-\theta(\lambda_x + 1)} \theta^{z + \alpha_x - 1} d\theta \end{aligned}$$

Do you know how to take this integral in one step, without knowing any integration techniques? Just notice that we are integrating a function $e^{-\theta(\lambda_x+1)}\theta^{z+\alpha_x-1}$ that is very similar to a density of Gamma distribution. It just needs a normalizing constant to be a full density, and then we know that the integral of a density is 1. All right, we can insert the needed constant, whose form we can find in the inventory of distributions in Appendix:

$$\begin{aligned}
 p(z | \mathbf{X}) &\sim \frac{1}{z!} \int_0^\infty e^{-\theta(\lambda_x+1)} \theta^{z+\alpha_x-1} d\theta \\
 &= \frac{\Gamma(z + \alpha_x)}{z! (\lambda_x + 1)^{z+\alpha_x}} \underbrace{\int_0^\infty \frac{(\lambda_x + 1)^{z+\alpha_x}}{\Gamma(z + \alpha_x)} e^{-\theta(\lambda_x+1)} \theta^{z+\alpha_x-1} d\theta}_{\text{Integral of any density over its full range equals 1}} \\
 &= \frac{\Gamma(z + \alpha_x)}{z! (\lambda_x + 1)^{z+\alpha_x}} \\
 &\sim \frac{\Gamma(z + \alpha_x)}{z!} \cdot \frac{1}{(\lambda_x + 1)^z}, \text{ for } z = 0, 1, 2, \dots
 \end{aligned}$$

Last step. Look through the list of discrete distributions to find one that looks like this $p(z | \mathbf{X})$. It contains a factorial $z!$, a Gamma function that is also a factorial $\Gamma(z + \alpha_x) = (z + \alpha_x - 1)!$ (see the Calculus review in Appendix), and a small number to the power of z . Also, the range of Z is $z \in \{0, 1, 2, \dots\}$ because Z is Poisson, given θ .

That's it! Have you recognized the *Negative Binomial* distribution? Let us write $p(z | \mathbf{X})$ this way,

$$\begin{aligned}
 p(z | \mathbf{X}) &\sim \frac{(z + \alpha_x - 1)!}{z!} \cdot \frac{1}{(\lambda_x + 1)^z} \\
 &\sim \frac{(z + \alpha_x - 1)!}{z! (\alpha_x - 1)!} \cdot \left(\frac{1}{\lambda_x + 1} \right)^z \left(1 - \frac{1}{\lambda_x + 1} \right)^{\alpha_x} \\
 &= \binom{z + \alpha_x - 1}{z} p^z (1 - p)^{\alpha_x}
 \end{aligned}$$

where $p = \frac{1}{\lambda_x + 1}$. Not exactly the Negative Binomial distribution as it is given in Appendix? Well, this is the distribution of *the number of failures* before the α_x -th success.

We conclude that the predictive distribution of Z , the number of accidents on a new segment of a highway, is *Negative Binomial* (the number of failures before the given number of successes) with parameters

$$\alpha_x = \alpha + \sum X_i \quad \text{and} \quad p = \frac{1}{\lambda_x + 1} = \frac{1}{\lambda + n}$$

and the probability mass function

$$p(z | \mathbf{X}) = \frac{(z + \alpha + \sum X_i - 1)!}{z! (\alpha + \sum X_i - 1)!} \left(\frac{1}{\lambda + n} \right)^z \left(1 - \frac{1}{\lambda + n} \right)^{\alpha + \sum X_i} \quad \text{for } z = 0, 1, 2, \dots$$

◇

Example 7.15 (PREDICTION IN LINEAR REGRESSION). Consider a no-intercept regression model

$$Y_i = \beta x_i + \varepsilon_i, \quad \varepsilon_i \sim \text{i.i.d. Normal}(0, \sigma^2).$$

Independent variables x_i are fixed, responses Y_i are observed, and the slope β is an unknown parameter. Our goal is to predict a response Z for a new regressor x_0 . Assume a non-informative prior $\pi(\beta) \equiv 1$.

From regression, we might know that the slope is estimated by

$$b = \frac{\sum x_i Y_i}{\sum x_i^2}.$$

Given β , the distribution of b is $\text{Normal}(\beta, \sigma^2 / \sum x_i^2)$,

$$f(b | \beta) \sim e^{-(b-\beta)^2/2(\sigma^2/\sum x_i^2)}.$$

Then, as in sec. 4.4, the posterior distribution of β is $\text{Normal}(b, \sigma^2 / \sum x_i^2)$,

$$\pi(\beta | \mathbf{Y}) = \pi(\beta | b) \sim f(b | \beta) \cdot 1 \sim e^{-(\beta-b)^2/2(\sigma^2/\sum x_i^2)}.$$

Also, for a new value x_0 , the response Z is $\text{Normal}(\beta x_0, \sigma^2)$, according to our regression model,

$$g(z | \beta) \sim \text{Normal}(\beta x_0, \sigma^2) \sim e^{-(z-\beta x_0)^2/2\sigma^2}.$$

Following the formula for the predictive density, we have to integrate

$$p(z | \mathbf{Y}) = \int g(z | \beta) \pi(\beta | \mathbf{Y}) d\beta \sim \int_{-\infty}^{\infty} e^{-(z-\beta x_0)^2/2\sigma^2} e^{-(\beta-b)^2/2(\sigma^2/\sum x_i^2)} d\beta,$$

which is certainly possible but long. Instead, as a shortcut, we notice that the joint density $g(z, \beta | \mathbf{Y})$ that we are integrating here is Bivariate Normal. Therefore, the predictive distribution of Z is Normal, and we only need to find its parameters:

$$\begin{aligned} \mathbf{E}\{Z | b\} &= \mathbf{E}(\mathbf{E}\{Z | \beta\} | b) = \mathbf{E}(\beta x_0 | b) = b x_0, \\ \text{Var}\{Z | b\} &= \mathbf{E}(\text{Var}\{Z | \beta\} | b) + \text{Var}(\mathbf{E}\{Z | \beta\} | b) \\ &= \mathbf{E}(\sigma^2 | b) + \text{Var}(\beta x_0 | b) \\ &= \sigma^2 + x_0^2 \frac{\sigma^2}{\sum x_i^2} = \sigma^2 \left(1 + \frac{x_0^2}{\sum x_i^2}\right). \end{aligned}$$

Now we can use this predictive distribution to test hypotheses about the new response Z , construct an HPD predictive credible set for it, etc. $\diamond$

Chapter 8

Monte Carlo methods

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In this chapter, we discuss the general use of Monte Carlo methodologies. In general, people use random computer-generated simulations in order to estimate certain quantities: probabilities, expectations, variances, and sometimes the entire densities.

Computer simulations refer to a regeneration of a process by writing a suitable computer program and observing its results. *Monte Carlo methods* are those based on computer simulations involving random numbers.

The main purpose of simulations is estimating such quantities whose direct computation is complicated, risky, consuming, expensive, or impossible. For example, suppose a complex device or machine is to be built and launched. Before it happens, its performance is simulated, and this allows experts to evaluate its adequacy and associated risks carefully and safely. For example, one surely prefers to evaluate reliability and safety of a new module of a space station by means of computer simulations rather than during the actual mission.

Monte Carlo methods are mostly used for the computation of probabilities, expected values, and other distribution characteristics. Recall that probability can be defined as a *long-run proportion*. With the help of random number generators, computers can actually simulate a *long run*. Then, probability can be estimated by a mere computation of the associated observed frequency. The longer run is simulated, the more accurate result is obtained. Similarly, one can estimate expectations, variances, and other distribution characteristics from a long run of simulated random variables.

Monte Carlo methods inherit their name from Europe's most famous *Monte Carlo casino* (Figure 8.1) located in Principality of Monaco in Europe since the 1850s. Probability distributions involved in gambling are often complicated, but they can be assessed via simulations. In early times, probabilists generated extra income by estimating vital probabilities, devising optimal gambling strategies, and selling them to professional gamblers.



FIGURE 8.1 *Casino Monte Carlo in Principality of Monaco.*

8.1 Applications and examples

Let us briefly look at a few examples, where distributions are rather complicated, and thus, Monte Carlo simulation appears simpler than any direct computation. Notice that in these examples, instead of generating the *actual* devices, computer systems, networks, viruses, and so on, we only simulate the associated *random variables*. For the study of probabilities and distributions, this is entirely sufficient.

Example 8.1 (FORECASTING). Given just a basic distribution model, it is often very difficult to make reasonably *remote predictions*. Often a one-day development depends on the results obtained during all the previous days. Then prediction for tomorrow may be straightforward whereas computation of a one-month forecast is already problematic.

On the other hand, *simulation* of such a process can be easily performed day by day (or even minute by minute). Based on present results, we simulate the next day. Now we “know it,” and thus, we can simulate the day after that, etc. For every time n , we simulate X_{n+1} based on already known $X_1, X_2, \dots, X_n$. Controlling the length of this do-loop, we obtain forecasts for the next days, weeks, or months. Such simulations result, for example, in beautiful animated weather maps that meteorologists often show to us on TV news. They help predict future paths of storms and hurricanes as well as possibilities of flash floods.

Simulation of future failures reflects reliability of devices and systems. Simulation of future stock and commodity prices plays a crucial role in finance, as it allows valuations of options and other financial deals. $\diamond$

Example 8.2 (PERCOLATION). Consider a network of nodes. Some nodes are connected, say, with transmission lines, others are not (mathematicians would call such a network a

graph). A signal is sent from a certain node. Once a node k receives a signal, it sends it along each of its output lines with some probability p_k . After a certain period of time, one desires to estimate the proportion of nodes that received a signal, the probability for a certain node to receive it, etc.

This general *percolation* model describes the way many phenomena may spread. The role of a signal may be played by a computer virus spreading from one computer to another, or by rumors spreading among people, or by fire spreading through a forest, or by a disease spreading between residents.

Technically, simulation of such a network reduces to generating Bernoulli random variables with parameters p_i . Line i transmits if the corresponding generated variable $X_i = 1$. In the end, we simply count the number of nodes that got the signal, or verify whether the given node received it. $\diamond$

Example 8.3 (QUEUEING). A queuing system is described by a number of random variables. It involves spontaneous arrivals of jobs, their random waiting time, assignment to servers, and finally, their random service time and departure. In addition, some jobs may exit prematurely, others may not enter the system if it appears full, and also, intensity of the incoming traffic and the number of servers on duty may change during the day.

When designing a queuing system or a server facility, it is important to evaluate its vital performance characteristics. This will include the job's average waiting time, the average length of a queue, the proportion of customers who had to wait, the proportion of "unsatisfied customers" (that exit prematurely or cannot enter), the proportion of jobs spending, say, more than one hour in the system, the expected usage of each server, the average number of available (idle) servers at the time when a job arrives, and so on. $\diamond$

Example 8.4 (MARKOV CHAIN MONTE CARLO). There is a modern technique of generating random variables from rather complex, often intractable distributions, as long as *conditional distributions* have a reasonably simple form. In semiconductor industry, for example, the joint distribution of good and defective chips on a produced wafer has a rather complicated correlation structure. As a result, it can only be written explicitly for rather simplified artificial models. On the other hand, the quality of each chip is predictable based on the quality of the surrounding, neighboring chips. Given its neighborhood, conditional probability for a chip to fail can be written, and thus, its quality can be simulated by generating a corresponding Bernoulli random variable with $X_i = 1$ indicating a failure.

According to the *Markov chain Monte Carlo* (MCMC) methodology, a long sequence of random variables is generated from conditional distributions. A wisely designed MCMC will then produce random variables that have the desired *unconditional* distribution, no matter how complex it is. $\diamond$

In all the examples, we saw how different types of phenomena can be computer-simulated. However, one simulation is not enough for estimating probabilities and expectations. After we understand how to program the given phenomenon once, we can embed it in a do-loop and repeat similar simulations a large number of times, generating a *long run*. Since the simulated variables are random, we will generally obtain a number of different *realizations*, from which we calculate probabilities and expectations as long-run frequencies and averages.

8.2 Estimating probabilities

This section discusses the most basic and most typical application of Monte Carlo methods. Keeping in mind that probabilities are long-run proportions, we generate a long run of experiments and compute the proportion of times when our event occurred.

For a random variable X , the probability $p = \mathbf{P}\{X \in A\}$ is estimated by

$$\hat{p} = \hat{\mathbf{P}}\{X \in A\} = \frac{\text{number of } X_1, \dots, X_N \in A}{N},$$

where N is the size of a Monte Carlo experiment, and $X_1, \dots, X_N$ are generated random variables with the same distribution as X .

How accurate is this method? To answer this question, compute $\mathbf{E}(\hat{p})$ and $\text{Std}(\hat{p})$. Since the number of $X_1, \dots, X_N$ that fall within set A has Binomial(N, p) distribution with expectation (Np) and variance $Np(1-p)$, we obtain

$$\begin{aligned}\mathbf{E}(\hat{p}) &= \frac{1}{N}(Np) = p, \text{ and} \\ \text{Std}(\hat{p}) &= \frac{1}{N}\sqrt{Np(1-p)} = \sqrt{\frac{p(1-p)}{N}}.\end{aligned}$$

The first result, $\mathbf{E}(\hat{p}) = p$, shows that our Monte Carlo estimator of p is *unbiased*, so that over a long run, it will on the average return the desired quantity p .

The second result, $\text{Std}(\hat{p}) = \sqrt{p(1-p)/N}$, indicates that the standard deviation of our estimator $\hat{p}$ decreases with N at the rate of $1/\sqrt{N}$. Larger Monte Carlo experiments produce more accurate results. A 100-fold increase in the number of generated variables reduces the standard deviation (therefore, enhancing accuracy) by a factor of 10.

Accuracy of a Monte Carlo study

In practice, how does it help to know the standard deviation of $\hat{p}$?

First, we can assess the accuracy of our results. For large N , we use Normal approximation

of the Binomial distribution of $N\hat{p}$. According to it,

$$\frac{N\hat{p} - Np}{\sqrt{Np(1-p)}} = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{N}}} \approx \text{Normal}(0, 1),$$

therefore,

$$\mathbf{P}\{|\hat{p} - p| > \varepsilon\} = \mathbf{P}\left\{\frac{|\hat{p} - p|}{\sqrt{\frac{p(1-p)}{N}}} > \frac{\varepsilon}{\sqrt{\frac{p(1-p)}{N}}}\right\} \approx 2\Phi\left(-\frac{\varepsilon\sqrt{N}}{\sqrt{p(1-p)}}\right), \quad (8.1)$$

where Φ is the Standard Normal cdf.

Second, we can design a Monte Carlo study that attains desired accuracy. That is, we can choose some small ε and α and conduct a Monte Carlo study of such a size N that will guarantee an error not exceeding ε with high probability $(1 - \alpha)$. In other words, we can find such N that

$$\mathbf{P}\{|\hat{p} - p| > \varepsilon\} \leq \alpha. \quad (8.2)$$

If we knew the value of p , we could have equated the right-hand side of (8.1) to α and could have solved the resulting equation for N . This would have shown how many Monte Carlo simulations are needed in order to achieve the desired accuracy with the desired high probability. However, p is unknown (if p is known, why do we need a Monte Carlo study to estimate it?). Then, we have two possibilities:

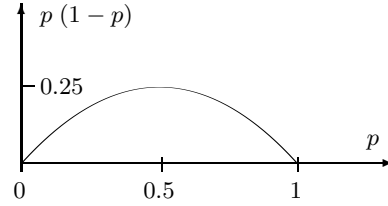


FIGURE 8.2 Function $p(1-p)$ attains its maximum at $p = 0.5$.

1. Use an “intelligent guess” (preliminary estimate) of p , if it is available.
2. Bound $p(1-p)$ by its largest possible value (see Figure 8.2),

$$p(1-p) \leq 0.25 \quad \text{for } 0 \leq p \leq 1.$$

In the first case, if p^* is an “intelligent guess” of p , then we solve the inequality

$$2\Phi\left(-\frac{\varepsilon\sqrt{N}}{\sqrt{p^*(1-p^*)}}\right) \leq \alpha$$

in terms of N . In the second case, we solve

$$2\Phi\left(-2\varepsilon\sqrt{N}\right) \leq \alpha$$

to find the conservatively sufficient size of the Monte Carlo study.

Solutions of these inequalities give us the following rule.

Size of a Monte Carlo study

In order to guarantee that $\mathbf{P}\{|\hat{p} - p| > \varepsilon\} \leq \alpha$, one needs to simulate

$$N \geq p^*(1 - p^*) \left(\frac{z_{\alpha/2}}{\varepsilon} \right)^2$$

random variables, where p^* is a preliminary estimator of p , or

$$N \geq 0.25 \left(\frac{z_{\alpha/2}}{\varepsilon} \right)^2$$

random variables, if no such estimator is available.

Recall that $z_\alpha = \Phi^{-1}(1 - \alpha)$ is such a value of a Standard Normal variable Z that can be exceeded with probability α . The area under the Normal curve to the right of z_α equals α (Figure 8.3).

If this formula returns N that is too small for the Normal approximation, we can use Chebyshev's inequality. We then obtain that

$$N \geq \frac{p^*(1 - p^*)}{\alpha \varepsilon^2}, \quad (8.3)$$

if an “intelligent guess” p^* is available for p , and

$$N \geq \frac{1}{4\alpha \varepsilon^2}, \quad (8.4)$$

otherwise, satisfying the desired condition (8.2).

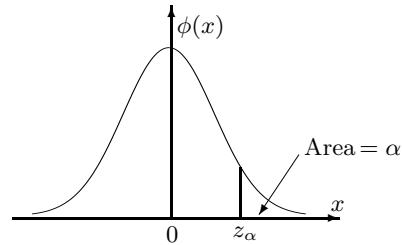


FIGURE 8.3 Critical value z_α .

8.3 Estimating means and standard deviations

Estimation of means, standard deviations, and other distribution characteristics is based on the same principle. We generate a Monte Carlo sequence of random variables $X_1, \dots, X_N$ and compute the necessary long-run averages. The mean $\mathbf{E}(X)$ is estimated by

$$\bar{X} = \frac{1}{N} (X_1 + \dots + X_N),$$

and we know that this estimator $\bar{X}$ is *unbiased* for estimating μ . Its standard deviation is $\text{Std}(\bar{X}) = \sigma/\sqrt{N}$, and it decreases like $1/\sqrt{N}$. For large N , we can use the Central Limit

Theorem again to assess accuracy of our results and to design a Monte Carlo study that attains the desired accuracy (the latter is possible if we have a “guess” about σ).

Also, naturally, variance of X will be estimated by the Monte Carlo sample variance,

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2.$$

8.4 Forecasting

Forecasting the future is often an attractive but uneasy task. In order to predict what happens in t days, one usually has to study events occurring every day between now and t days from now. It is often found that tomorrow’s events depend on today, yesterday, etc. In such common situations, exact computation of probabilities may be long and difficult. However, Monte Carlo forecasts are feasible.

The currently known information can be used to generate random variables for tomorrow. Then, they can be considered “known” information, and we can generate random variables for the day after tomorrow, and so on, until day t .

The rest is similar to the previous sections. We generate N Monte Carlo runs for the next t days and estimate probabilities, means, and standard deviations of day t variables by long-run averaging.

In addition to forecasts for a specific day t , we can predict *how long* a certain process will last, *when* a certain event will occur, and *how many* events will occur.

8.5 Monte Carlo integration

We can extend the Monte Carlo method to estimating *definite integrals* and estimating areas below or above the graphs of corresponding functions. An **R** code for estimating an integral

$$\mathcal{I} = \int_0^1 g(x)dx$$

is

```
N = 10000;           # Number of simulations
U <- runif(N,0,1);    # (U,V) is a random point
V <- runif(N,0,1);    # in the bounding box
I <- mean( V < g(U) ); # Estimator of integral I
```

Expression $\mathbf{V} < \mathbf{g}(\mathbf{U})$ returns an $N \times 1$ vector. Each component equals 1 if the inequality holds for the given pair (U_i, V_i) , hence this point appears below the graph of $g(x)$, and 0 otherwise. The average of these 0s and 1s, calculated by `mean`, is the proportion of 1s, and this is the Monte Carlo estimator of integral I .

Remark: This code assumes that the function g is already defined by the user in a file named “g.m.” Otherwise, its full expression should be written in place of $\mathbf{g}(\mathbf{U})$.

Remark: We also assumed that $0 \leq x \leq 1$ and $0 \leq g(x) \leq 1$. If not, we transform U and V into non-standard Uniform variables $X = a + (b - a)U$ and $Y = cV$, as in the rejection method, then the obtained integral should be multiplied by the box area $c(b - a)$.

Accuracy of results

So far, we estimated lengths, areas, volumes, and integrals by long-run proportions, the method described in Section 8.2. As we noted there, our estimates are unbiased, and their standard deviation is

$$\text{Std}(\hat{\mathcal{I}}) = \sqrt{\frac{\mathcal{I}(1 - \mathcal{I})}{N}}, \quad (8.5)$$

where $\mathcal{I}$ is the actual quantity of interest.

Turns out, there are Monte Carlo integration methods that can beat this rate. Next, we derive an unbiased area estimator with a lower standard deviation. Also, it will not be restricted to an interval $[0, 1]$ or even $[a, b]$.

Improved Monte Carlo integration method

First, we notice that a definite integral

$$\mathcal{I} = \int_a^b g(x)dx = \frac{1}{b-a} \int_a^b (b-a)g(x)dx = \mathbf{E}\{(b-a)g(X)\}$$

equals the expectation of $(b-a)g(X)$ for a $\text{Uniform}(a, b)$ variable X . Hence, instead of using proportions, we can estimate $\mathcal{I}$ by averaging $(b-a)g(X_i)$ for some large number of $\text{Uniform}(a, b)$ variables $X_1, \dots, X_N$.

Furthermore, with a proper adjustment, we can use *any continuous distribution* in place of $\text{Uniform}(a, b)$. To do this, we choose some density $f(x)$ and write $\mathcal{I}$ as

$$\mathcal{I} = \int_a^b g(x)dx = \int_a^b \frac{g(x)}{f(x)}f(x)dx = \mathbf{E}\left(\frac{g(X)}{f(X)}\right),$$

where X has density $f(x)$. It remains to generate a long run of variables $X_1, \dots, X_N$ with this density and compute the average of $g(X_i)/f(X_i)$.

In particular, we are no longer limited to a finite interval $[a, b]$. For example, by choosing $f(x)$ to be a Standard Normal density, we can perform Monte Carlo integration from

$a = -\infty$ to $b = +\infty$:

```
N <- 10000;           # Number of simulations
Z <- dnorm(N,0,1);     # Standard Normal variables
f <- dnorm(Z,0,1);     # Standard Normal density
Iest <- mean( g(Z)/f ); # Estimator of  $\int_{-\infty}^{\infty} g(x) dx$ 
```

Accuracy of the improved method

As we already know, using long-run averaging returns an *unbiased* estimator $\hat{\mathcal{I}}$, hence $\mathbf{E}(\hat{\mathcal{I}}) = \mathcal{I}$. We also know that

$$\text{Std}(\hat{\mathcal{I}}) = \frac{\sigma}{\sqrt{N}},$$

where σ is the standard deviation of a random variable $R = g(X)/f(X)$. Hence, the estimator $\hat{\mathcal{I}}$ is more reliable if σ is small and N is large. Small σ can be obtained by choosing a density $f(x)$ that is approximately proportional to $g(x)$ making R nearly a constant with $\text{Std}(R) \approx 0$. However, generating variables from such a density will usually be just as difficult as computing the integral $\mathcal{I}$.

In fact, using a simple Standard Uniform distribution of X , we already obtain a lower standard deviation than in (8.5). Indeed, suppose that $0 \leq g(x) \leq 1$ for $0 \leq x \leq 1$. For a $\text{Uniform}(0,1)$ variable X , we have $f(X) = 1$ so that

$$\sigma^2 = \text{Var } R = \text{Var } g(X) = \mathbf{E}g^2(X) - \mathbf{E}^2g(X) = \int_0^1 g^2(x)dx - \mathcal{I}^2 \leq \mathcal{I} - \mathcal{I}^2,$$

because $g^2 \leq g$ for $0 \leq g \leq 1$. We conclude that for this method,

$$\text{Std}(\hat{\mathcal{I}}) \leq \sqrt{\frac{\mathcal{I} - \mathcal{I}^2}{N}} = \sqrt{\frac{\mathcal{I}(1 - \mathcal{I})}{N}}.$$

Comparing with (8.5), we see that with the same number of simulations N , the latter method gives more accurate results. We can also say that to attain the same desired accuracy, the second method requires fewer simulations.

This can be extended to an arbitrary interval $[a, b]$ and any function $g \in [0, c]$.

8.6 Markov Chains Monte Carlo

Markov chain Monte Carlo (MCMC) allows to generate samples from an unknown and computationally unavailable distribution. This situation typically occurs in problems with

multidimensional parameters, and where the prior distributions are not conjugate to the model, so the posterior does not have a tractable form.

MCMC is an iterative algorithm that generates a sequence of samples according to a wisely designed *Markov chain* that converges to the desired limiting, *steady-state distribution*.

Several popular algorithms exist for this purpose.

8.7 Gibbs Sampler

Consider a problem with several parameters of interest, $\theta_1, \dots, \theta_d$. Each of them has a prior distribution, and the entire set $\boldsymbol{\theta} = (\theta_1, \dots, \theta_d)$ has a joint prior $\pi(\boldsymbol{\theta})$. These parameters have different meanings - they represent means, variances, shapes, slopes, etc., and so, their joint posterior distribution is intractable. Its exact computation requires multiple integration, and the resulting multivariate density does not have any analytic form.

Example 8.5 (NORMAL DISTRIBUTION WITH UNKNOWN MEAN AND VARIANCE). As one of the simplest examples illustrating this situation, consider the model

$$X_1, \dots, X_n \sim \text{Normal}(\theta, \sigma^2)$$

with both θ and σ^2 being unknown parameters.

In the very best case, suppose that a priori, θ and σ^2 are distributed independently of each other, and according to their corresponding conjugate priors,

$$\begin{aligned}\theta &\sim \text{Normal}(\mu, \tau^2) \\ \sigma^2 &\sim \text{Inverse Gamma}(\alpha, \beta).\end{aligned}$$

Alternatively (and it is not harder at all), we can assume the improper joint prior

$$\pi(\theta, \sigma^2) = \frac{1}{\sigma^2}, \quad \theta \in \mathbf{R}, \sigma > 0,$$

which is $1 \cdot \frac{1}{\sigma^2}$, meaning that θ and σ are still independent of each other, and each of them has its natural non-informative distribution.

If we fix one parameter and vary the other, we have our old known problem.

Let us fix σ^2 . Then the posterior distribution of θ , given data $\mathbf{X}$, is Normal with posterior parameters

$$\mu_x = \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} \quad \text{and} \quad \tau_x^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}$$

or $\text{Normal}(\bar{X}, \sigma^2/n)$ in case of the improper prior. But this is a *posterior distribution conditioned on σ^2* .

Conversely, if we fix θ , then the posterior distribution of σ^2 is Inverse Gamma with posterior parameters

$$\alpha_x = \frac{n}{2} - 1 \text{ and } \beta_x = \frac{2}{\sum (X_i - \theta)^2}.$$

This is conditioned on θ , which is an unknown quantity. For example, we can use this posterior density to calculate

$$\mathbf{P} \{ \sigma^2 > 5 \mid \mathbf{X}, \theta \},$$

but unconditionally from θ , so far we cannot compute $\mathbf{P} \{ \sigma^2 > 5 \mid \mathbf{X} \}$, needed for testing a hypothesis of $\sigma^2 > 5$. We cannot even compute the posterior mean or median of σ^2 directly.

For the posterior analysis (which is Bayesian analysis!) of θ and σ^2 , we need their marginal and/or joint posterior distributions that is independent of any unknown quantities. However, even according to the introduced *rather simple* two-parameter model, the joint posterior distribution

$$\pi(\theta, \sigma^2 \mid \mathbf{X})$$

is intractable. ◇

Gibbs sampling will solve the problem in this Example. It cannot derive the closed form of the posterior density, but it is able to produce any number of pairs (θ, σ^2) coming from that very joint posterior distribution.

Gibbs sampler is a type of MCMC, a computer-generated Markov chain, where transitions occur according to the conditional distributions of one parameter, given all the other quantities in the problem.

Step 0, initialization. Start with some possible parameter values $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, \dots, \theta_d^{(0)})$.

Step 1, a cycle of first iterations.

(1.1) Generate $\theta_1^{(1)}$ from the conditional posterior distribution $\pi(\theta_1 \mid \mathbf{X}, \theta_2^{(0)}, \dots, \theta_d^{(0)})$.

(1.2) Generate $\theta_2^{(1)}$ from the conditional posterior distribution $\pi(\theta_2 \mid \mathbf{X}, \theta_1^{(1)}, \theta_3^{(0)}, \dots, \theta_d^{(0)})$.

(1.2) Generate $\theta_3^{(1)}$ from the conditional posterior distribution $\pi(\theta_3 \mid \mathbf{X}, \theta_1^{(1)}, \theta_2^{(1)}, \theta_4^{(0)}, \dots, \theta_d^{(0)})$.

... ..

.

(1.d) Generate $\theta_d^{(1)}$ from the conditional posterior distribution $\pi(\theta_d \mid \mathbf{X}, \theta_1^{(1)}, \dots, \theta_{d-1}^{(1)})$.

This concludes the 1st round of iterations. As a result, we have a set of parameters $\boldsymbol{\theta}^{(1)} = (\theta_1^{(1)}, \dots, \theta_d^{(1)})$ generated from $\boldsymbol{\theta}^{(0)}$.

Step 2, the 2nd cycle of iterations.

(2.1) Generate $\theta_1^{(2)}$ from the conditional posterior distribution $\pi(\theta_1 | \mathbf{X}, \theta_2^{(1)}, \dots, \theta_d^{(1)})$.

... ..

During each simulation, the most recent generated parameter values are being used.

Iterations continue until MCMC “converges”, i.e., it *stabilizes*. It does not converge to a constant. During each step, the new parameters are still generated according to the conditional distributions, but the overall joint distribution of $\boldsymbol{\theta}^{(N)}$ must become stable after a sufficient number of steps.

When ranges of parameters are independent of each other, the MCMC will converge to the correct joint posterior distribution.

Example 8.6 (TWO-PARAMETER NORMAL MODEL, CONTINUED). The following WinBUGS code can be used for the problem in the previous example. Gibbs sampler converges to the joint posterior distribution of θ and σ^2 .

Notice that the 2nd parameter of the Normal distribution in WinBUGS is “precision”, which is $1/\sigma$. Accordingly, the Gamma prior distribution is used for it instead of Inverse Gamma.

```
model{
  for (i in 1:n) { X[i] ~ dnorm(theta,sigma)}
  theta ~ dnorm(mu,tau)
  sigma ~ dgamma(alpha,beta)
}

list( n=3, X = c(-1.4,0.3,1.9), mu=0, tau=10, alpha=5, beta=0.8 )
```

After convergence, resulting posterior parameter estimates are in the table,

node	mean	sd	MC error	2.5%	median	97.5%	start	sample
sigma	1.768	0.7007	0.007619	0.6789	1.672	3.434	1	10000
theta	0.08807	0.2592	0.002648	-0.4173	0.08963	0.588	1	10000

The marginal posterior densities (unconditional, independent of the other parameter!) are on Fig. 9.5. ◇

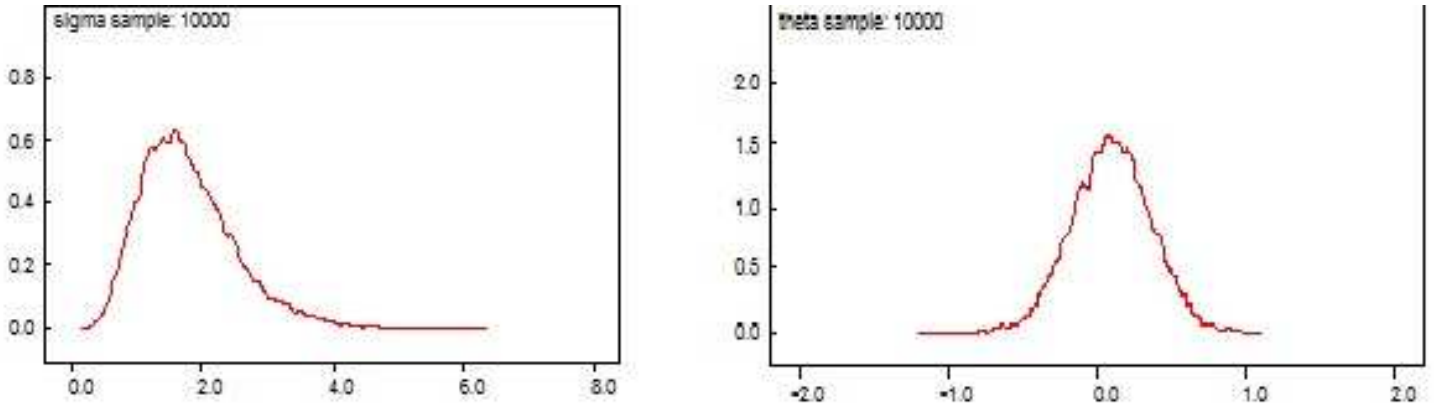


FIGURE 8.4: Estimated posterior densities $\pi(\theta|\mathbf{X})$ and $\pi(\sigma|\mathbf{X})$ for the two-parameter Normal distribution problem.

8.8 Central Limit Theorem for the Posterior Distribution

Another popular way of managing complex posterior structures is finding a working approximation of the posterior distribution. Typically, by a Normal distribution.

We've seen that Gamma, Inverse Gamma, Binomial, Negative Binomial, and some other distributions are approximately Normal when the corresponding parameter is sufficiently large. In fact, we can prove a general result that the posterior $\pi(\theta | \mathbf{X})$ is approximately Normal when the densities are sufficiently smooth, and the sample size is sufficiently large.

8.8.1 Preparation

Recall a few facts from Mathematical Statistics.

1. **Maximum likelihood estimation.** Suppose we observe a sample $\mathbf{X} = (X_1, \dots, X_n)$ from the joint density or probability mass function $f(\mathbf{x} | \theta)$. The maximum likelihood estimator (MLE) of θ is the parameter value $\hat{\theta}$ that maximizes $f(\mathbf{X} | \theta)$ for the given data set $\mathbf{X}$.

Under *regularity conditions* (essentially meaning that the density $f(\mathbf{x} | \theta)$ is twice differentiable in θ for any $\mathbf{x}$, and differentiation under the sign of integral is justified), we have

$$\mathbf{E}_\theta(\hat{\theta}) \rightarrow \theta \text{ and } \text{Var}_\theta(\hat{\theta}) \rightarrow 0, \text{ as } n \rightarrow \infty,$$

and therefore, $\hat{\theta} \rightarrow \theta$ in probability,

$$P\left\{|\hat{\theta} - \theta| > \varepsilon\right\} \rightarrow 0, \text{ as } n \rightarrow \infty, \text{ for } \forall \varepsilon > 0.$$

In short, it means that the MLE $\hat{\theta}$ is *consistent*.

2. **Fisher information.** The Fisher information number is defined as the expected square of the joint density's log-derivative,

$$I(\theta) := \mathbf{E}_{\theta} \left(\frac{\partial \ln f(\mathbf{X} | \theta)}{\partial \theta} \right)^2.$$

Denoting the derivative $\frac{\partial \ln f(\mathbf{X} | \theta)}{\partial \theta}$ by f' and using regularity conditions, we get an alternative formula for $I(\theta)$,

$$\begin{aligned} I(\theta) &= \mathbf{E}_{\theta} \left(\frac{f'}{f} \right)^2 = \int f \left(\frac{f'}{f} \right)^2 = \int \frac{(f')^2}{f} = \int \left(\frac{(f')^2}{f} - f'' \right) \\ &\quad \text{(because by regularity conditions, } \int f'' = \left(\int f \right)' = (1)' = 0) \\ &= \int \frac{(f')^2 - f''f}{f} = -\mathbf{E}_{\theta} \frac{f''f - (f')^2}{f^2} = -\mathbf{E}_{\theta} \frac{\partial(f'/f)}{\partial \theta} = -\mathbf{E}_{\theta} \frac{\partial^2}{\partial \theta^2} \ln f(\mathbf{X} | \theta). \end{aligned}$$

So, $I(\theta) = -\mathbf{E}(\ln f)''$.

For a sample of size n , $I_{\mathbf{X}}(\theta) = nI(\theta)$.

3. **Cramer-Rao inequality.** This is where you probably saw a reference to Fisher information - for any estimator δ of θ , with bias $b(\theta)$ and variance $v(\theta)$ (under regularity conditions)

$$\text{Var}_{\theta}(\delta) \geq \frac{(b'(\theta))^2}{I(\theta)}.$$

8.8.2 Main result, asymptotic normality

Theorem 2 Let $\mathbf{X} = (X_1, \dots, X_n)$ be a sample of independent, identically distributed with the density or probability mass function $f(x | \theta)$ that satisfies the regularity conditions.

Then, the posterior distribution of θ is asymptotically Normal,

$$\pi(\theta | \mathbf{X}) \approx \text{Normal} \left(\hat{\theta}, \frac{1}{nI(\hat{\theta})} \right), \text{ as } n \rightarrow \infty.$$

Proof.

This is a rather classical proof. We write

$$f(\mathbf{X} | \theta) = e^{\ln f(\mathbf{X} | \theta)}$$

and apply the Taylor expansion of $\ln f$ at the point $\theta = \hat{\theta}$,

$$\ln f(\mathbf{X} | \theta) = \ln f(\mathbf{X} | \hat{\theta}) + (\ln f(\mathbf{X} | \hat{\theta}))' \Big|_{\theta=\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2} (\ln f(\mathbf{X} | \hat{\theta}))'' \Big|_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + O(\theta - \hat{\theta})^3.$$

In this formula, $(\ln f(\mathbf{X} | \hat{\theta}))' \Big|_{\theta=\hat{\theta}} = 0$ because $\hat{\theta}$, the MLE, is the point of maximum, and the derivative equals 0 there.

Also, $(\ln f(\mathbf{X} | \hat{\theta}))'' \Big|_{\theta=\hat{\theta}}$ converges in probability to $\mathbf{E}_{\theta=\hat{\theta}}(\ln f(\mathbf{X} | \hat{\theta}))'' = -I(\hat{\theta})$, because $\text{Var}(\ln f(\mathbf{X} | \hat{\theta}))'' \rightarrow 0$ as long as $\text{Var}(\hat{\theta}) \rightarrow 0$.

Plugging these results into the model and then, the posterior distribution, we get

$$f(\mathbf{X} | \theta) \approx e^{\ln f(\mathbf{X} | \hat{\theta}) - \frac{1}{2}I(\hat{\theta})(\theta - \hat{\theta})^2} = f(\mathbf{X} | \hat{\theta})e^{-\frac{1}{2}I_{\mathbf{X}}(\hat{\theta})(\theta - \hat{\theta})^2}$$

(already looking like a Normal density of θ) and

$$\pi(\theta | \mathbf{X}) \approx \frac{f(\mathbf{X} | \hat{\theta})e^{-\frac{1}{2}I_{\mathbf{X}}(\hat{\theta})(\theta - \hat{\theta})^2} \pi(\theta)}{\int f(\mathbf{X} | \hat{\theta})e^{-\frac{1}{2}I_{\mathbf{X}}(\hat{\theta})(\theta - \hat{\theta})^2} \pi(\theta) d\theta}$$

Why did I cancel $\pi(\theta)$? In the region where $|\theta - \hat{\theta}| \leq \varepsilon$, $\pi(\theta) \approx \pi(\hat{\theta})$, a constant of θ , and it cancels. On the other hand, in the region where $|\theta - \hat{\theta}| > \varepsilon$, we have $e^{-\frac{1}{2}I_{\mathbf{X}}(\hat{\theta})(\theta - \hat{\theta})^2} \rightarrow 0$ because $I_{\mathbf{X}}(\hat{\theta}) = nI(\hat{\theta}) \rightarrow \infty$.

All right, now we obtained

$$\pi(\theta | \mathbf{X}) \sim e^{-\frac{1}{2}I_{\mathbf{X}}(\hat{\theta})(\theta - \hat{\theta})^2},$$

a Normal density with mean $\hat{\theta}$ and variance $I_{\mathbf{X}}^{-1}(\hat{\theta}) = \frac{1}{nI(\hat{\theta})}$.



Chapter 9

Computer Generation of Random Variables

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As we see, implementation of Monte Carlo methods reduces to generation of random variables from given distributions. Hence, it remains to design algorithms for generating random variables and vectors that will have the desired distributions.

Statistics software packages like SAS, R, Splus, SPSS, Minitab, and others have built-in procedures for the generation of random variables from the most common discrete and continuous distributions. Similar tools are found in recent versions of **MATLAB**, Microsoft Excel, and some libraries.

Majority of computer languages have a *random number generator* that returns only *Uniformly distributed* independent random variables. This section discusses general methods of transforming Uniform random variables, obtained by a standard random number generator, into variables with desired distributions.

9.1 Random number generators

Obtaining a good random variable is not a simple task. How do we know that it is “truly random” and does not have any undesired patterns? For example, quality random number generation is so important in coding and password creation that people design special tests to verify the “randomness” of generated numbers.

In the fields sensitive to good random numbers, their generation is typically related to an accurate measurement of some physical variable, for example, the computer time or noise. Certain transformations of this variable will generally give the desired result.

More often than not, a *pseudo-random number generator* is utilized. This is nothing but a very long list of numbers. A user specifies a *random number seed* that points to the location from which this list will be read. It should be noted that if the same computer code is executed with the same seed, then the same random numbers get generated, leading to identical results. Often each seed is generated within the system, which of course improves the quality of random numbers.

Instead of a computer, a **table of random numbers** may be used for small-size studies.

Can a table adequately replace a computer random number generator? There are at least two issues that one should be aware of.

1. Generating a *random number seed*. A rule of thumb suggests to close your eyes and put your finger “somewhere” on the table. Start reading random numbers from this point, either horizontally, or vertically, or even diagonally, etc. Either way, this method is not pattern-free, and it does not guarantee “perfect” randomness.
2. Using the same table more than once. Most likely, we cannot find a new table of random numbers for each Monte Carlo study. As soon as we use the same table again, we may no longer consider our results independent of each other. Should we be concerned about it? It depends on the situation. For totally unrelated projects, this should not be a problem.

One way or another, a random number generator or a table of random numbers delivers to us Uniform random variables $U_1, U_2, \dots \in (0, 1)$. The next three subsections show how to transform them into a random variable (vector) X with the given desired distribution $F(x)$.

$$\text{NOTATION} \quad \left\| U; U_1, U_2, \dots \right\| = \begin{array}{l} \text{generated Uniform(0,1)} \\ \text{random variables} \end{array} \quad \left\| \right.$$

9.2 Discrete methods

At this point, we have obtained one or several independent Uniform(0,1) random variables by means of a random number generator or a table of random numbers. Variables from certain simple distributions can be immediately generated from this.

Example 9.1 (BERNOULLI). First, simulate a Bernoulli trial with probability of success p . For a Standard Uniform variable U , define

$$X = \begin{cases} 1 & \text{if } U < p \\ 0 & \text{if } U \geq p \end{cases}$$

We call it “a success” if $X = 1$ and “a failure” if $X = 0$. Using the Uniform distribution of U , we find that

$$P\{\text{success}\} = P\{U < p\} = p.$$

Thus, we have generated a Bernoulli trial, and X has Bernoulli distribution with the desired probability p , see Figure 9.1a.

A **MATLAB** code for this scheme is

```
U = rand;
X = (U < p)
```

The value of p should be specified prior to this program. $\diamond$

Example 9.2 (BINOMIAL). Once we know how to generate Bernoulli variables, we can obtain a Binomial variable as a sum of n independent Bernoulli. For this purpose, we start with n Uniform random numbers, for example:

```
n = 20; p = 0.68;
U = rand(n,1);
X = sum(U < p)
```

$\diamond$

Example 9.3 (GEOMETRIC). A while-loop of Bernoulli trials will generate a Geometric random variable, literally according to its definition on p. ???. We run the loop of trials until the first success occurs. Variable X counts the number of failures:

```
X = 1;           % Need at least one trial
while rand > p;  % Continue while there are failures
    X = X+1;
end;             % Stop at the first success
X
```

The percentage sign (%) is used to separate MATLAB statements from comments. $\diamond$

Example 9.4 (NEGATIVE BINOMIAL). Just as in Example 9.2, once we know how to generate a Geometric variable, we can generate a number of them and obtain a Negative Binomial(k, p) variable as a sum of k independent Geometric(p) variables. $\diamond$

Arbitrary discrete distribution

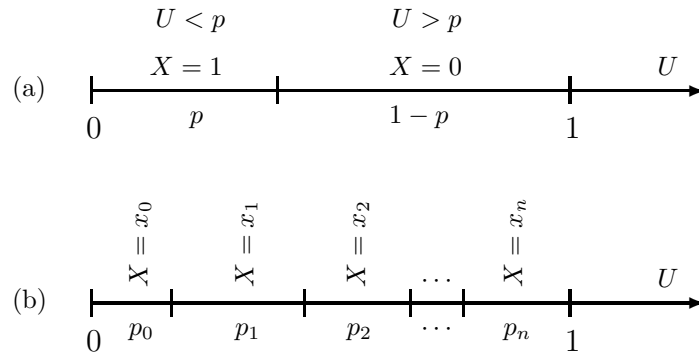


FIGURE 9.1: Generating discrete random variables. The value of X is determined by the region where the generated value of U belongs.

Example 9.1 can be extended to any arbitrary discrete distribution. In this Example, knowing that the random number generator returns a number between 0 and 1, we divided the interval $[0, 1]$ into two parts, p and $(1 - p)$ in length. Then we determined the value of X according to the part where the generated value of U fell, as in Figure 9.1a.

Now consider an arbitrary discrete random variable X that takes values $x_0, x_1, \dots$ with probabilities $p_0, p_1, \dots$,

$$p_i = \mathbf{P}\{X = x_i\}, \quad \sum_i p_i = 1.$$

The scheme similar to Example 9.1 can be applied as follows.

Algorithm 9.1 (Generating discrete variables)

1. Divide the interval $[0, 1]$ into subintervals as shown in Figure 9.1b,

$$\begin{aligned} A_0 &= [0, p_0) \\ A_1 &= [p_0, p_0 + p_1) \\ A_2 &= [p_0 + p_1, p_0 + p_1 + p_2) \\ &\text{etc.} \end{aligned}$$

Subinterval A_i will have length p_i ; there may be a finite or infinite number of them, according to possible values of X .

2. Obtain a Standard Uniform random variable from a random number generator or a table of random numbers.
3. If U belongs to A_i , let $X = x_i$.

From the Uniform distribution, it follows again that

$$\mathbf{P}\{X = x_i\} = \mathbf{P}\{U \in A_i\} = p_i.$$

Hence, the generated variable X has the desired distribution.

Notice that contrary to Examples 9.2 and 9.4, this algorithm is economic as it requires only one Uniform random number for each generated variable.

Values x_i can be written in any order, but they have to correspond to their probabilities p_i .

Example 9.5 (POISSON). Let us use Algorithm 9.1 to generate a Poisson variable with parameter $\lambda = 5$.

Recall that a Poisson variable takes values $x_0 = 0, x_1 = 1, x_2 = 2, \dots$ with probabilities

$$p_i = \mathbf{P}\{X = x_i\} = e^{-\lambda} \frac{\lambda^i}{i!} \text{ for } i = 0, 1, 2, \dots$$

Following the algorithm, we generate a Uniform random number U and find the set A_i containing U , so that

$$p_0 + \dots + p_{i-1} \leq U < p_0 + \dots + p_{i-1} + p_i,$$

or, in terms of the cumulative distribution function,

$$F(i-1) \leq U < F(i).$$

This can be done in **MATLAB** by means of a while-loop:

```
lambda = 5;                % Parameter
U       = rand;            % Generated Uniform variable
i       = 0;               % Initial value
F       = exp(-lambda);    % Initial value, F(0)
while (U >= F);            % The loop ends when U < F(i)
    F = F + exp(-lambda) * lambda^i / gamma(i+1);
    i = i + 1;
end;
X=i
```

◇

Remark: Statistical toolbox of **MATLAB** has built-in tools for generating random variables from certain given distributions. Generating Binomial, Geometric, Poisson, and some other variables can be done in one line `random(name,parameters)`, where “name” indicates the desired family of distributions, or by commands `binornd`, `poissrnd`, etc., listed in our distribution inventory. This chapter discusses general methods of generating random variables from arbitrary distributions. The codes given here are general too – they can be used as flow-charts and be directly translated into other languages.

9.3 Inverse transform method

As Algorithm 9.1 applies to all discrete distributions, we now turn attention to the *generation of continuous random variables*. The method will be based on the following simple yet surprising fact.

Theorem 3 *Let X be a continuous random variable with cdf $F_X(x)$. Define a random variable $U = F_X(X)$. The distribution of U is $\text{Uniform}(0,1)$.*

PROOF: First, we notice that $0 \leq F(x) \leq 1$ for all x , therefore, values of U lie in $[0, 1]$. Second, for any $u \in [0, 1]$, find the cdf of U ,

$$\begin{aligned} F_U(u) &= \mathbf{P}\{U \leq u\} \\ &= \mathbf{P}\{F_X(X) \leq u\} \\ &= \mathbf{P}\{X \leq F_X^{-1}(u)\} && \text{(solve the inequality for } X\text{)} \\ &= F_X(F_X^{-1}(u)) && \text{(by definition of cdf)} \\ &= u && (F_X \text{ and } F_X^{-1} \text{ cancel}) \end{aligned}$$

In the case if F_X is not invertible, let $F_X^{-1}(u)$ denote the smallest x such that $F_X(x) = u$.

We see that U has cdf $F_U(u) = u$ and density $f_U(u) = F'_U(u) = 1$ for $0 \leq u \leq 1$. This is the $\text{Uniform}(0,1)$ density, hence, U has $\text{Uniform}(0,1)$ distribution. $\square$

Regardless of the initial distribution of X , it becomes $\text{Uniform}(0,1)$, once X is substituted into its own cumulative distribution function!

Arbitrary continuous distribution

In order to generate variable X with the given continuous cdf F , let us revert the formula $U = F(X)$. Then X can be obtained from a generated Standard Uniform variable U as $X = F^{-1}(U)$.

Algorithm 9.2 (Generating continuous variables)

1. Obtain a Standard Uniform random variable from a random number generator.
2. Compute $X = F^{-1}(U)$. In other words, solve the equation $F(X) = U$ for X .

Example 9.6 (EXPONENTIAL). How shall we generate an Exponential variable with parameter λ ? According to Algorithm 9.2, we start by generating a Uniform random variable U . Then we recall that the Exponential cdf is $F(x) = 1 - e^{-\lambda x}$ and solve the equation

$$1 - e^{-\lambda X} = U.$$

The result is

$$X = -\frac{1}{\lambda} \ln(1 - U). \quad (9.1)$$

Can this formula be simplified? By any rule of algebra, the answer is “no.” On the other hand, $(1 - U)$ has the same distribution as U , Standard Uniform. Therefore, we can replace U by $(1 - U)$, and variable

$$X_1 = -\frac{1}{\lambda} \ln(U), \quad (9.2)$$

although different from X , will also have the desired Exponential(λ) distribution.

Let us check if our generated variables, X and X_1 , have a suitable range of values. We know that $0 < U < 1$ with probability 1, hence $\ln(U)$ and $\ln(1 - U)$ are negative numbers, so that both X and X_1 are positive, as Exponential variables should be. Just a check. $\diamond$

Example 9.7 (GAMMA). Gamma cdf has a complicated integral form, not mentioning F^{-1} , needed for Algorithm 9.2. There are numerical methods of solving the equation $F(X) = U$, but alternatively, much simpler, for any integer α , a Gamma variable can be generated as a sum of α independent Exponential variables:

```
X = sum( -1/lambda * log(rand(alpha,1)) )
```

$\diamond$

Discrete distributions revisited

Algorithm 9.2 is not directly applicable to discrete distributions because the inverse function F^{-1} does not exist in the discrete case. In other words, the key equation $F(X) = U$ may have either infinitely many roots or no roots at all (see Figure 9.2 on p. 122).

Moreover, a discrete variable X has a finite or countable range of possible values, and so does $F(X)$. The probability that U coincidentally equals one of these values is 0. We conclude that the equation $F(X) = U$ has no roots with probability 1.

Let us modify the scheme in order to accommodate discrete variables. Instead of solving

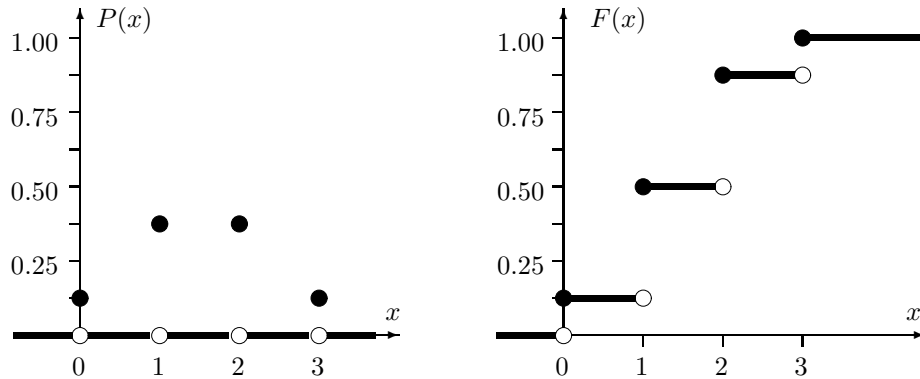


FIGURE 9.2: The probability mass function $P(x)$ and the cumulative distribution function $F(x)$ of the Binomial(3,0.5) distribution. White circles denote excluded points.

$F(x) = U$ exactly, which is impossible, we solve it approximately by finding x , the smallest possible value of X such that $F(x) > U$.

The resulting algorithm is described below.

Algorithm 9.3 (Generating discrete variables, revisited)

1. Obtain a Standard Uniform random variable from a random number generator or a table of random numbers.
2. Compute $X = \min \{x \in S \text{ such that } F(x) > U\}$, where S is a set of possible values of X .

Algorithms 9.1 and 9.3 are equivalent if values x_i are arranged in their increasing order in Algorithm 9.1.

Example 9.8 (GEOMETRIC REVISITED). Applying Algorithm 9.3 to Geometric cdf

$$F(x) = 1 - (1 - p)^x,$$

we need to find the smallest integer x satisfying the inequality (solve it for x)

$$1 - (1 - p)^x > U, \quad (1 - p)^x < 1 - U, \quad x \ln(1 - p) < \ln(1 - U),$$

$$x > \frac{\ln(1 - U)}{\ln(1 - p)} \quad (\text{changing the sign because } \ln(1 - p) < 0)$$

The smallest such integer is the ceiling¹ of $\ln(1 - U)/\ln(1 - p)$, so that

$$X = \left\lceil \frac{\ln(1 - U)}{\ln(1 - p)} \right\rceil \quad (9.3)$$

¹Remark: Ceiling function $\lceil x \rceil$ is defined as the smallest integer that is no less than x .

9.4 Rejection method

Besides a good computer, one thing is needed to generate continuous variables using the inverse transform method of Section 9.3. It is a reasonably *simple form of the cdf* $F(x)$ that allows direct computation of $X = F^{-1}(U)$.

When $F(x)$ has a complicated form but a density $f(x)$ is available, random variables with this density can be generated by *rejection method*.

Consider a point (X, Y) chosen at random from under the graph of density $f(x)$, as shown in Figure 9.3. What is the distribution of X , its first coordinate?

Theorem 4 *Let a pair (X, Y) have Uniform distribution over the region*

$$A = \{(x, y) \mid 0 \leq y \leq f(x)\}$$

for some density function f . Then f is the density of X .

PROOF: Uniform distribution has a constant density. In case of a pair (X, Y) , this density equals 1 because

$$\iint_A 1 \, dy \, dx = \int_x \left(\int_{y=0}^{f(x)} 1 \, dy \right) dx = \int_x f(x) \, dx = 1$$

because $f(x)$ is a density.

The marginal density of X obtains from the joint density by integration,

$$f_X(x) = \int f_{(X,Y)}(x, y) \, dy = \int_{y=0}^{f(x)} 1 \, dy = f(x).$$

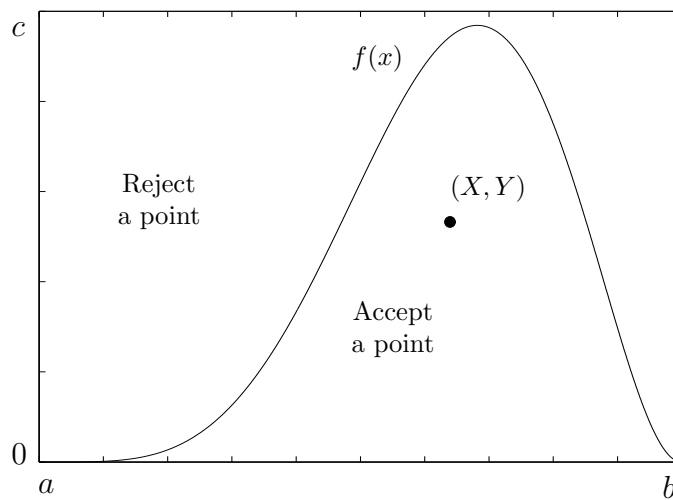


FIGURE 9.3: *Rejection method. A pair (X, Y) should have a Uniform distribution in the region under the graph of $f(x)$.*

□

It remains to generate Uniform points in the region A . For this purpose, we select a *bounding box* around the graph of $f(x)$ as shown in Figure 9.3, generate random point in this rectangle, and reject all the points not belonging to A . The remaining points are Uniformly distributed in A .

Algorithm 9.4 (*Rejection method*)

1. Find such numbers a , b , and c that $0 \leq f(x) \leq c$ for $a \leq x \leq b$. The bounding box stretches along the x -axis from a to b and along the y -axis from 0 to c .
2. Obtain Standard Uniform random variables U and V from a random number generator or a table of random numbers.
3. Define $X = a + (b - a)U$ and $Y = cV$. Then X has Uniform(a, b) distribution, Y is Uniform(0, c), and the point (X, Y) is Uniformly distributed in the bounding box.
4. If $Y > f(X)$, reject the point and return to step 2. If $Y \leq f(X)$, then X is the desired random variable having the density $f(x)$.

Example 9.9 (REJECTION METHOD FOR BETA DISTRIBUTION). Beta distribution has density

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \text{ for } 0 \leq x \leq 1.$$

For $\alpha = 5.5$ and $\beta = 3.1$, this density is graphed in Figure 9.3. It never exceeds 2.5, therefore we choose the bounding box with $a = 0$, $b = 1$, and $c = 2.5$. **MATLAB** commands for generating a Beta(α, β) random variable by the rejection method are:

```
alpha=5.5; beta=3.1; a=0; b=1; c=2.5;
X=0; Y=c; % Initial values
while Y > gamma(alpha+beta)/gamma(alpha)/gamma(beta)...
    * X.^(alpha-1) .* (1-X).^(beta-1);
    U=rand; V=rand; X=a+(b-a)*U; Y=c*V;
end; X
```

In this code, dot operations (preceded by a dot “.”) are applied separately to each element of a matrix (as opposed to operations with whole matrices).

A histogram of 10,000 random variables generated (in a loop) by this algorithm is shown in Figure 9.4. Compare its shape with the graph of the density $f(x)$ in Figure 9.3. It is not as smooth as $f(x)$ because of randomness and a finite sample size, but if the simulation algorithm is designed properly, the shapes should be similar. ◇

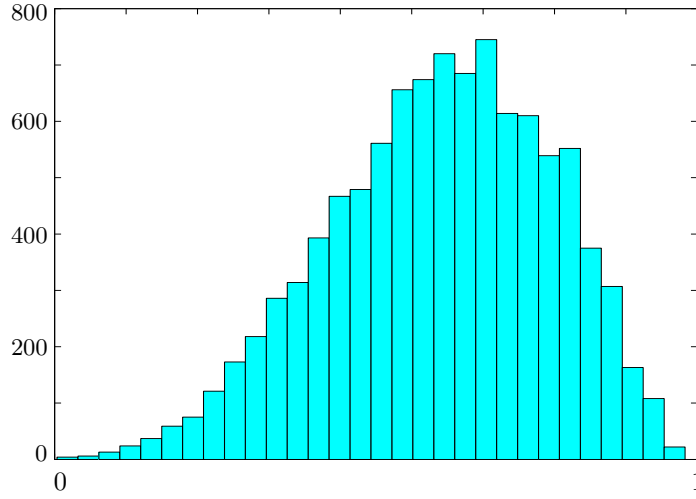


FIGURE 9.4: A histogram of Beta random variables generated by rejection method. Compare with Figure 9.3.

9.5 Generation of random vectors

Along the same lines, we can use rejection method to generate *random vectors* having desired joint densities. A bounding box now becomes a multidimensional cube, where we generate a Uniformly distributed random point $(X_1, X_2, \dots, X_n, Y)$, accepted only if $Y \leq f(X_1, \dots, X_n)$. Then, the generated vector $(X_1, \dots, X_n)$ has the desired joint density

$$\begin{aligned} f_{X_1, \dots, X_n}(x_1, \dots, x_n) &= \int f_{X_1, \dots, X_n, Y}(x_1, \dots, x_n, y) dy \\ &= \int_0^{f(x_1, \dots, x_n)} 1 dy = f(x_1, \dots, x_n). \end{aligned}$$

The inverse transform method can also be used to generate random vectors. Let us generate X_1 with the marginal cdf F_{X_1} . Observe its value, x_1 , and generate X_2 from the *conditional cdf*

$$F_{X_2|X_1}(x_2|x_1) = \begin{cases} \frac{\sum_{x \leq x_2} P_{X_1, X_2}(x_1, x)}{P_{X_1}(x_1)} & \text{in the discrete case} \\ \frac{\int_{-\infty}^{x_2} f_{X_1, X_2}(x_1, x) dx}{f_{X_1}(x_1)} & \text{in the continuous case} \end{cases}$$

Then generate X_3 from the cdf $F_{X_3|X_1, X_2}(x_3|x_1, x_2)$, etc.

The inverse transform method requires tractable expressions for the cdf F and conditional cdf's. Rejection method requires only the joint density; however, it needs more random

number generations due to rejections. This is not a significant disadvantage for relatively small Monte Carlo studies or relatively tight bounding boxes.

As we see, implementation of Monte Carlo methods reduces to generation of random variables from given distributions. Hence, it remains to design algorithms for generating random variables and vectors that will have the desired distributions.

Statistics software packages like SAS, R, Splus, SPSS, Minitab, and others have built-in procedures for the generation of random variables from the most common discrete and continuous distributions. Similar tools are found in recent versions of **MATLAB**, Microsoft Excel, and some libraries.

Majority of computer languages have a *random number generator* that returns only *Uniformly distributed* independent random variables. This section discusses general methods of transforming Uniform random variables, obtained by a standard random number generator, into variables with desired distributions.

9.6 Gibbs Sampler

Consider a problem with several parameters of interest, $\theta_1, \dots, \theta_d$. Each of them has a prior distribution, and the entire set $\boldsymbol{\theta} = (\theta_1, \dots, \theta_d)$ has a joint prior $\pi(\boldsymbol{\theta})$. These parameters have different meanings - they represent means, variances, shapes, slopes, etc., and so, their joint posterior distribution is intractable. Its exact computation requires multiple integration, and the resulting multivariate density does not have any analytic form.

Example 9.10 (NORMAL DISTRIBUTION WITH UNKNOWN MEAN AND VARIANCE). As one of the simplest examples illustrating this situation, consider the model

$$X_1, \dots, X_n \sim \text{Normal}(\theta, \sigma^2)$$

with both θ and σ^2 being unknown parameters.

In the very best case, suppose that a priori, θ and σ^2 are distributed independently of each other, and according to their corresponding conjugate priors,

$$\begin{aligned}\theta &\sim \text{Normal}(\mu, \tau^2) \\ \sigma^2 &\sim \text{Inverse Gamma}(\alpha, \beta).\end{aligned}$$

Alternatively (and it is not harder at all), we can assume the improper joint prior

$$\pi(\theta, \sigma^2) = \frac{1}{\sigma^2}, \quad \theta \in \mathbf{R}, \sigma > 0,$$

which is $1 \cdot \frac{1}{\sigma^2}$, meaning that θ and σ are still independent of each other, and each of them has its natural non-informative distribution.

If we fix one parameter and vary the other, we have our old known problem.

Let us fix σ^2 . Then the posterior distribution of θ , given data $\mathbf{X}$, is Normal with posterior parameters

$$\mu_x = \frac{n\bar{X}/\sigma^2 + \mu/\tau^2}{n/\sigma^2 + 1/\tau^2} \text{ and } \tau_x^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}$$

or $\text{Normal}(\bar{X}, \sigma^2/n)$ in case of the improper prior. But this is a *posterior distribution conditioned on σ^2* .

Conversely, if we fix θ , then the posterior distribution of σ^2 is Inverse Gamma with posterior parameters

$$\alpha_x = \frac{n}{2} - 1 \text{ and } \beta_x = \frac{2}{\sum (X_i - \theta)^2}.$$

This is conditioned on θ , which is an unknown quantity. For example, we can use this posterior density to calculate

$$\mathbf{P}\{\sigma^2 > 5 \mid \mathbf{X}, \theta\},$$

but unconditionally from θ , so far we cannot compute $\mathbf{P}\{\sigma^2 > 5 \mid \mathbf{X}\}$, needed for testing a hypothesis of $\sigma^2 > 5$. We cannot even compute the posterior mean or median of σ^2 directly.

For the posterior analysis (which is Bayesian analysis!) of θ and σ^2 , we need their marginal and/or joint posterior distributions that is independent of any unknown quantities. However, even according to the introduced *rather simple* two-parameter model, the join posterior distribution

$$\pi(\theta, \sigma^2 \mid \mathbf{X})$$

is intractable. ◇

Gibbs sampling will solve the problem in this Example. It cannot derive the closed form of the posterior density, but it is able to produce any number of pairs (θ, σ^2) coming from that very joint posterior distribution.

Gibbs sampler is a type of MCMC, a computer-generated Markov chain, where transitions occur according to the conditional distributions of one parameter, given all the other quantities in the problem.

Step 0, initialization. Start with some possible parameter values $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, \dots, \theta_d^{(0)})$.

Step 1, a cycle of first iterations.

(1.1) Generate $\theta_1^{(1)}$ from the conditional posterior distribution $\pi(\theta_1 \mid \mathbf{X}, \theta_2^{(0)}, \dots, \theta_d^{(0)})$.

(1.2) Generate $\theta_2^{(1)}$ from the conditional posterior distribution $\pi(\theta_2 \mid \mathbf{X}, \theta_1^{(1)}, \theta_3^{(0)}, \dots, \theta_d^{(0)})$.

(1.2) Generate $\theta_3^{(1)}$ from the conditional posterior distribution $\pi(\theta_3 | \mathbf{X}, \theta_1^{(1)}, \theta_2^{(1)}, \theta_4^{(0)}, \dots, \theta_d^{(0)})$.

... ..

.

(1.d) Generate $\theta_d^{(1)}$ from the conditional posterior distribution $\pi(\theta_d | \mathbf{X}, \theta_1^{(1)}, \dots, \theta_{d-1}^{(1)})$.

This concludes the 1st round of iterations. As a result, we have a set of parameters $\boldsymbol{\theta}^{(1)} = (\theta_2^{(1)}, \dots, \theta_d^{(1)})$ generated from $\boldsymbol{\theta}^{(0)}$.

Step 2, the 2nd cycle of iterations.

(2.1) Generate $\theta_1^{(2)}$ from the conditional posterior distribution $\pi(\theta_1 | \mathbf{X}, \theta_2^{(1)}, \dots, \theta_d^{(1)})$.

... ..

.

During each simulation, the most recent generated parameter values are being used.

Iterations continue until MCMC “converges”, i.e., it *stabilizes*. It does not converge to a constant. During each step, the new parameters are still generated according to the conditional distributions, but the overall joint distribution of $\boldsymbol{\theta}^{(N)}$ must become stable after a sufficient number of steps.

When ranges of parameters are independent of each other, the MCMC will converge to the correct joint posterior distribution.

Example 9.11 (TWO-PARAMETER NORMAL MODEL, CONTINUED). The following WinBUGS code can be used for the problem in the previous example. Gibbs sampler converges to the joint posterior distribution of θ and σ^2 .

Notice that the 2nd parameter of the Normal distribution in WinBUGS is “precision”, which is $1/\sigma$. Accordingly, the Gamma prior distribution is used for it instead of Inverse Gamma.

```
model{
  for (i in 1:n) { X[i] ~ dnorm(theta,sigma)}
  theta ~ dnorm(mu,tau)
  sigma ~ dgamma(alpha,beta)
}

list( n=3, X = c(-1.4,0.3,1.9), mu=0, tau=10, alpha=5, beta=0.8 )
```

After convergence, resulting posterior parameter estimates are in the table,

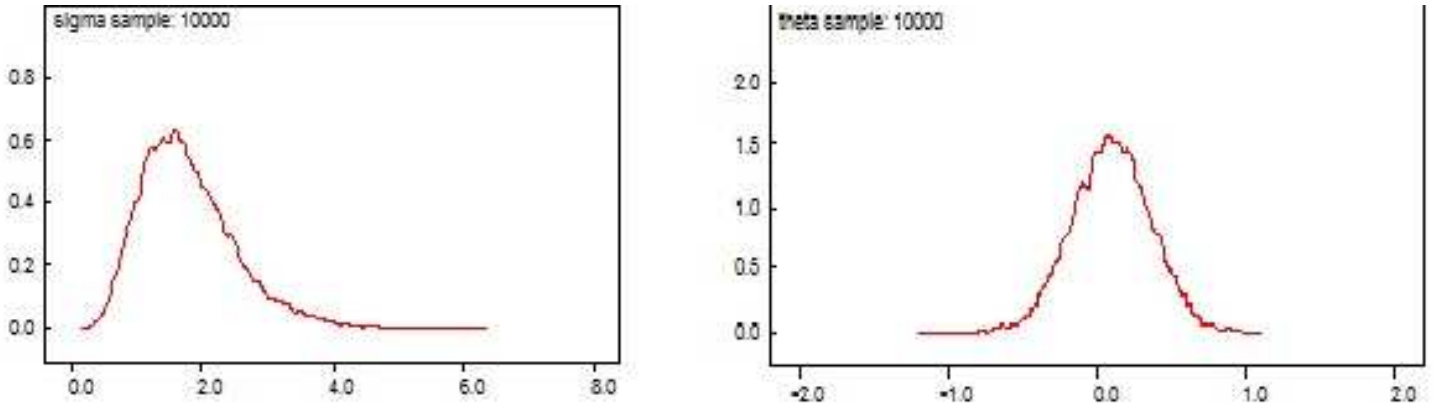


FIGURE 9.5: Estimated posterior densities $\pi(\theta|\mathbf{X})$ and $\pi(\sigma|\mathbf{X})$ for the two-parameter Normal distribution problem.

node	mean	sd	MC error	2.5%	median	97.5%	start	sample
sigma	1.768	0.7007	0.007619	0.6789	1.672	3.434	1	10000
theta	0.08807	0.2592	0.002648	-0.4173	0.08963	0.588	1	10000

The marginal posterior densities (unconditional, independent of the other parameter!) are on Fig. 9.5. ◇

9.7 The Metropolis Algorithm

This algorithm can effectively be used to simulate samples of θ from its posterior distribution $\pi(\theta | \mathbf{X})$ whose closed-form expression is difficult to derive.

The algorithm only requires a function $g(\theta)$ that is proportional to the posterior density, $g(\theta) \sim \pi(\theta | \mathbf{X})$. Also, we need to choose a [proposal distribution](#) $P(\theta | \eta)$, a symmetric density that is easy to simulate from. By symmetry we mean $P(\theta | \eta) = P(\eta_2 | \theta)$. A popular choice is $P(\theta | \eta) \sim \text{Normal}(\eta, \sigma^2)$.

1. Choose a starting point θ_0 , which may be fixed or randomly drawn from a distribution.
2. Generate a candidate θ_1 from the distribution $P(\theta_1 | \theta_0)$.
3. For this generated number, compute the [acceptance ratio](#)

$$r(\theta_1) = \frac{g(\theta_1)}{g(\theta_0)} = \frac{\pi(\theta_1 | \mathbf{X})}{\pi(\theta_0 | \mathbf{X})}$$

(recall that $g(\theta)$ is proportional to $\pi(\theta | \mathbf{X})$).

4. Generate a standard Uniform variable $U \sim \text{Uniform}(0,1)$.

If $U \leq r(\theta_1)$ then accept the candidate and move to generating θ_2
If $U > r(\theta_1)$ then reject the candidate, let $\theta_1 = \theta_0$, and move to generating θ_2
5. Repeat steps 2-4 a large number of times. Every time, generate a candidate θ_{t+1} from the distribution $P(\theta_{t+1} | \theta_t)$, calculate the acceptance ratio r , generate a Uniform variable, compare it with r , and decide on accepting or rejecting the candidate.
6. Remove the first batch of iterations (100 to 1000 values, the “burn-in” sequence), to avoid too much dependence on the initial value.

After the burn-in period, the algorithm will generate random variables θ_t from approximately the posterior distribution $\pi(\theta | \mathbf{X})$.

The Metropolis algorithm can be used for any distribution, and it is particularly handy for simulating θ from its posterior. In this case, we do have a function $g(\theta)$ that is proportional to $\pi(\theta | \mathbf{X})$,

$$g(\theta) = f(\mathbf{X} | \theta)\pi(\theta) \sim \pi(\theta | \mathbf{X}).$$

Often it is difficult to calculate the normalizing constant, the one that we have been dropping all the time, especially in the case of multivariate parameters. This constant, the marginal distribution of $\mathbf{X}$, requires integration over θ , which may not even be feasible in non-trivial cases, when the prior $\pi(\theta)$ is not conjugate to the model $f(\mathbf{X} | \theta)$.

9.8 The Metropolis-Hastings Algorithm

The Metropolis-Hastings Algorithm is a generalization of the Metropolis Algorithm, in which the proposal distribution does not have to be symmetric. To correct for the asymmetry in the proposal distribution, the acceptance ratio is defined as

$$r(\theta_{t+1}) = \frac{g(\theta_{t+1})/P(\theta_{t+1} | \theta_t)}{g(\theta_t)/P(\theta_t | \theta_{t+1})}$$

Chapter 10

Review of Markov Chains

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WinBUGS general methodology for generating samples is called **Markov chain Monte Carlo**. Based on the distributions that we introduce in the “model”, WinBUGS constructs a Markov chain of simulations that converges to a distribution of interest.

What is a **Markov chain** and how does it converge? A review of the main concepts is in this chapter.

10.1 Main definitions

DEFINITION 10.1

A **stochastic process** is a random variable that also depends on time. It is therefore a function of two arguments, $X(t, \omega)$, where:

- $t \in \mathcal{T}$ is time, with $\mathcal{T}$ being a set of possible times, usually $[0, \infty)$, $(-\infty, \infty)$, $\{0, 1, 2, \dots\}$, or $\{\dots, -2, -1, 0, 1, 2, \dots\}$;
- $\omega \in \Omega$, as before, is an outcome of an experiment, with Ω being the whole sample space.

Values of $X(t, \omega)$ are called *states*.

DEFINITION 10.2

Stochastic process $X(t)$ is **Markov** if for any $t_1 < \dots < t_n < t$ and any sets $A; A_1, \dots, A_n$

$$\begin{aligned} P\{X(t) \in A \mid X(t_1) \in A_1, \dots, X(t_n) \in A_n\} \\ = P\{X(t) \in A \mid X(t_n) \in A_n\}. \end{aligned} \quad (10.1)$$

Let us look at the equation (10.1). It means that the conditional distribution of $X(t)$ is the same under two different conditions,

- (1) given observations of the process X at several moments in the past;
- (2) given only *the latest* observation of X .

If a process is Markov, then its future behavior is the same under conditions (1) and (2). In other words, knowing the present, we get no information from the past that can be used to predict the future,

$$P \{ \text{future} \mid \text{past, present} \} = P \{ \text{future} \mid \text{present} \}$$

Then, for the future development of a Markov process, only its present state is important, and it does not matter *how* the process arrived to this state.

Some processes satisfy the Markov property, and some don't.

Example 10.1 (INTERNET CONNECTIONS). Let $X(t)$ be the total number of internet connections registered by some internet service provider by the time t . Typically, people connect to the internet at random times, regardless of how many connections have already been made. Therefore, the number of connections in a minute will only depend on the current number. For example, if 999 connections have been registered by 10 o'clock, then their total number will exceed 1000 during the next minute regardless of *when* and *how* these 999 connections were made in the past. This process is *Markov*. $\diamond$

Example 10.2 (STOCK PRICES). Let $Y(t)$ be the value of some stock or some market index at time t . If we know $Y(t)$, do we also want to know $Y(t-1)$ in order to predict $Y(t+1)$? One may argue that if $Y(t-1) < Y(t)$, then the market is rising, therefore, $Y(t+1)$ is likely (but not certain) to exceed $Y(t)$. On the other hand, if $Y(t-1) > Y(t)$, we may conclude that the market is falling and may expect $Y(t+1) < Y(t)$. It looks like knowing the past *in addition* to the present did help us to predict the future. Then, this process is *not Markov*. $\diamond$

Due to a well-developed theory and a number of simple techniques available for Markov processes, it is important to know whether the process is Markov or not. The idea of Markov dependence was proposed and developed by *Andrei Markov* (1856–1922) who was a student of P. Chebyshev at St. Petersburg University in Russia.

10.2 Markov chains

DEFINITION 10.3

A **Markov chain** is a discrete-time, discrete-state Markov stochastic process.

Introduce a few convenient simplifications. The time is discrete, so let us define the time set as $\mathcal{T} = \{0, 1, 2, \dots\}$. We can then look at a Markov chain as a random sequence

$$\{X(0), X(1), X(2), \dots\}.$$

The state set is also discrete, so let us enumerate the states as $1, 2, \dots, n$. Sometimes we'll start enumeration from state 0, and sometimes we'll deal with a Markov chain with infinitely many (discrete) states, then we'll have $n = \infty$.

The *Markov* property means that only the value of $X(t)$ matters for predicting $X(t+1)$, so the conditional probability

$$\begin{aligned} p_{ij}(t) &= \mathbf{P}\{X(t+1) = j \mid X(t) = i\} \\ &= \mathbf{P}\{X(t+1) = j \mid X(t) = i, X(t-1) = h, X(t-2) = g, \dots\} \end{aligned} \quad (10.2)$$

depends on i, j , and t only and equals the probability for the Markov chain X to make a *transition* from state i to state j at time t .

DEFINITION 10.4

Probability $p_{ij}(t)$ in (10.2) is called a **transition probability**. Probability

$$p_{ij}^{(h)}(t) = \mathbf{P}\{X(t+h) = j \mid X(t) = i\}$$

of moving from state i to state j by means of h transitions is an **h -step transition probability**.

DEFINITION 10.5

A Markov chain is **homogeneous** if all its transition probabilities are independent of t . Being homogeneous means that transition from i to j has the same probability at any time. Then $p_{ij}(t) = p_{ij}$ and $p_{ij}^{(h)}(t) = p_{ij}^{(h)}$.

Characteristics of a Markov chain

What do we need to know to describe a Markov chain?

By the Markov property, each next state should be predicted from the previous state only. Therefore, it is sufficient to know the distribution of its initial state $X(0)$ and the mechanism of transitions from one state to another.

The distribution of a Markov chain is completely determined by the initial distribution P_0 and one-step transition probabilities p_{ij} . Here P_0 is the probability mass function of X_0 ,

$$P_0(x) = \mathbf{P}\{X(0) = x\} \text{ for } x \in \{1, 2, \dots, n\}$$

Based on this data, we would like to compute:

- h -step transition probabilities $p_{ij}^{(h)}$;
- P_h , the distribution of states at time h , which is our forecast for $X(h)$;
- the limit of $p_{ij}^{(h)}$ and P_h as $h \rightarrow \infty$, which is our long-term forecast.

Indeed, when making forecasts for many transitions ahead, computations will become rather lengthy, and thus, it will be more efficient to take the limit.

<u>NOTATION</u>	p_{ij}	=	$P\{X(t+1) = j \mid X(t) = i\},$ transition probability
	$p_{ij}^{(h)}$	=	$P\{X(t+h) = j \mid X(t) = i\},$ h -step transition probability
	$P_t(x)$	=	$P\{X(t) = x\},$ distribution of $X(t),$ distribution of states at time t
	$P_0(x)$	=	$P\{X(0) = x\},$ initial distribution

Example 10.3 (WEATHER FORECASTS). In some town, each day is either sunny or rainy. A sunny day is followed by another sunny day with probability 0.7, whereas a rainy day is followed by a sunny day with probability 0.4.

It rains on Monday. Make forecasts for Tuesday, Wednesday, and Thursday.

Solution. Weather conditions in this problem represent a homogeneous Markov chain with 2 states: state 1 = “sunny” and state 2 = “rainy.” Transition probabilities are:

$$p_{11} = 0.7, \quad p_{12} = 0.3, \quad p_{21} = 0.4, \quad p_{22} = 0.6,$$

where p_{12} and p_{22} were computed by the complement rule.

If it rains on Monday, then Tuesday is sunny with probability $p_{21} = 0.4$ (making a transition from a rainy to a sunny day), and Tuesday is rainy with probability $p_{22} = 0.6$. We can predict a 60% chance of rain.

Wednesday forecast requires 2-step transition probabilities, making one transition from Monday to Tuesday, $X(0)$ to $X(1)$, and another one from Tuesday to Wednesday, $X(1)$ to $X(2)$. We'll have to condition on the weather situation on Tuesday and use the Law of Total Probability from p. 17,

$$\begin{aligned}
 p_{21}^{(2)} &= \mathbf{P}\{\text{Wednesday is sunny} \mid \text{Monday is rainy}\} \\
 &= \sum_{i=1}^2 \mathbf{P}\{X(1) = i \mid X(0) = 2\} \mathbf{P}\{X(2) = 1 \mid X(1) = i\} \\
 &= \mathbf{P}\{X(1) = 1 \mid X(0) = 2\} \mathbf{P}\{X(2) = 1 \mid X(1) = 1\} \\
 &\quad + \mathbf{P}\{X(1) = 2 \mid X(0) = 2\} \mathbf{P}\{X(2) = 1 \mid X(1) = 2\} \\
 &= p_{21}p_{11} + p_{22}p_{21} = (0.4)(0.7) + (0.6)(0.4) = 0.52.
 \end{aligned}$$

By the Complement Rule, $p_{22}^{(2)} = 0.48$, and thus, we predict a 52% chance of sun and a 48% chance of rain on Wednesday.

For the Thursday forecast, we need to compute 3-step transition probabilities $p_{ij}^{(3)}$ because it takes 3 transitions to move from Monday to Thursday. We have to use the Law of Total Probability conditioning on *both* Tuesday and Wednesday. For example, going from rainy Monday to sunny Thursday means going from rainy Monday to either rainy or sunny Tuesday, then to either rainy or sunny Wednesday, and finally, to sunny Thursday,

$$p_{21}^{(3)} = \sum_{i=1}^2 \sum_{j=1}^2 p_{2i}p_{ij}p_{j1}.$$

This corresponds to a sequence of states $2 \rightarrow i \rightarrow j \rightarrow 1$. However, we have already computed 2-step transition probabilities $p_{21}^{(2)}$ and $p_{22}^{(2)}$, describing transition from Monday to Wednesday. It remains to add one transition to Thursday, hence,

$$p_{21}^{(3)} = p_{21}^{(2)}p_{11} + p_{22}^{(2)}p_{21} = (0.52)(0.7) + (0.48)(0.4) = 0.556.$$

So, we predict a 55.6% chance of sun on Thursday and a 44.4% chance of rain.

◇

The following *transition diagram* (Figure 10.1) reflects the behavior of this Markov chain. Arrows represent all possible one-step transitions, along with the corresponding probabilities. Check this diagram against the transition probabilities stated in Example 10.3. To obtain, say, a 3-step transition probability $p_{21}^{(3)}$, find all 3-arrow paths from state 2 “rainy” to state 1 “sunny.” Multiply probabilities along each path and add over all 3-step paths.

Example 10.4 (WEATHER, CONTINUED). Suppose now that it does not rain yet, but meteorologists predict an 80% chance of rain on Monday. How does this affect our forecasts?

In Example 10.3, we have computed forecasts under the condition of rain on Monday. Now,

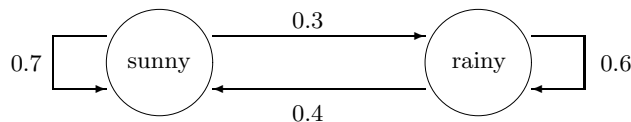


FIGURE 10.1: Transition diagram for the Markov chain in Example 10.3.

a sunny Monday (state 1) is also possible. Therefore, in addition to probabilities $p_{2j}^{(h)}$ we also need to compute $p_{1j}^{(h)}$ (say, using the transition diagram, see Figure 10.1),

$$\begin{aligned} p_{11}^{(2)} &= (0.7)(0.7) + (0.3)(0.4) = 0.61, \\ p_{11}^{(3)} &= (0.7)^3 + (0.7)(0.3)(0.4) + (0.3)(0.4)(0.7) + (0.3)(0.6)(0.4) = 0.583. \end{aligned}$$

The initial distribution $P_0(x)$ is given as

$$P_0(1) = \mathbf{P} \{ \text{sunny Monday} \} = 0.2, \quad P_0(2) = \mathbf{P} \{ \text{rainy Monday} \} = 0.8.$$

Then, for each forecast, we use the Law of Total Probability, conditioning on the weather on Monday,

$$\begin{aligned} P_1(1) &= P \{ X(1) = 1 \} = P_0(1)p_{11} + P_0(2)p_{21} = 0.46 && \text{for Tuesday} \\ P_2(1) &= P \{ X(2) = 1 \} = P_0(1)p_{11}^{(2)} + P_0(2)p_{21}^{(2)} = 0.538 && \text{for Wednesday} \\ P_3(1) &= P \{ X(3) = 1 \} = P_0(1)p_{11}^{(3)} + P_0(2)p_{21}^{(3)} = 0.5614 && \text{for Thursday} \end{aligned}$$

These are probabilities of a sunny day (state 1), respectively, on Tuesday, Wednesday, and Thursday. Then, the chance of rain (state 2) on these days is $P_1(2) = 0.54$, $P_2(2) = 0.462$, and $P_3(2) = 0.4386$. $\diamond$

Noticeably, more remote forecasts require more lengthy computations. For a t -day ahead forecast, we have to account for all t -step paths on diagram Figure 10.1. Or, we use the Law of Total Probability, conditioning on *all* the intermediate states $X(1), X(2), \dots, X(t-1)$.

To simplify the task, we shall employ *matrices*.

10.3 Matrix approach

All one-step transition probabilities p_{ij} can be conveniently written in an $n \times n$ *transition probability matrix*

$$P = \begin{array}{c|cccc} & \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix} & \begin{array}{c} \text{From} \\ \text{state:} \\ 1 \\ 2 \\ \vdots \\ n \end{array} \\ \hline \begin{array}{c} \text{To state:} \\ 1 \quad 2 \quad \cdots \quad n \end{array} & & & & \end{array}$$

The entry on the intersection of the i -th row and the j -th column is p_{ij} , the transition probability from state i to state j .

From each state, a Markov chain makes a transition to one and only one state. States-destinations are disjoint and exhaustive events, therefore, *each row total equals 1*,

$$p_{i1} + p_{i2} + \cdots + p_{in} = 1. \quad (10.3)$$

We can also say that probabilities $p_{i1}, p_{i2}, \dots, p_{in}$ form the *conditional distribution* of $X(1)$, given $X(0)$, so they have to add to 1.

In general, this does not hold for column totals. Some states may be “more favorable” than others, then they are visited more often the others, thus their column total will be larger. Matrices with property (10.3) are called *stochastic*.

Similarly, h -step transition probabilities can be written in an *h -step transition probability matrix*

$$P^{(h)} = \begin{pmatrix} p_{11}^{(h)} & p_{12}^{(h)} & \cdots & p_{1n}^{(h)} \\ p_{21}^{(h)} & p_{22}^{(h)} & \cdots & p_{2n}^{(h)} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1}^{(h)} & p_{n2}^{(h)} & \cdots & p_{nn}^{(h)} \end{pmatrix}$$

This matrix is also stochastic because each row represents the conditional distribution of $X(h)$, given $X(0)$ (which is a good way to check our results if we compute $P^{(h)}$ by hand).

Computing h -step transition probabilities

A simple matrix formula connects matrices $P^{(h)}$ and P . Let's start with the 2-step transition probabilities.

By the Law of Total Probability, conditioning and adding over all values of $k = X(1)$,

$$\begin{aligned}
 p_{ij}^{(2)} &= \mathbf{P}\{X(2) = j \mid X(0) = i\} \\
 &= \sum_{k=1}^n \mathbf{P}\{X(1) = k \mid X(0) = i\} \mathbf{P}\{X(2) = j \mid X(1) = k\} \\
 &= \sum_{k=1}^n p_{ik} p_{kj} = (p_{i1}, \dots, p_{in}) \begin{pmatrix} p_{1j} \\ \vdots \\ p_{nj} \end{pmatrix}.
 \end{aligned}$$

Each probability $p_{ij}^{(2)}$ is computed as a sum of $\mathbf{P}\{i \rightarrow k \rightarrow j\}$, over all 2-step paths leading from state i to state j (also, see Example 10.3). As a result, each $p_{ij}^{(2)}$ is a product of the i -th row and the j -th column of matrix P . Hence, the entire 2-step transition probability matrix is

**2-step transition
probability matrix**

$$P^{(2)} = P \cdot P = P^2$$

Further, h -step transition probabilities can be obtained from $(h-1)$ -step transition probabilities by conditioning on $X(h-1) = k$,

$$\begin{aligned}
 p_{ij}^{(h)} &= \mathbf{P}\{X(h) = j \mid X(0) = i\} \\
 &= \sum_{k=1}^n \mathbf{P}\{X(h-1) = k \mid X(0) = i\} \mathbf{P}\{X(h) = j \mid X(h-1) = k\} \\
 &= \sum_{k=1}^n p_{ik}^{(h-1)} p_{kj} = (p_{i1}^{(h-1)}, \dots, p_{in}^{(h-1)}) \begin{pmatrix} p_{1j} \\ \vdots \\ p_{nj} \end{pmatrix}.
 \end{aligned}$$

This time, we consider all h -step paths from i to j . They consist of all $(h-1)$ -step paths from i to some state k , followed by one step from k to j . Now, the result is the i -th row of matrix $P^{(h-1)}$ multiplied by the j -th column of P . Hence, $P^{(h)} = P^{(h-1)} \cdot P$, and we have a general formula

**h -step transition
probability matrix**

$$P^{(h)} = \underbrace{P \cdot P \cdot \dots \cdot P}_{h \text{ times}} = P^h$$

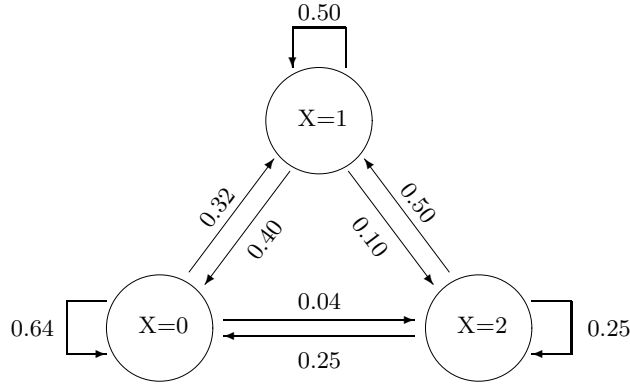


FIGURE 10.2: Transition diagram for the Markov chain in Example 10.5.

Example 10.5 (SHARED DEVICE). A computer is shared by 2 users who send tasks to a computer remotely and work independently. At any minute, any connected user may disconnect with probability 0.5, and any disconnected user may connect with a new task with probability 0.2. Let $X(t)$ be the number of concurrent users at time t (minutes). This is a Markov chain with 3 states: 0, 1, and 2.

Compute transition probabilities. Suppose $X(0) = 0$, i.e., there are no users at time $t = 0$. Then $X(1)$ is the number of new connections within the next minute. It has Binomial(2,0.2) distribution, therefore,

$$p_{00} = (.8)^2 = .64, \quad p_{01} = 2(.2)(.8) = .32, \quad p_{02} = (.2)^2 = .04.$$

Next, suppose $X(0) = 1$, i.e., one user is connected, and the other is not. The number of new connections is Binomial(1,0.2), and the number of disconnections is Binomial(1,0.5). Considering all the possibilities, we obtain (verify)

$$p_{10} = (.8)(.5) = .40, \quad p_{11} = (.2)(.5) + (.8)(.5) = .50, \quad p_{12} = (.2)(.5) = .10.$$

Finally, when $X(0) = 2$, no new users can connect, and the number of disconnections is Binomial(2,0.5), so that

$$p_{20} = .25, \quad p_{21} = .50, \quad p_{22} = .25.$$

We obtain the following transition probability matrix,

$$P = \begin{pmatrix} .64 & .32 & .04 \\ .40 & .50 & .10 \\ .25 & .50 & .25 \end{pmatrix}.$$

The transition diagram corresponding to this matrix is shown in Figure 10.2.

The 2-step transition probability matrix is then computed as

$$P^2 = \begin{pmatrix} .64 & .32 & .04 \\ .40 & .50 & .10 \\ .25 & .50 & .25 \end{pmatrix} \begin{pmatrix} .64 & .32 & .04 \\ .40 & .50 & .10 \\ .25 & .50 & .25 \end{pmatrix} = \begin{pmatrix} .5476 & .3848 & .0676 \\ .4810 & .4280 & .0910 \\ .4225 & .4550 & .1225 \end{pmatrix}.$$

For example, if both users are connected at 10:00, then at 10:02 there will be no users with probability $p_{20}^{(2)} = 0.4225$, one user with probability $p_{21}^{(2)} = 0.4550$, and two users with probability $p_{22}^{(2)} = 0.1225$.

Check that both matrices P and P^2 are stochastic. $\diamond$

Computing the distribution of $X(h)$

The distribution of states after h transitions, or the probability mass function of $X(h)$, can be written in a $1 \times n$ matrix,

$$P_h = (P_h(1), \dots, P_h(n)).$$

By the Law of Total Probability, this time conditioning on $X(0) = k$, we compute

$$\begin{aligned} P_h(j) &= \mathbf{P}\{X(h) = j\} = \sum_k \mathbf{P}\{X(0) = k\} \mathbf{P}\{X(h) = j \mid X(0) = k\} \\ &= \sum_k P_0(k) p_{kj}^{(h)} = \begin{pmatrix} P_0(1) & \cdots & P_0(n) \end{pmatrix} \begin{pmatrix} p_{1j}^{(h)} \\ \vdots \\ p_{1n}^{(h)} \end{pmatrix}. \end{aligned}$$

Each probability $P_h(j)$ is obtained when the entire row P_0 (initial distribution of X) is multiplied by the entire j -th column of matrix P . Hence,

$$\begin{array}{c} \text{Distribution} \\ \text{of } X(h) \end{array} \quad \boxed{P_h = P_0 P^h} \quad (10.4)$$

Example 10.6 (SHARED DEVICE, CONTINUED). If we know that there are 2 users connected at 10:00, we can write the initial distribution as

$$P_0 = (0, 0, 1).$$

Then the distribution of the number of users at 10:02, after $h = 2$ transitions, is computed as

$$P_2 = P_0 P^2 = (0, 0, 1) \begin{pmatrix} .5476 & .3848 & .0676 \\ .4810 & .4280 & .0910 \\ .4225 & .4550 & .1225 \end{pmatrix} = (.4225, .4550, .1225),$$

just as we concluded in the end of Example 10.5.

Suppose that all states are equally likely at 10:00. How can we compute the probability of 1 connected user at 10:02? Here, the initial distribution is

$$P_0 = (1/3, 1/3, 1/3).$$

The distribution of $X(2)$ is

$$P_2 = P_0 P^2 = (1/3, 1/3, 1/3) \begin{pmatrix} .5476 & .3848 & .0676 \\ .4810 & .4280 & .0910 \\ .4225 & .4550 & .1225 \end{pmatrix} = (\dots, .4226, \dots).$$

Thus, $\mathbf{P}\{X(2) = 1\} = 0.4226$. We only computed $P_2(1)$. The question did not require computation of the entire distribution P_2 . $\diamond$

In practice, knowing the distribution of the number of users at 10:00, one would rarely be interested in the distribution at 10:02. How should one figure the distribution at 11:00, the next day, or the next month?

In other words, how does one compute the distribution of $X(h)$ for large h ? A direct solution is $P_h = P_0 \cdot P^h$, which can be computed for moderate n and h . For example, the distribution of the number of users at 11:00, $h = 60$ transitions after 10:00, can be computed in **MATLAB** as

```
P = [ .64 .32 .04
      .40 .50 .10
      .25 .50 .25 ];
P0 = [ 1/3 1/3 1/3 ];
h = 60;
P0*P^h
```

For large h , one would really like to take a limit of P_h as $h \rightarrow \infty$ instead of a tedious computation of $P_0 \cdot P^h$. We learn how to do it in the next section.

10.4 Steady-state distribution

This section discusses the distribution of states of a Markov chain after a large number of transitions.

DEFINITION 10.6

A collection of limiting probabilities

$$\pi_x = \lim_{h \rightarrow \infty} P_h(x)$$

is called a **steady-state distribution** of a Markov chain $X(t)$.

When this limit exists, it can be used as a forecast of the distribution of X after *many* transitions. A fast system (say, a central processing unit with a clock rate of several gigahertz), will go through a very large number of transitions very quickly. Then, its distribution of states at virtually any time is the steady-state distribution.

Computing the steady-state distribution

When a steady-state distribution π exists, it can be computed as follows. We notice that π is a limit of not only P_h but also P_{h+1} . The latter two are related by the formula

$$P_h P = P_0 P^h P = P_0 P^{h+1} = P_{h+1}.$$

Taking the limit of P_h and P_{h+1} , as $h \rightarrow \infty$, we obtain

$$\pi P = \pi. \quad (10.5)$$

Then, solving (10.5) for π , we get the steady-state distribution of a Markov chain with a transition probability matrix P .

The system of steady-state equations (10.5) consists of n equations with n unknowns which are probabilities of n states. However, this system is *singular*; it has infinitely many solutions. That is because both sides of the system (10.5) can be multiplied by any constant C , and thus, any multiple of π is also a solution of (10.5).

There is yet one more condition that we have not used. Being a distribution, all the probabilities π_x must add to 1,

$$\pi_1 + \dots + \pi_n = 1.$$

Only one solution of (10.5) can satisfy this condition, and this is the **steady-state distribution**.

**Steady-state
distribution**

$\pi = \lim_{h \rightarrow \infty} P_h$
is computed as a solution of

$$\begin{cases} \pi P &= \pi \\ \sum_x \pi_x &= 1 \end{cases}$$

Example 10.7 (WEATHER, CONTINUED). In Example 10.3 on p. 134, the transition probability matrix of sunny and rainy days is

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}.$$

The steady-state equation for this Markov chain is

$$\pi P = \pi,$$

or

$$(\pi_1, \pi_2) = (\pi_1, \pi_2) \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} = (0.7\pi_1 + 0.4\pi_2, 0.3\pi_1 + 0.6\pi_2).$$

We obtain a system of equations,

$$\begin{cases} 0.7\pi_1 + 0.4\pi_2 = \pi_1 \\ 0.3\pi_1 + 0.6\pi_2 = \pi_2 \end{cases} \Leftrightarrow \begin{cases} 0.4\pi_2 = 0.3\pi_1 \\ 0.3\pi_1 = 0.4\pi_2 \end{cases} \Leftrightarrow \pi_2 = \frac{3}{4}\pi_1.$$

We see that two equations in our system reduced to one. This will always happen: *one equation will follow from the others*, and this is because the system $\pi P = \pi$ is singular. It remains to use the *normalizing equation* $\sum \pi_x = 1$,

$$\pi_1 + \pi_2 = \pi_1 + \frac{3}{4}\pi_1 = \frac{7}{4}\pi_1 = 1,$$

from where

$$\underline{\pi_1 = 4/7} \quad \text{and} \quad \underline{\pi_2 = 3/7}.$$

In a long history of this city, $4/7 \approx 57\%$ of days are sunny, and $3/7 \approx 43\%$ are rainy. $\diamond$

Example 10.8 (SHARED DEVICE, CONTINUED). In Example 10.5 on p. 139, the transition probability matrix for the number of concurrent users is

$$P = \begin{pmatrix} .64 & .32 & .04 \\ .40 & .50 & .10 \\ .25 & .50 & .25 \end{pmatrix}.$$

Let us find the steady-state distribution. It will automatically serve as our forecast for the number of users next day or next month. Thus, it will answer the question posed in the end of Section 10.3.

The steady-state equations are:

$$\begin{cases} .64\pi_0 + .40\pi_1 + .25\pi_2 = \pi_0 \\ .32\pi_0 + .50\pi_1 + .50\pi_2 = \pi_1 \\ .04\pi_0 + .10\pi_1 + .25\pi_2 = \pi_2 \end{cases} ; \quad \begin{cases} -.36\pi_0 + .40\pi_1 + .25\pi_2 = 0 \\ .32\pi_0 - .50\pi_1 + .50\pi_2 = 0 \\ .04\pi_0 + .10\pi_1 - .75\pi_2 = 0 \end{cases}$$

Solving this system by method of elimination, we express π_2 from the first equation and substitute into the other equations,

$$\begin{cases} \pi_2 = 1.44\pi_0 - 1.6\pi_1 \\ .32\pi_0 - .50\pi_1 + .50(1.44\pi_0 - 1.6\pi_1) = 0 \\ .04\pi_0 + .10\pi_1 - .75(1.44\pi_0 - 1.6\pi_1) = 0 \end{cases} ; \quad \begin{cases} \pi_2 = 1.44\pi_0 - 1.6\pi_1 \\ 1.04\pi_0 - 1.3\pi_1 = 0 \\ -1.04\pi_0 + 1.3\pi_1 = 0 \end{cases}$$

The last two equations are equivalent. This was anticipated; one equation should always follow from the others, so we are “probably” on a right track.

Express π_1 from the second equation and substitute into the first one,

$$\begin{cases} \pi_2 &= 1.44\pi_0 - 1.6(.8\pi_0) \\ \pi_1 &= .8\pi_0 \end{cases} ; \quad \begin{cases} \pi_2 &= .16\pi_0 \\ \pi_1 &= .8\pi_0 \end{cases}$$

Finally, use the normalizing equation,

$$\pi_0 + \pi_1 + \pi_2 = \pi_0 + .8\pi_0 + .16\pi_0 = 1.96\pi_0 = 1,$$

from where we compute the answer,

$$\begin{cases} \pi_0 &= 1/1.96 &= .5102 \\ \pi_1 &= .8(.5102) &= .4082 \\ \pi_2 &= .16(.5102) &= .0816 \end{cases}$$

◇

The limit of P^h

Then, what will the h -step transition probabilities be for large h ? It turns out that matrix $P^{(h)}$ also has a limit, as $h \rightarrow \infty$, and the limiting matrix has the form

$$\Pi = \lim_{h \rightarrow \infty} P^{(h)} = \begin{pmatrix} \pi_1 & \pi_2 & \cdots & \pi_n \\ \pi_1 & \pi_2 & \cdots & \pi_n \\ \vdots & \vdots & \cdots & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_n \end{pmatrix}.$$

All the rows of the limiting matrix Π are equal, and they consist of the steady-state probabilities π_x !

How can this phenomenon be explained? First, the forecast for some very remote future should not depend on the current state $X(0)$. Say, our weather forecast for the next century should not depend on the weather we have today. Therefore, $p_{ik} = p_{jk}$ for all i, j, k , and this is why all rows of Π coincide.

Second, our forecast, independent of $X(0)$, will just be given in terms of long-term proportions, that is, π_x . Indeed, if your next-year vacation is in August, then the best weather forecast you can get will likely be given as the historical average for this time of the year.

Steady state

What is the steady state of a Markov chain? Suppose the system has reached its steady state, so that the current distribution of states is $P_t = \pi$. A system makes one more transition, and the distribution becomes $P_{t+1} = \pi P$. But $\pi P = \pi$, and thus, $P_t = P_{t+1}$. We see that *in a steady state, transitions do not affect the distribution*. A system may go from one state to another, but the distribution of states does not change. In this sense, it is *steady*.

Existence of a steady-state distribution. Regular Markov chains

As we know from Calculus, there are situations when a limit simply does not exist. Similarly, there are Markov chains with no steady-state distribution.

Example 10.9 (PERIODIC MARKOV CHAIN, NO STEADY-STATE DISTRIBUTION). In chess, a knight can only move to a field of different color, from white to black, and from black to white. Then, the transition probability matrix of the color of its field is

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Computing P^2 , P^3 , etc., we find that

$$P^{(h)} = P^h = \begin{cases} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{for all odd } h \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \text{for all even } h \end{cases}$$

Indeed, after any odd number of moves, the knight's field will change its color, and after any even number of moves, it will return to the initial color. Thus, there is no limit of P_h and $P^{(h)}$. $\diamond$

The Markov chain in Example 10.9 is *periodic* with period 2 because $X(t) = X(t+2)$ for all t with probability 1. Periodic Markov chains cannot be regular; from any state, some h -step transitions are possible and some are not, depending on whether or not h is divisible by the period.

There are other situations when steady-state probabilities cannot be found. What we need is a criterion for the existence of a steady-state distribution.

DEFINITION 10.7

A Markov chain is **regular** if

$$p_{ij}^{(h)} > 0$$

for some h and all i, j . That is, for some h , matrix $P^{(h)}$ has only non-zero entries, and h -step transitions from any state to any state are possible.

Any regular Markov chain has a steady-state distribution.

Example 10.10. Markov chains in Examples 10.3 and 10.5 are *regular* because all transitions are possible for $h = 1$ already, and matrix P does not contain any zeros. $\diamond$

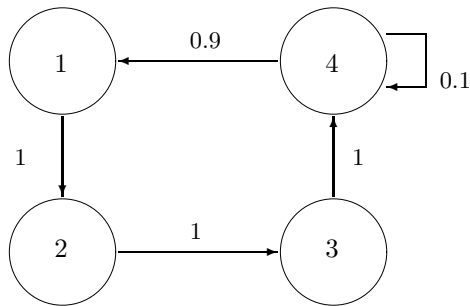


FIGURE 10.3: Transition diagram for a regular Markov chain in Example 10.11.

Example 10.11. A Markov chain with transition probability matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0.9 & 0 & 0 & 0.1 \end{pmatrix}.$$

is also regular. Matrix P contains zeros, and so do P^2 , P^3 , P^4 , and P^5 . The 6-step transition probability matrix

$$P^{(6)} = \begin{pmatrix} .009 & .090 & .900 & .001 \\ .001 & .009 & .090 & .900 \\ .810 & .001 & .009 & .180 \\ .162 & .810 & .001 & .027 \end{pmatrix}$$

contains no zeros and proves regularity of this Markov chain.

In fact, computation of all P^h up to $h = 6$ is not required in this problem. Regularity can also be seen from the transition diagram in Figure 10.3. Based on this diagram, any state j can be reached in 6 steps from any state i . Indeed, moving counterclockwise through this Figure, we can reach state 4 from any state i in ≤ 3 steps. Then, we can reach state j from state 4 in ≤ 3 additional steps, for the total of ≤ 6 steps. If we can reach state i from state j in fewer than 6 steps, we just use the remaining steps circling around state 4. For example, state 2 is reached from state 1 in 6 steps as follows:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 4 \rightarrow 1 \rightarrow 2.$$

Notice that we don't have to compute $p_{ij}^{(h)}$. We only need to verify that they are all positive for some h . $\diamond$

Example 10.12 (IRREGULAR MARKOV CHAIN, ABSORBING STATES). When there is a state i with $p_{ii} = 1$, a Markov chain cannot be regular. There is no exit from state i ; therefore, $p_{ij}^{(h)} = 0$ for all h and all $j \neq i$. Such a state is called *absorbing*. For example, state 4 in Figure 10.4a is absorbing, therefore, the Markov chain is irregular.

There may be several absorbing states or an entire absorbing zone, from which the remaining

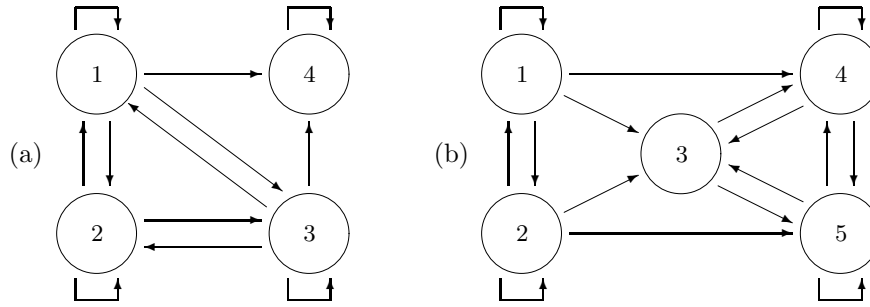


FIGURE 10.4: Absorbing states and absorbing zones (Example 10.12).

states can never be reached. For example, states 3, 4, and 5 in Figure 10.4b form an *absorbing zone*, some kind of a Bermuda triangle. When this process finds itself in the set $\{3, 4, 5\}$, there is no route from there to the set $\{1, 2\}$. As a result, probabilities $p_{31}^{(h)}$, $p_{52}^{(h)}$ and some others equal 0 for all h . Although all p_{ij} may be less than 1, such Markov chains are still irregular.

Notice that both Markov chains do have steady-state distributions. The first process will eventually reach state 4 and will stay there for good. Therefore, the limiting distribution of $X(h)$ is $\pi = \lim P_h = (0, 0, 0, 1)$. The second Markov chain will eventually leave states 1 and 2 for good, thus its limiting (steady-state) distribution has the form $\pi = (0, 0, \pi_3, \pi_4, \pi_5)$. $\diamond$

Conclusion

This section gives us an important method of analyzing rather complicated stochastic systems. Once the Markov property of a process is established, it only remains to find its one-step transition probabilities. Then, the steady-state distribution can be computed, and thus, we obtain the distribution of the process at any time, after a sufficient number of transitions.



Chapter 11

Empirical and Hierarchical Bayes Analysis

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In this chapter, we discuss the usual situations when the prior distribution is not perfectly known. Several approaches are proposed to handle these cases:

- **Empirical Bayes rules** estimate the prior distribution from data. There are
 - **Parametric Empirical Bayes rules** where a family of prior distributions is assumed, and its parameters are estimated from data;
 - **Nonparametric Empirical Bayes rules** where no model is assumed for the prior distribution. Instead, the marginal distribution of data is estimated and used to construct the Bayes rule bypassing the posterior distribution.
- **Hierarchical Bayes rules** where unknown parameters of the prior distribution (called **hyper-parameters**) are assumed to have their own prior distributions (called **hyper-priors**). Then, these unknown hyper-parameters are integrated out.
- **Bayesian robustness** not covered in this course but covered in chap. 4 of Berger, 1985. Its purpose is to find such decision rules that perform uniformly well within a class of prior distributions.

11.1 Parametric Empirical Bayes Approach

Suppose that it is too ambitious or unrealistic to claim that we know the prior distribution $\pi(\theta)$ completely. Instead, assume that the prior belongs to a given family of distributions $\{\pi(\theta \mid \alpha, \beta)\}$ such as Normal or Gamma, probably, the conjugate family, but its parameters α and β are unknown. Parameters of the prior distribution are called **hyper-parameters**. Parametric empirical Bayes rules will then use estimators of these parameters.

A **naïve** parametric empirical Bayes procedure

- integrates θ out to obtain the distribution of data given hyper-parameters but free of the main parameter $\theta - \pi(\mathbf{X} \mid \alpha, \beta)$;
- estimates the hyper-parameters from the distribution $\pi(\mathbf{X} \mid \alpha, \beta)$ using some standard method;
- simply plugs estimates $\hat{\alpha}$ and $\hat{\beta}$ into the prior distribution $\pi(\theta \mid \alpha, \beta)$ and “pretends” that these were the true hyper-parameters.

Example 11.1. Assume the data coming from Normal distribution with unknown mean θ and known variance σ^2 , where θ has a $\text{Normal}(\mu, \tau^2)$ prior distribution, with both μ and τ^2 being unknown. These μ and τ^2 are our hyper-parameters.

Then the distribution of $X_1, \dots, X_n$ given μ and τ^2 is also Normal, with parameters

$$\begin{aligned}\mathbf{E}(X \mid \mu, \tau^2) &= \mu \\ \text{Var}(X \mid \mu, \tau^2) &= \sigma^2 + \tau^2\end{aligned}$$

The standard estimates of $\mathbf{E}(X \mid \mu, \tau^2)$ and $\text{Var}(X \mid \mu, \tau^2)$ are $\bar{X}$ and s_X^2 , the sample mean and sample variance. From this, we can estimate the hyper-parameters as

$$\begin{aligned}\hat{\mu} &= \bar{X} \\ \hat{\tau}^2 &= s_X^2 - \sigma^2\end{aligned}$$

Then, with these estimates, the *estimated* prior distribution $\pi(\theta \mid \hat{\mu}, \hat{\tau}^2)$ is used instead of the unknown $\pi(\theta \mid \mu, \tau^2)$, and we proceed as in the usual Bayesian analysis, computing the posterior and doing further inference. $\diamond$

Some obvious criticism of this naive approach is that we use estimated hyper-parameters as if they were true values. In fact, these estimates obtained from the data will fit the data “too well”.

Often historical data are used to estimate the hyper-parameters.

Also, besides this naive approach, there are other empirical Bayes techniques that compensate for the uncertainty in these estimators.

11.2 Nonparametric Empirical Bayes Approach

As always in Nonparametric Statistics, here we assume no model for the prior distribution. We simply do not pretend to know anything about the prior. However, we choose not to

place a non-informative prior on the parameter θ either, because we believe that θ does not have such a high degree of uncertainty. It has a proper prior distribution which is just unknown to us.

Nonparametric empirical Bayes procedures make use of the marginal distribution of data. This marginal distribution $m(\mathbf{X})$ is estimated from the data and then used to construct the Bayes rule bypassing the posterior distribution. That is, this first time around, we will not be constructing the posterior distribution (for which we would have needed the form of the prior which we don't have). We'll go to the Bayes rule directly.

Example 11.2. For example, suppose we have observed an Exponential variable X with parameter θ and density $f(x | \theta) = \theta e^{-\theta x}$.

The marginal distribution of X is then $m(x) = \mathbf{E}^{\pi(\theta)}(\theta e^{-\theta x})$ whereas the Bayes rule, under the squared-error loss is

$$\delta(X) = \mathbf{E}(\theta | X) = \frac{\theta^2 e^{-\theta x}}{m(x)} = -\frac{m'(x)}{m(x)},$$

where $m'(x)$ is the derivative of the marginal density of X (check that this is true!).

Then we estimate the marginal density (for example, with a histogram) or estimate the cdf with an empirical cdf. The nonparametric empirical Bayes rule is then

$$\delta_{NPEB}(X) = -\frac{\hat{m}'(x)}{\hat{m}(x)}$$

◇

Example 11.3. As another example, consider Poisson distribution with parameter θ , whose prior is unknown.

If we knew the prior, the Bayes rule would have been

$$\begin{aligned} \delta &= \mathbf{E}(\theta | X) \\ &= \int \theta \pi(\theta | X) d\theta \\ &= \frac{\int \theta f(X | \theta) \pi(\theta) d\theta}{m(X)} \\ &= \frac{\int \theta e^{-\theta} \theta^X / X! \cdot \pi(\theta) d\theta}{m(X)} \\ &= \frac{1}{X! m(X)} \int \theta X + 1 e^{-\theta} \pi(\theta) d\theta \\ &= \frac{(X+1)!}{X! m(X)} \int f(X+1 | \theta) \pi(\theta) d\theta \\ &= \frac{(X+1) m(X+1)}{m(X)} \end{aligned}$$

Again, we estimate the marginal density $m(X)$ and construct the nonparametric empirical Bayes rule as

$$\delta_{NPEB}(X) = \frac{(X+1)\hat{m}(X+1)}{\hat{m}(X)}$$

◇

11.3 Hierarchical Bayes Approach

Hierarchical Bayes approach is similar to the empirical Bayes, however, the unknown hyper-parameters are treated in a perfectly Bayesian way. They are assumed to have their own prior distributions, **hyper-priors**. Then, these unknown hyper-parameters are integrated out, and the posterior distribution $\pi(\theta | \mathbf{X})$ is derived.

In general, consider the following model

$$\begin{aligned} \text{Data } X_1, \dots, X_n &\sim f(x | \theta), \text{ the model} \\ \text{Parameter } \theta &\sim \pi(\theta | \lambda), \text{ the prior} \\ \text{Hyper-parameter } \lambda &\sim p(\lambda) \text{ or } p(\lambda | \alpha, \beta), \text{ the hyper-prior} \end{aligned}$$

Doing hierarchical Bayes analysis manually, on paper, usually requires several steps of integration. The posterior distribution is

$$\begin{aligned} \pi(\theta | \mathbf{X}) &= \int \pi(\theta(\lambda, \mathbf{X})p(\lambda | \mathbf{X})d\lambda \\ &= \int \frac{f(\mathbf{X} | \theta)\pi(\theta | \lambda)}{m(\mathbf{X} | \lambda)} \cdot \frac{m(\mathbf{X} | \lambda)p(\lambda)}{m(\mathbf{X})}d\lambda \\ &= \frac{\int f(\mathbf{X} | \theta)\pi(\theta | \lambda)p(\lambda)d\lambda}{\int \int f(\mathbf{X} | \theta)\pi(\theta | \lambda)p(\lambda)d\theta d\lambda} \end{aligned}$$

Practical applications often involve rather advanced modeling with multiple parameters and several levels of hyper-priors. While it may be difficult to handle analytically, these models are easy to enter in WinBUGS or other Gibbs sampling software that will then produce a large number of generated values $\theta_1, \dots, \theta_N$ from the posterior distribution $\pi(\theta | \mathbf{X})$.

Berger 4.5-4.6; Christensen 4.10-4.13, 9.5

Chapter 12

Minimax Decisions and Game Theory

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12.1 General concepts

As we already know, decision δ is **minimax** if it minimizes $\max_{\theta} R(\theta, \delta)$, the worst frequentist risk over all $\theta \in \Theta$. That is,

$$\max_{\theta} R(\theta, \delta_{\text{minimax}}) = \min_{\delta \in \mathcal{A}} \max_{\theta} R(\theta, \delta)$$

This would be the best strategy in **statistical games**, playing against an intelligent opponent who controls θ and delivers the worst case scenario as long as it exists.

The resulting value of the risk is called **the upper value of the game**,

$$\overline{V} = \min_{\delta \in \mathcal{A}} \max_{\theta} R(\theta, \delta)$$

Symmetrically, since our “opponents” are playing with θ , their best strategy is called **maximin**, if it maximizes our risk, assuming that we play with our best strategy. That is,

$$\min_{\delta} R(\theta_{\text{maximin}, \delta}) = \max_{\theta} \min_{\delta} R(\theta, \delta).$$

The resulting value of the risk is called **the lower value of the game**,

$$\underline{V} = \max_{\theta} \min_{\delta} R(\theta, \delta)$$

It is not hard to see that $\underline{V} \leq \overline{V}$ all the time. Often (under certain conditions), $\underline{V} = \overline{V}$, in which case it is said that **the game has a value**, and the value of the game is

$$V = \underline{V} \leq \overline{V}$$

A game is called a **zero-sum game** if the loss of one player is the gain of the other player. Then, the sum of risks of all the players is 0, which means that (1) nobody outside of this

game can gain or lose from it, and (2) the players have no incentive to collaborate. By helping your opponent, you will always hurt yourself.

Only the simplest games are played with “deterministic” strategies δ and θ . The interesting and most reasonable games appear when players use **randomized strategies** - choosing a whole prior distribution $\pi(\theta)$ instead of choosing just θ and choosing a **randomized rule** δ^* , which is a distribution of decisions on the action space $\mathcal{A}$.

Then, a randomized decision δ^* is **minimax** if

$$\max_{\pi(\theta)} r(\pi, \delta_{\text{minimax}}^*) = \min_{\delta^*} \max_{\pi(\theta)} r(\pi, \delta^*) = \min_{\delta^*} \max_{\theta} r(\pi, \delta^*),$$

and the prior distribution $\pi(\theta)$ is **maximin**, if it maximizes the minimum risk,

$$\min_{\delta^*} r(\pi_{\text{maximin}, \delta^*}) = \max_{\pi(\theta)} \min_{\delta^*} r(\pi, \delta^*) = \max_{\pi(\theta)} \min_{\delta} r(\pi, \delta^*).$$

The **Minimax Theorem** establishes when a zero-sum game has a value, in which case the maximin strategy of Player II is the worst (least favorable) prior distribution for Player I, and the minimax strategy of Player I is Bayes with respect to this least favorable prior. Symmetrically, the randomized minimax strategy of Player I is the worst case scenario for Player II, and the maximin strategy of Player II is Bayes with respect to this case. Also, when **equalizer** strategies exist for both players, those rules with constant risks, these equalizers will be the minimax and maximin strategies.

12.2 Ann, Jim, and Mark problem

Here is a nice example of finding minimax and maximin strategies that are **equalizers**.

Ann arrives home from work at a random time, uniformly distributed between 4 and 5. She always goes out to dinner at 5. Mark and Jim would each like to take her for dinner. Unfortunately, they each have only one coin for a phone call. Whoever calls first, while Ann is at home, will get the dinner date. The loss function for both Mark and Jim is

$$\begin{aligned} L(\text{taking Ann for dinner}) &= -1 \\ L(\text{other man takes Ann for dinner}) &= 1 \\ L(\text{neither man takes Ann for dinner}) &= 0 \end{aligned}$$

Find the minimax and maximin strategies for Jim and Mark regarding the time to make that telephone call.

SOLUTION.

Let δ and θ be the strategies for Jim and Mark, respectively. Obviously, calling before 4 pm

or after 5 pm is inadmissible, so let δ and θ denote the time, the number of hours after 4 pm, with $\delta, \theta \in [0, 1]$.

The problem involves a random variable $X \sim \text{Uniform}(0,1)$, the time (also, the number of hours after 4 pm) when Ann comes home. Given X , the loss function for Jim (playing with δ) is

$$L(\theta, \delta) = \begin{cases} -1 & \text{if } \theta < X < \delta \quad \text{or} \quad X < \delta < \theta \quad (\text{Mark calls too early or Jim talks to Ann first}) \\ 0 & \text{if } \delta < X \quad \text{and} \quad \theta < X \quad (\text{both Jim and Mark call too early}) \\ 1 & \text{if } \delta < X < \theta \quad \text{or} \quad X < \theta < \delta \quad (\text{Jim calls too early or Mark talks to Ann first}) \end{cases}$$

Then, unconditionally, the loss function is

$$L(\theta, \delta) = \begin{cases} \mathbf{P}\{\delta < X < \theta\} - \mathbf{P}\{X < \delta\} & = (\theta - \delta) - \delta = \theta - 2\delta & \text{if } \delta < \theta \\ \mathbf{P}\{X < \theta\} - \mathbf{P}\{\theta < X < \delta\} & = \theta - (\delta - \theta) = 2\theta - \delta & \text{if } \theta < \delta \end{cases}$$

Playing with a randomized decision δ^* , Jim is looking for a density function of δ , say, $f(t)$. There is no obviously best choice of f , so let's search for an **equalizer**.

The risk for Jim is

$$L(\delta^*, \theta) = \int_0^\theta (\theta - 2t)f(t)dt + \int_\theta^1 (2\theta - t)f(t)dt = \text{const}(\theta).$$

We want this risk to be constant in terms of θ to make δ^* an equalizer. However, the choice of $f(t)$ that achieves this goal is still not clear.

Let's take a derivative! It is simpler than an integral, right? To have a constant risk, the derivative must equal 0.

Differentiating integrals with variable limits¹, we get

$$\begin{aligned} L'(\delta^*, \theta) &= (\theta - 2\theta)f(\theta) + \int_0^\theta f(t)dt - (2\theta - \theta)f(\theta) + 2 \int_\theta^1 f(t)dt \\ &= -2\theta f(\theta) + \int_0^\theta f(t)dt + 2 \int_\theta^1 f(t)dt = 0. \end{aligned}$$

Still not clear how to solve this... Let's take another derivative - all the integrals will then disappear.

$$\begin{aligned} L''(\delta^*, \theta) &= -2\theta f'(\theta) - 2f(\theta) + f(\theta) - 2f(\theta) \\ &= -2\theta f'(\theta) - 3f(\theta) = 0. \end{aligned}$$

Bingo. This is a differential equation with separating variables that we can now easily solve

¹Do you remember how to do it? $\frac{d}{d\theta} \int_0^\theta g(t, \theta)dt = g(\theta, \theta) + \int_0^\theta \frac{d}{d\theta} g(t, \theta)dt$. If θ comes in the lower limit of an integral, then the sign is negative.

in terms of the function $f(\theta)$. Separating variables means we can move f and f' to one side and θ to the other side. This is a standard ODE (ordinary differential equations) exercise,

$$\begin{aligned} -2\theta f'(\theta) &= 3f(\theta) &\Rightarrow & \frac{f'(\theta)}{f(\theta)} = -\frac{3}{2\theta} \\ \Rightarrow & \int \frac{f'(\theta)}{f(\theta)} d\theta = -\frac{3}{2} \int \frac{1}{\theta} d\theta + C_1 \\ \Rightarrow & \ln f(\theta) = -\frac{3}{2} \ln \theta + C_1 &\Rightarrow & \underline{f(\theta) = C_2 \theta^{-3/2}}, \quad 0 < \theta < 1. \end{aligned}$$

All right! We found the density $f(t)$ of the time when Jim should call, in order to play with an equalizer strategy. We just need to find a suitable normalizing constant C_2 to make it a density... Wait! Oh, no. This $f(\theta)$ is not integrable at 0! The integral diverges because of a singularity at 0,

$$\int_0^1 f(\theta) d\theta = C \int_0^1 \theta^{-3/2} d\theta = \infty.$$

Jim (Berger) is a very smart statistician. He realizes that since the problem is at 0, it means that some neighborhood of 0 should be removed from the set $\mathcal{A}$ of his possible decisions.

Let's try that. Assume that the density $f(t)$ is defined on the interval $[K, 1]$, for some $0 < K < 1$. We need to redo our derivations, but essentially, we just replace $\int_0^\theta$ with $\int_K^\theta$. This will have no effect on the second derivative. After taking two derivatives and then solving the differential equation, we'll again arrive to the density

$$f(\theta) = C\theta^{-3/2}, \quad K < \theta < 1.$$

To make it a density, the normalizing constant C is

$$C = \frac{1}{\int_K^1 \theta^{-3/2} d\theta} = \frac{1}{2 \left(\frac{1}{\sqrt{K}} - 1 \right)}.$$

It remains to plug this solution into the risk function $L(\delta^*, \theta)$ and see what K makes it constant, so δ^* becomes an equalizer.

Notice that Jim and Mark are in absolutely similar conditions, so the best strategy for Jim is also the best strategy for Mark. Then both δ and θ should be generated from the interval $[K, 1]$.

Substituting the obtained density into the risk, we get

$$\begin{aligned}
L(\delta^*, \theta) &= \int_K^\theta (\theta - 2t)f(t)dt + \int_\theta^1 (2\theta - t)f(t)dt \\
&= C \int_K^\theta (\theta - 2t)t^{-3/2}dt + C \int_\theta^1 (2\theta - t)t^{-3/2}dt \\
&= C\theta \int_K^\theta t^{-3/2}dt - 2C \int_K^\theta t^{-1/2}dt + 2C\theta \int_\theta^1 t^{-3/2}dt - C \int_\theta^1 t^{-1/2}dt \\
&= -2C\theta t^{-1/2} \Big|_K^\theta - 4Ct^{1/2} \Big|_K^\theta - 4C\theta t^{-1/2} \Big|_\theta^1 - 2Ct^{1/2} \Big|_\theta^1 \\
&= -2C\theta^{1/2} + 2C\theta K^{-1/2} - 4C\theta^{1/2} + 4CK^{1/2} - 4C\theta + 4C\theta^{1/2} - 2C + 2C\theta^{1/2} \\
&= \theta(2CK^{-1/2} - 4C) + 4CK^{1/2} - 2C = \text{const}
\end{aligned}$$

This risk is a linear function of θ . Of course, the second derivative erased the linear term, so we did not even notice it. Now we see that the risk will be constant (making δ^* an equalizer) iff $2CK^{-1/2} - 4C = 0$, i.e.,

$$K = \frac{1}{4}.$$

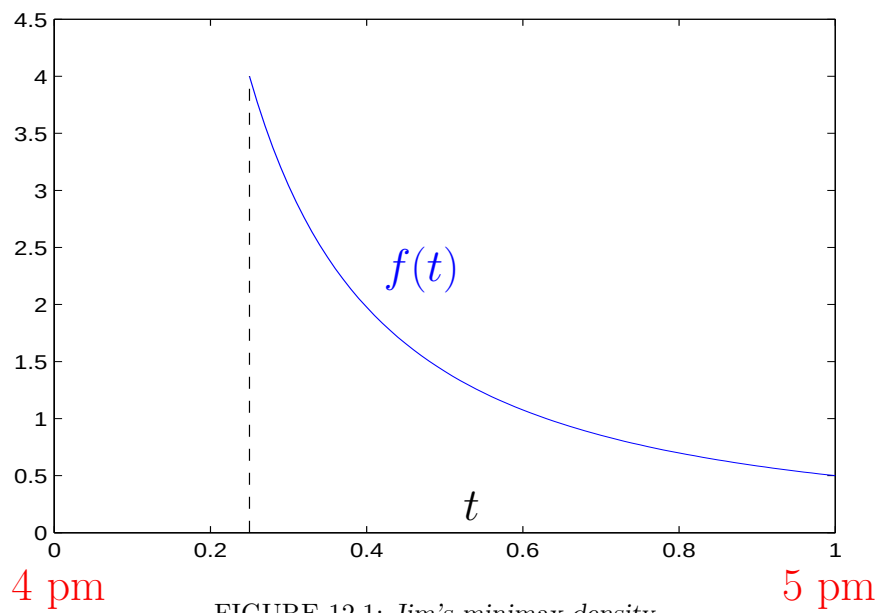
OK, we found it. The equalizer solution (for both Jim and Mark) is given by the density

$$f(t) = Ct^{-3/2}, \quad \text{for } \frac{1}{4} < t < 1, \quad \text{where } C = \frac{1}{2\left(\frac{1}{\sqrt{1/4}} - 1\right)} = \frac{1}{2}.$$

So, each man should wait 1/4 hour until 4:15 pm and then make a call at a random time, generated from the density

$$f(t) = \frac{1}{2}t^{-3/2}, \quad \frac{1}{4} < t < 1.$$

Both Jim and Mark have equalizer strategies. Therefore, these are in fact **minimax** and **maximin** strategies.

FIGURE 12.1: *Jim's minimax density.*

Chapter 13

Appendix

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13.1 Inventory of distributions

13.1.1 Discrete families

Bernoulli(p)

Generic description:	0 or 1, success or failure, result of one Bernoulli trial
Range of values:	$x = 0, 1$
Parameter:	$p \in (0, 1)$, probability of success
Probability mass function:	$P(x) = p^x(1 - p)^{1-x}$
Expectation:	$\mu = p$
Variance:	$\sigma^2 = p(1 - p)$
Relations:	Special case of <i>Binomial</i> (n, p) when $n = 1$ A sum of n independent Bernoulli(p) variables is <i>Binomial</i> (n, p)

Binomial(n, p)

Generic description:	Number of successes in n independent Bernoulli trials
Range of values:	$x = 0, 1, 2, \dots, n$
Parameters:	$n = 1, 2, 3, \dots$, number of Bernoulli trials $p \in (0, 1)$, probability of success
Probability mass function:	$P(x) = \binom{n}{x} p^x(1 - p)^{n-x}$
Cdf:	Table A2

Expectation:	$\mu = np$
Variance:	$\sigma^2 = np(1 - p)$
Relations:	Binomial(n, p) is a sum of n independent <i>Bernoulli</i> (p) variables Binomial($1, p$) = <i>Bernoulli</i> (p)

Geometric(p)

Generic description:	Number of Bernoulli trials until the first success
Range of values:	$x = 1, 2, 3, \dots$
Parameter:	$p \in (0, 1)$, probability of success
Probability mass function:	$P(x) = p(1 - p)^{x-1}$
Cdf:	$1 - (1 - p)^x$
Expectation:	$\mu = \frac{1}{p}$
Variance:	$\sigma^2 = \frac{1 - p}{p^2}$
Relations:	Special case of <i>Negative Binomial</i> (k, p) when $k = 1$ A sum of n independent Geometric(p) variables is <i>Negative Binomial</i> (n, p)

Negative Binomial(k, p)

Generic description:	Number of Bernoulli trials until the k -th success
Range of values:	$x = k, k + 1, k + 2, \dots$
Parameters:	$k = 1, 2, 3, \dots$, number of successes $p \in (0, 1)$, probability of success
Probability mass function:	$P(x) = \binom{x-1}{k-1} (1-p)^{x-k} p^k$
Expectation:	$\mu = \frac{k}{p}$
Variance:	$\sigma^2 = \frac{k(1-p)}{p^2}$
Relations:	Negative Binomial(k, p) is a sum of n independent <i>Geometric</i> (p) variables Negative Binomial($1, p$) = <i>Geometric</i> (p)

Poisson(λ)

Generic description:	Number of “rare events” during a fixed time interval
Range of values:	$x = 0, 1, 2, \dots$
Parameter:	$\lambda \in (0, \infty)$, frequency of “rare events”
Probability mass function:	$P(x) = e^{-\lambda} \frac{\lambda^x}{x!}$

Cdf:	Table A3
Expectation:	$\mu = \lambda$
Variance:	$\sigma^2 = \lambda$
Relations:	Limiting case of <i>Binomial</i> (n, p) when $n \rightarrow \infty$, $p \rightarrow 0$, $np \rightarrow \lambda$

13.1.2 Continuous families

Beta(α, β)

Generic description:	In a sample from Standard Uniform distribution, it is the distribution of the k^{th} smallest observation
Range of values:	$0 < x < 1$
Parameter:	$\alpha, \beta \in (0, \infty)$, frequency of events, inverse scale parameter
Density:	$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$
Expectation:	$\mu = \alpha/(\alpha + \beta)$
Variance:	$\sigma^2 = \alpha\beta/(\alpha + \beta)^2(\alpha + \beta + 1)$
Relations:	As a prior distribution, Beta family is conjugate to the <i>Binomial</i> model. Beta(1,1) distribution is <i>Uniform</i> (0,1).

Chi-square $\chi^2(\nu)$

Generic description:	Sum of squares of ν iid Normal variables
Range of values:	$x > 0$
Parameter:	$\nu \in (0, \infty)$, degrees of freedom
Density:	$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$
Expectation:	$\mu = \nu$
Variance:	$\sigma^2 = 2\nu$
Relations:	Special case of <i>Gamma</i> (α, λ) when $\alpha = \nu/2$ and $\lambda = 1/2$. Ratio of two independent χ^2 variables, divided by their respective degrees of freedom, has <i>F-distribution</i> .

Exponential(λ)

Generic description:	In a Poisson process, time between consecutive events
Range of values:	$x > 0$
Parameter:	$\lambda \in (0, \infty)$, frequency of events, inverse scale parameter

Density:	$f(x) = \lambda e^{-\lambda x}$
Cdf:	$F(x) = 1 - e^{-\lambda x}$
Expectation:	$\mu = 1/\lambda$
Variance:	$\sigma^2 = 1/\lambda^2$
Relations:	Special case of $\text{Gamma}(\alpha, \lambda)$ when $\alpha = 1$ and <i>chi-square</i> with $\nu = 2$ degrees of freedom. A sum of α independent $\text{Exponential}(\lambda)$ variables is $\text{Gamma}(\alpha, \lambda)$

Fisher-Snedecor $F(\nu_1, \nu_2)$

Generic description:	Ratio of independent mean squares
Range of values:	$0 < x < +\infty$
Parameters:	$\nu_1 > 0$, numerator degrees of freedom $\nu_2 > 0$, denominator degrees of freedom
Density function:	$f(x) = \frac{\Gamma\left(\frac{\nu_1 + \nu_2}{2}\right)}{x \Gamma\left(\frac{\nu_1}{2}\right) \Gamma\left(\frac{\nu_2}{2}\right)} \sqrt{\frac{(\nu_1 x)^{\nu_1} \nu_2^{\nu_2}}{(\nu_1 x + \nu_2)^{\nu_1 + \nu_2}}}$
Expectation:	$\frac{\nu_2}{\nu_2 - 2}$ for $\nu_2 > 2$
Variance:	$\frac{2\nu^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)}$ for $\nu_2 > 4$
Relations:	For independent $\chi^2(\nu_1)$ and $\chi^2(\nu_2)$, the ratio $\frac{\chi^2(\nu_1)/\nu_1}{\chi^2(\nu_2)/\nu_2}$ is $F(\nu_1, \nu_2)$ Special case: a square of $t(\nu)$ variable is $F(1, \nu)$

Gamma(α, λ)

Generic description:	In a Poisson process, the time of the α -th event
Range of values:	$x > 0$
Parameters:	$\alpha \in (0, \infty)$, shape parameter $\lambda \in (0, \infty)$, frequency of events, inverse scale parameter
Density function:	$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$
Expectation:	$\mu = \alpha/\lambda$
Variance:	$\sigma^2 = \alpha/\lambda^2$
Relations:	For integer α , $\text{Gamma}(\alpha, \lambda)$ is a sum of α independent $\text{Exponential}(\lambda)$ variables $\text{Gamma}(1, \lambda) = \text{Exponential}(\lambda)$ $\text{Gamma}(\nu/2, 1/2) = \text{Chi-square}(\nu)$ As a prior distribution, Gamma family is conjugate to the $\text{Poisson}(\theta)$ model.

Normal(μ, σ)

Generic description:	Often used as distribution of errors, measurements, sums, averages, etc.
Range of values:	$-\infty < x < +\infty$
Parameters:	$\mu \in (-\infty, \infty)$, expectation, location parameter $\sigma \in (0, \infty)$, standard deviation, scale parameter
Density function:	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}$
Cdf:	$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$, use Table A1 for $\Phi(z)$
Expectation:	μ
Variance:	σ^2
Relations:	Limiting case of standardized sums of random variables, including <i>Binomial</i> (n, p), <i>Negative Binomial</i> (k, p), and <i>Gamma</i> (α, λ), as $n, k, \alpha \rightarrow \infty$

Pareto(θ, σ)

Generic description:	A heavy-tail distribution often used to model amount of internet traffic, various financial and sociological data
Range of values:	$x > \sigma$
Parameters:	$\theta \in (0, \infty)$, shape parameter $\sigma \in (0, \infty)$, scale parameter
Density function:	$\theta\sigma^\theta x^{-\theta-1}$
Cdf:	$F(x) = 1 - \left(\frac{x}{\sigma}\right)^{-\theta}$
Expectation:	$\frac{\theta\sigma}{\theta-1}$ for $\theta > 1$; does not exist for $\theta \leq 1$
Variance:	$\frac{\theta\sigma^2}{(\theta-1)^2(\theta-2)}$ for $\theta > 2$; does not exist for $\theta \leq 2$
Relations:	If X is <i>Exponential</i> (θ) then $Y = \sigma e^X$ is <i>Pareto</i> (θ, σ)

These commands are for the Generalized Pareto distribution.
For *Pareto*(θ, σ), use `gpcdf(x, 1/theta, sigma/theta/sigma)`

Student's T(ν)

Generic description:	Standardized statistic with an estimated standard deviation
Range of values:	$-\infty < x < +\infty$
Parameter:	$\nu > 0$, number of degrees of freedom

Density function:	$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$
Expectation:	0
Variance:	$\frac{\nu}{\nu-2}$ for $\nu > 2$
Relations:	Converges to Normal distribution as $\nu \rightarrow \infty$

Uniform(a, b)

Generic description:	A number selected “at random” from a given interval
Range of values:	$a < x < b$
Parameters:	$-\infty < a < b < +\infty$, ends of the interval
Density function:	$f(x) = \frac{1}{b-a}$
Expectation:	$\mu = \frac{a+b}{2}$
Variance:	$\sigma^2 = \frac{(b-a)^2}{12}$
Relations:	Uniform(0,1) distribution is <i>Beta</i> (1,1)

13.2 Distribution tables

Table A1. Standard Normal distribution

$$\Phi(z) = P\{Z \leq z\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$

<i>z</i>	-0.09	-0.08	-0.07	-0.06	-0.05	-0.04	-0.03	-0.02	-0.01	-0.00
-(3.9+)	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
-3.8	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.7	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.6	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0002
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003
-3.3	.0003	.0004	.0004	.0004	.0004	.0004	.0004	.0005	.0005	.0005
-3.2	.0005	.0005	.0005	.0006	.0006	.0006	.0006	.0006	.0007	.0007
-3.1	.0007	.0007	.0008	.0008	.0008	.0008	.0009	.0009	.0009	.0010
-3.0	.0010	.0010	.0011	.0011	.0011	.0012	.0012	.0013	.0013	.0013
-2.9	.0014	.0014	.0015	.0015	.0016	.0016	.0017	.0018	.0018	.0019
-2.8	.0019	.0020	.0021	.0021	.0022	.0023	.0023	.0024	.0025	.0026
-2.7	.0026	.0027	.0028	.0029	.0030	.0031	.0032	.0033	.0034	.0035
-2.6	.0036	.0037	.0038	.0039	.0040	.0041	.0043	.0044	.0045	.0047
-2.5	.0048	.0049	.0051	.0052	.0054	.0055	.0057	.0059	.0060	.0062
-2.4	.0064	.0066	.0068	.0069	.0071	.0073	.0075	.0078	.0080	.0082
-2.3	.0084	.0087	.0089	.0091	.0094	.0096	.0099	.0102	.0104	.0107
-2.2	.0110	.0113	.0116	.0119	.0122	.0125	.0129	.0132	.0136	.0139
-2.1	.0143	.0146	.0150	.0154	.0158	.0162	.0166	.0170	.0174	.0179
-2.0	.0183	.0188	.0192	.0197	.0202	.0207	.0212	.0217	.0222	.0228
-1.9	.0233	.0239	.0244	.0250	.0256	.0262	.0268	.0274	.0281	.0287
-1.8	.0294	.0301	.0307	.0314	.0322	.0329	.0336	.0344	.0351	.0359
-1.7	.0367	.0375	.0384	.0392	.0401	.0409	.0418	.0427	.0436	.0446
-1.6	.0455	.0465	.0475	.0485	.0495	.0505	.0516	.0526	.0537	.0548
-1.5	.0559	.0571	.0582	.0594	.0606	.0618	.0630	.0643	.0655	.0668
-1.4	.0681	.0694	.0708	.0721	.0735	.0749	.0764	.0778	.0793	.0808
-1.3	.0823	.0838	.0853	.0869	.0885	.0901	.0918	.0934	.0951	.0968
-1.2	.0985	.1003	.1020	.1038	.1056	.1075	.1093	.1112	.1131	.1151
-1.1	.1170	.1190	.1210	.1230	.1251	.1271	.1292	.1314	.1335	.1357
-1.0	.1379	.1401	.1423	.1446	.1469	.1492	.1515	.1539	.1562	.1587
-0.9	.1611	.1635	.1660	.1685	.1711	.1736	.1762	.1788	.1814	.1841
-0.8	.1867	.1894	.1922	.1949	.1977	.2005	.2033	.2061	.2090	.2119
-0.7	.2148	.2177	.2206	.2236	.2266	.2296	.2327	.2358	.2389	.2420
-0.6	.2451	.2483	.2514	.2546	.2578	.2611	.2643	.2676	.2709	.2743
-0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981	.3015	.3050	.3085
-0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336	.3372	.3409	.3446
-0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707	.3745	.3783	.3821
-0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090	.4129	.4168	.4207
-0.1	.4247	.4286	.4325	.4364	.4404	.4443	.4483	.4522	.4562	.4602
-0.0	.4641	.4681	.4721	.4761	.4801	.4840	.4880	.4920	.4960	.5000

Table A1, continued. Standard Normal distribution

[illegible]

Table A2. Binomial distribution

$$F(x) = P\{X \leq x\} = \sum_{k=0}^x \binom{n}{k} p^k (1-p)^{n-k}$$

n	x	p																		
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950
1	0	.950	.900	.850	.800	.750	.700	.650	.600	.550	.500	.450	.400	.350	.300	.250	.200	.150	.100	.050
2	0	.903	.810	.723	.640	.563	.490	.423	.360	.303	.250	.203	.160	.123	.090	.063	.040	.023	.010	.003
	1	.998	.990	.978	.960	.938	.910	.878	.840	.798	.750	.698	.640	.578	.510	.438	.360	.278	.190	.098
3	0	.857	.729	.614	.512	.422	.343	.275	.216	.166	.125	.091	.064	.043	.027	.016	.008	.003	.001	.000
	1	.993	.972	.939	.896	.844	.784	.718	.648	.575	.500	.425	.352	.282	.216	.156	.104	.061	.028	.007
	2	1.0	.999	.997	.992	.984	.973	.957	.936	.909	.875	.834	.784	.725	.657	.578	.488	.386	.271	.143
4	0	.815	.656	.522	.410	.316	.240	.179	.130	.092	.063	.041	.026	.015	.008	.004	.002	.001	.000	.000
	1	.986	.948	.890	.819	.738	.652	.563	.475	.391	.313	.241	.179	.126	.084	.051	.027	.012	.004	.000
	2	1.0	.996	.988	.973	.949	.916	.874	.821	.759	.688	.609	.525	.437	.348	.262	.181	.110	.052	.014
	3	1.0	1.0	.999	.998	.996	.992	.985	.974	.959	.938	.908	.870	.821	.760	.684	.590	.478	.344	.185
5	0	.774	.590	.444	.328	.237	.168	.116	.078	.050	.031	.018	.010	.005	.002	.001	.000	.000	.000	.000
	1	.977	.919	.835	.737	.633	.528	.428	.337	.256	.188	.131	.087	.054	.031	.016	.007	.002	.000	.000
	2	.999	.991	.973	.942	.896	.837	.765	.683	.593	.500	.407	.317	.235	.163	.104	.058	.027	.009	.001
	3	1.0	1.0	.998	.993	.984	.969	.946	.913	.869	.813	.744	.663	.572	.472	.367	.263	.165	.081	.023
	4	1.0	1.0	1.0	1.0	.999	.998	.995	.990	.982	.969	.950	.922	.884	.832	.763	.672	.556	.410	.226
6	0	.735	.531	.377	.262	.178	.118	.075	.047	.028	.016	.008	.004	.002	.001	.000	.000	.000	.000	.000
	1	.967	.886	.776	.655	.534	.420	.319	.233	.164	.109	.069	.041	.022	.011	.005	.002	.000	.000	.000
	2	.998	.984	.953	.901	.831	.744	.647	.544	.442	.344	.255	.179	.117	.070	.038	.017	.006	.001	.000
	3	1.0	.999	.994	.983	.962	.930	.883	.821	.745	.656	.558	.456	.353	.256	.169	.099	.047	.016	.002
	4	1.0	1.0	1.0	.998	.995	.989	.978	.959	.931	.891	.836	.767	.681	.580	.466	.345	.224	.114	.033
	5	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.984	.972	.953	.925	.882	.822	.738	.623	.469	.265
7	0	.698	.478	.321	.210	.133	.082	.049	.028	.015	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000
	1	.956	.850	.717	.577	.445	.329	.234	.159	.102	.063	.036	.019	.009	.004	.001	.000	.000	.000	.000
	2	.996	.974	.926	.852	.756	.647	.532	.420	.316	.227	.153	.096	.056	.029	.013	.005	.001	.000	.000
	3	1.0	.997	.988	.967	.929	.874	.800	.710	.608	.500	.392	.290	.200	.126	.071	.033	.012	.003	.000
	4	1.0	1.0	.999	.995	.987	.971	.944	.904	.847	.773	.684	.580	.468	.353	.244	.148	.074	.026	.004
	5	1.0	1.0	1.0	1.0	.999	.996	.991	.981	.964	.938	.898	.841	.766	.671	.555	.423	.283	.150	.044
	6	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.985	.972	.951	.918	.867	.790	.679	.522	.302
8	0	.663	.430	.272	.168	.100	.058	.032	.017	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000	.000
	1	.943	.813	.657	.503	.367	.255	.169	.106	.063	.035	.018	.009	.004	.001	.000	.000	.000	.000	.000
	2	.994	.962	.895	.797	.679	.552	.428	.315	.220	.145	.088	.050	.025	.011	.004	.001	.000	.000	.000
	3	1.0	.995	.979	.944	.886	.806	.706	.594	.477	.363	.260	.174	.106	.058	.027	.010	.003	.000	.000
	4	1.0	1.0	.997	.990	.973	.942	.894	.826	.740	.637	.523	.406	.294	.194	.114	.056	.021	.005	.000
	5	1.0	1.0	1.0	1.0	.999	.996	.995	.975	.950	.912	.855	.780	.685	.572	.448	.321	.203	.105	.038
	6	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.982	.965	.937	.894	.831	.745	.633	.497	.343	.187	.057
	7	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.983	.968	.942	.900	.832	.728	.570	.337
9	0	.630	.387	.232	.134	.075	.040	.021	.010	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000
	1	.929	.775	.599	.436	.300	.196	.121	.071	.039	.020	.009	.004	.001	.000	.000	.000	.000	.000	.000
	2	.992	.947	.859	.738	.601	.463	.337	.232	.150	.090	.050	.025	.011	.004	.001	.000	.000	.000	.000
	3	.999	.992	.966	.914	.834	.730	.609	.483	.361	.254	.166	.099	.054	.025	.010	.003	.001	.000	.000
	4	1.0	.999	.994	.980	.951	.901	.828	.733	.621	.500	.379	.267	.172	.099	.049	.020	.006	.001	.000
	5	1.0	1.0	.999	.997	.990	.975	.946	.901	.834	.746	.639	.517	.391	.270	.166	.086	.034	.008	.001
	6	1.0	1.0	1.0	1.0	.999	.996	.989	.975	.950	.910	.850	.768	.663	.537	.399	.262	.141	.053	.008
	7	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.980	.961	.929	.879	.804	.700	.564	.401	.225	.071
	8	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.990	.979	.960	.925	.866	.768	.613	.370
10	0	.599	.349	.197	.107	.056	.028	.013	.006	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.914	.736	.544	.376	.244	.149	.086	.046	.023	.011	.005	.002	.001	.000	.000	.000	.000	.000	.000
	2	.988	.930	.820	.678	.526	.383	.262	.167	.100	.055	.027	.012	.005	.002	.000	.000	.000	.000	.000
	3	.999	.987	.950	.879	.776	.650	.514	.382	.266	.172	.102	.055	.026	.011	.004	.001	.000	.000	.000
	4	1.0	.998	.990	.967	.922	.850	.751	.633	.504	.377	.262	.166	.095	.047	.020	.006	.001	.000	.000
	5	1.0	1.0	.999	.994	.980	.953	.905	.834	.738	.623	.496	.367	.249	.150	.078	.033	.010	.002	.000
	6	1.0	1.0	1.0	.999	.996	.989	.974	.945	.898	.828	.734	.618	.486	.350	.224	.121	.050	.013	.001
	7	1.0	1.0	1.0	1.0	1.0	.998	.995	.988	.973	.945	.900	.833	.738	.617	.474	.322	.180	.070	.012
	8	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.989	.977	.954	.914	.851	.756	.624	.456	.264	.086
	9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.994	.987	.972	.944	.893	.803	.651	.401

Table A2, continued. Binomial distribution

n	x	p																		
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950
11	0	.569	.314	.167	.086	.042	.020	.009	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.898	.697	.492	.322	.197	.113	.061	.030	.014	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000
	2	.985	.910	.779	.617	.455	.313	.200	.119	.065	.033	.015	.006	.002	.001	.000	.000	.000	.000	.000
	3	.998	.981	.931	.839	.713	.570	.426	.296	.191	.113	.061	.029	.012	.004	.001	.000	.000	.000	.000
	4	1.0	.997	.984	.950	.885	.790	.668	.533	.397	.274	.174	.099	.050	.022	.008	.002	.000	.000	.000
	5	1.0	1.0	.997	.988	.966	.922	.851	.753	.633	.500	.367	.247	.149	.078	.034	.012	.003	.000	.000
	6	1.0	1.0	1.0	.998	.992	.978	.950	.901	.826	.726	.603	.467	.332	.210	.115	.050	.016	.003	.000
	7	1.0	1.0	1.0	1.0	.999	.996	.988	.971	.939	.887	.809	.704	.574	.430	.287	.161	.069	.019	.002
	8	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.985	.967	.935	.881	.800	.687	.545	.383	.221	.090	.015
	9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.986	.970	.939	.887	.803	.678	.508	.303	.102
10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.980	.958	.914	.833	.686	.431	
12	0	.540	.282	.142	.069	.032	.014	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.882	.659	.443	.275	.158	.085	.042	.020	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000
	2	.980	.889	.736	.558	.391	.253	.151	.083	.042	.019	.008	.003	.001	.000	.000	.000	.000	.000	.000
	3	.998	.974	.908	.795	.649	.493	.347	.225	.134	.073	.036	.015	.006	.002	.000	.000	.000	.000	.000
	4	1.0	.996	.976	.927	.842	.724	.583	.438	.304	.194	.112	.057	.026	.009	.003	.001	.000	.000	.000
	5	1.0	.999	.995	.981	.946	.882	.787	.665	.527	.387	.261	.158	.085	.039	.014	.004	.001	.000	.000
	6	1.0	1.0	.999	.996	.986	.961	.915	.842	.739	.613	.473	.335	.213	.118	.054	.019	.005	.001	.000
	7	1.0	1.0	1.0	.999	.997	.991	.974	.943	.888	.806	.696	.562	.417	.276	.158	.073	.024	.004	.000
	8	1.0	1.0	1.0	1.0	1.0	.998	.994	.985	.964	.927	.866	.775	.653	.507	.351	.205	.092	.026	.002
	9	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.981	.958	.917	.849	.747	.609	.442	.264	.111	.020
	10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.980	.958	.915	.842	.725	.557	.341	.118
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.986	.968	.931	.858	.718	.460
13	0	.513	.254	.121	.055	.024	.010	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.865	.621	.398	.234	.127	.064	.030	.013	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000
	2	.975	.866	.692	.502	.333	.202	.113	.058	.027	.011	.004	.001	.000	.000	.000	.000	.000	.000	.000
	3	.997	.966	.882	.747	.584	.421	.278	.169	.093	.046	.020	.008	.003	.001	.000	.000	.000	.000	.000
	4	1.0	.994	.966	.901	.794	.654	.501	.353	.228	.133	.070	.032	.013	.004	.001	.000	.000	.000	.000
	5	1.0	.999	.992	.970	.920	.835	.716	.574	.427	.291	.179	.098	.046	.018	.006	.001	.000	.000	.000
	6	1.0	1.0	.999	.993	.976	.938	.871	.771	.644	.500	.356	.229	.129	.062	.024	.007	.001	.000	.000
	7	1.0	1.0	1.0	.999	.994	.982	.954	.902	.821	.709	.573	.426	.284	.165	.080	.030	.008	.001	.000
	8	1.0	1.0	1.0	1.0	.999	.996	.987	.968	.930	.867	.772	.647	.499	.346	.206	.099	.034	.006	.000
	9	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.980	.954	.907	.831	.722	.579	.416	.253	.118	.034	.003
	10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.973	.942	.887	.798	.667	.498	.308	.134	.025
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.987	.970	.936	.873	.766	.602	.379	.135
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.990	.976	.945	.879	.746	.487
14	0	.488	.229	.103	.044	.018	.007	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.847	.585	.357	.198	.101	.047	.021	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.970	.842	.648	.448	.281	.161	.084	.040	.017	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000
	3	.996	.956	.853	.698	.521	.355	.220	.124	.063	.029	.011	.004	.001	.000	.000	.000	.000	.000	.000
	4	1.0	.991	.953	.870	.742	.584	.423	.279	.167	.090	.043	.018	.006	.002	.000	.000	.000	.000	.000
	5	1.0	.999	.988	.956	.888	.781	.641	.486	.337	.212	.119	.058	.024	.008	.002	.000	.000	.000	.000
	6	1.0	1.0	.998	.988	.962	.907	.816	.692	.546	.395	.259	.150	.075	.031	.010	.002	.000	.000	.000
	7	1.0	1.0	1.0	.998	.990	.969	.925	.850	.741	.605	.454	.308	.184	.093	.038	.012	.002	.000	.000
	8	1.0	1.0	1.0	1.0	.998	.992	.976	.942	.881	.788	.663	.514	.359	.219	.112	.044	.012	.001	.000
	9	1.0	1.0	1.0	1.0	1.0	.998	.994	.982	.957	.910	.833	.721	.577	.416	.258	.130	.047	.009	.000
	10	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.971	.937	.876	.780	.645	.479	.302	.147	.044	.004
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.983	.960	.916	.839	.719	.552	.352	.158	.030
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.979	.953	.899	.802	.643	.415	.153
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.993	.982	.956	.897	.771	.512
15	0	.463	.206	.087	.035	.013	.005	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.829	.549	.319	.167	.080	.035	.014	.005	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.964	.816	.604	.398	.236	.127	.062	.027	.011	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000
	3	.995	.944	.823	.648	.461	.297	.173	.091	.042	.018	.006	.002	.000	.000	.000	.000	.000	.000	.000
	4	.999	.987	.938	.836	.686	.515	.352	.217	.120	.059	.025	.009	.003	.001	.000	.000	.000	.000	.000
	5	1.0	.998	.983	.939	.852	.722	.564	.403	.261	.151	.077	.034	.012	.004	.001	.000	.000	.000	.000
	6	1.0	1.0	.996	.982	.943	.869	.755	.610	.452	.304	.182	.095	.042	.015	.004	.001	.000	.000	.000
	7	1.0	1.0	.999	.996	.983	.950	.887	.787	.654	.500	.346	.213	.113	.050	.017	.004	.001	.000	.000
	8	1.0	1.0	1.0	.999	.996	.985	.958	.905	.818	.696	.548	.390	.245	.131	.057	.018	.004	.000	.000
	9	1.0	1.0	1.0	1.0	.999	.996	.988	.966	.923	.849	.739	.597	.436	.278	.148	.061	.017	.002	.000
	10	1.0	1.0	1.0	1.0	1.0	.999	.997	.991	.975	.941	.880	.783	.648	.485	.314	.164	.062	.013	.001
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.982	.958	.909	.827	.703	.539	.352	.177	.056	.005
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.973	.938	.873	.764	.602	.396	.184	.036
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.995	.986	.965	.965	.920	.833	.681	.451	.171
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.995	.987	.965	.913	.794	.537

Table A2, continued. Binomial distribution

<i>n</i>	<i>x</i>	<i>p</i>																		
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950
16	1	.811	.515	.284	.141	.063	.026	.010	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.957	.789	.561	.352	.197	.099	.045	.018	.007	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000
	3	.993	.932	.790	.598	.405	.246	.134	.065	.028	.011	.003	.001	.000	.000	.000	.000	.000	.000	.000
	4	.999	.983	.921	.798	.630	.450	.289	.167	.085	.038	.015	.005	.001	.000	.000	.000	.000	.000	.000
	5	1.0	.997	.976	.918	.810	.660	.490	.329	.198	.105	.049	.019	.006	.002	.000	.000	.000	.000	.000
	6	1.0	.999	.994	.973	.920	.825	.688	.527	.366	.227	.124	.058	.023	.007	.002	.000	.000	.000	.000
	7	1.0	1.0	.999	.993	.973	.926	.841	.716	.563	.402	.256	.142	.067	.026	.007	.001	.000	.000	.000
	8	1.0	1.0	1.0	.999	.993	.974	.933	.858	.744	.598	.437	.284	.159	.074	.027	.007	.001	.000	.000
	9	1.0	1.0	1.0	1.0	.998	.993	.977	.942	.876	.773	.634	.473	.312	.175	.080	.027	.006	.001	.000
	10	1.0	1.0	1.0	1.0	1.0	.998	.994	.981	.951	.895	.802	.671	.510	.340	.190	.082	.024	.003	.000
	11	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.985	.962	.915	.833	.711	.550	.370	.202	.079	.017	.001
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.989	.972	.935	.866	.754	.595	.402	.210	.068	.007
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.993	.982	.955	.901	.803	.648	.439	.211	.043
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.990	.974	.937	.859	.716	.485	.189
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.990	.972	.926	.815	.560
18	1	.774	.450	.224	.099	.039	.014	.005	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.942	.734	.480	.271	.135	.060	.024	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	3	.989	.902	.720	.501	.306	.165	.078	.033	.012	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000
	4	.998	.972	.879	.716	.519	.333	.189	.094	.041	.015	.005	.001	.000	.000	.000	.000	.000	.000	.000
	5	1.0	.994	.958	.867	.717	.534	.355	.209	.108	.048	.018	.006	.001	.000	.000	.000	.000	.000	.000
	6	1.0	.999	.988	.949	.861	.722	.549	.374	.226	.119	.054	.020	.006	.001	.000	.000	.000	.000	.000
	7	1.0	1.0	.997	.984	.943	.859	.728	.563	.391	.240	.128	.058	.021	.006	.001	.000	.000	.000	.000
	8	1.0	1.0	.999	.996	.981	.940	.861	.737	.578	.407	.253	.135	.060	.021	.005	.001	.000	.000	.000
	9	1.0	1.0	1.0	.999	.995	.979	.940	.865	.747	.593	.422	.263	.139	.060	.019	.004	.001	.000	.000
	10	1.0	1.0	1.0	1.0	.999	.994	.979	.942	.872	.760	.609	.437	.272	.141	.057	.016	.003	.000	.000
	11	1.0	1.0	1.0	1.0	1.0	.999	.994	.980	.946	.881	.774	.626	.451	.278	.139	.051	.012	.001	.000
	12	1.0	1.0	1.0	1.0	1.0	1.0	.999	.994	.982	.952	.892	.791	.645	.466	.283	.133	.042	.006	.000
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.985	.959	.906	.811	.667	.481	.284	.121	.028	.002
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.988	.967	.922	.835	.694	.499	.280	.098	.011
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.976	.940	.865	.729	.520	.266	.058
	16	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.986	.961	.901	.776	.550	.226
20	1	.736	.392	.176	.069	.024	.008	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.925	.677	.405	.206	.091	.035	.012	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	3	.984	.867	.648	.411	.225	.107	.044	.016	.005	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	4	.997	.957	.830	.630	.415	.238	.118	.051	.019	.006	.002	.000	.000	.000	.000	.000	.000	.000	.000
	5	1.0	.989	.933	.804	.617	.416	.245	.126	.055	.021	.006	.002	.000	.000	.000	.000	.000	.000	.000
	6	1.0	.998	.978	.913	.786	.608	.417	.250	.130	.058	.021	.006	.002	.000	.000	.000	.000	.000	.000
	7	1.0	1.0	.994	.968	.898	.772	.601	.416	.252	.132	.058	.021	.006	.001	.000	.000	.000	.000	.000
	8	1.0	1.0	.999	.990	.959	.887	.762	.596	.414	.252	.131	.057	.020	.005	.001	.000	.000	.000	.000
	9	1.0	1.0	1.0	.997	.986	.952	.878	.755	.591	.412	.249	.128	.053	.017	.004	.001	.000	.000	.000
	10	1.0	1.0	1.0	.999	.996	.983	.947	.872	.751	.588	.409	.245	.122	.048	.014	.003	.000	.000	.000
	11	1.0	1.0	1.0	1.0	.999	.995	.980	.943	.869	.748	.586	.404	.238	.113	.041	.010	.001	.000	.000
	12	1.0	1.0	1.0	1.0	1.0	.999	.994	.979	.942	.868	.748	.584	.399	.228	.102	.032	.006	.000	.000
	13	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.979	.942	.870	.750	.583	.392	.214	.087	.022	.002	.000
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.979	.945	.874	.755	.584	.383	.196	.067	.011	.000
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.981	.949	.882	.762	.585	.370	.170	.043	.003
	16	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.984	.956	.893	.775	.589	.352	.133	.016
25	2	.873	.537	.254	.098	.032	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	3	.966	.764	.471	.234	.096	.033	.010	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	4	.993	.902	.682	.421	.214	.090	.032	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	5	.999	.967	.838	.617	.378	.193	.083	.029	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000
	6	1.0	.991	.930	.780	.561	.341	.173	.074	.026	.007	.002	.000	.000	.000	.000	.000	.000	.000	.000
	7	1.0	.998	.975	.891	.727	.512	.306	.154	.064	.022	.006	.001	.000	.000	.000	.000	.000	.000	.000
	8	1.0	1.0	.992	.953	.851	.677	.467	.274	.134	.054	.017	.004	.001	.000	.000	.000	.000	.000	.000
	9	1.0	1.0	.998	.983	.929	.811	.630	.425	.242	.115	.044	.013	.003	.000	.000	.000	.000	.000	.000
	10	1.0	1.0	1.0	.994	.970	.902	.771	.586	.384	.212	.096	.034	.009	.002	.000	.000	.000	.000	.000
	11	1.0	1.0	1.0	.998	.989	.956	.875	.732	.543	.345	.183	.078	.025	.006	.001	.000	.000	.000	.000
	12	1.0	1.0	1.0	1.0	.997	.983	.940	.846	.694	.500	.306	.154	.060	.017	.003	.000	.000	.000	.000
	13	1.0	1.0	1.0	1.0	.999	.994	.975	.922	.817	.655	.457	.268	.125	.044	.011	.002	.000	.000	.000
	14	1.0	1.0	1.0	1.0	1.0	.998	.991	.966	.904	.788	.616	.414	.229	.098	.030	.006	.000	.000	.000
	15	1.0	1.0	1.0	1.0	1.0	1.0	.997	.987	.956	.885	.758	.575	.370	.189	.071	.017	.002	.000	.000
	16	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.983	.946	.866	.726	.533	.323	.149	.047	.008	.000	.000
	17	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.994	.978	.936	.846	.694	.488	.273	.109	.025	.002	.000
	18	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.993	.974	.926	.827	.659	.439	.220	.070	.009	.000
	19	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.991	.971	.917	.807	.622	.383	.162	.033	.001
	20	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.991	.968	.910	.786	.579	.318	.098	.007

Table A3. Poisson distribution

$$F(x) = P\{X \leq x\} = \sum_{k=0}^x \frac{e^{-\lambda} \lambda^k}{k!}$$

x	λ														
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5
0	.905	.819	.741	.670	.607	.549	.497	.449	.407	.368	.333	.301	.273	.247	.223
1	.995	.982	.963	.938	.910	.878	.844	.809	.772	.736	.699	.663	.627	.592	.558
2	1.00	.999	.996	.992	.986	.977	.966	.953	.937	.920	.900	.879	.857	.833	.809
3	1.00	1.00	1.00	.999	.998	.997	.994	.991	.987	.981	.974	.966	.957	.946	.934
4	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.996	.995	.992	.989	.986	.981
5	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.998	.997	.996
6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999
7	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

x	λ														
	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
0	.202	.183	.165	.150	.135	.122	.111	.100	.091	.082	.074	.067	.061	.055	.050
1	.525	.493	.463	.434	.406	.380	.355	.331	.308	.287	.267	.249	.231	.215	.199
2	.783	.757	.731	.704	.677	.650	.623	.596	.570	.544	.518	.494	.469	.446	.423
3	.921	.907	.891	.875	.857	.839	.819	.799	.779	.758	.736	.714	.692	.670	.647
4	.976	.970	.964	.956	.947	.938	.928	.916	.904	.891	.877	.863	.848	.832	.815
5	.994	.992	.990	.987	.983	.980	.975	.970	.964	.958	.951	.943	.935	.926	.916
6	.999	.998	.997	.997	.995	.994	.993	.991	.988	.986	.983	.979	.976	.971	.966
7	1.00	1.00	.999	.999	.999	.999	.998	.997	.997	.996	.995	.993	.992	.990	.988
8	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.999	.999	.998	.998	.997	.996
9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.999	.999
10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

x	λ														
	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0	10.5
0	.030	.018	.011	.007	.004	.002	.002	.001	.001	.000	.000	.000	.000	.000	.000
1	.136	.092	.061	.040	.027	.017	.011	.007	.005	.003	.002	.001	.001	.000	.000
2	.321	.238	.174	.125	.088	.062	.043	.030	.020	.014	.009	.006	.004	.003	.002
3	.537	.433	.342	.265	.202	.151	.112	.082	.059	.042	.030	.021	.015	.010	.007
4	.725	.629	.532	.440	.358	.285	.224	.173	.132	.100	.074	.055	.040	.029	.021
5	.858	.785	.703	.616	.529	.446	.369	.301	.241	.191	.150	.116	.089	.067	.050
6	.935	.889	.831	.762	.686	.606	.527	.450	.378	.313	.256	.207	.165	.130	.102
7	.973	.949	.913	.867	.809	.744	.673	.599	.525	.453	.386	.324	.269	.220	.179
8	.990	.979	.960	.932	.894	.847	.792	.729	.662	.593	.523	.456	.392	.333	.279
9	.997	.992	.983	.968	.946	.916	.877	.830	.776	.717	.653	.587	.522	.458	.397
10	.999	.997	.993	.986	.975	.957	.933	.901	.862	.816	.763	.706	.645	.583	.521
11	1.00	.999	.998	.995	.989	.980	.966	.947	.921	.888	.849	.803	.752	.697	.639
12	1.00	1.00	.999	.998	.996	.991	.984	.973	.957	.936	.909	.876	.836	.792	.742
13	1.00	1.00	1.00	.999	.998	.996	.993	.987	.978	.966	.949	.926	.898	.864	.825
14	1.00	1.00	1.00	1.00	.999	.999	.997	.994	.990	.983	.973	.959	.940	.917	.888
15	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.995	.992	.986	.978	.967	.951	.932
16	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.996	.993	.989	.982	.973	.960
17	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.997	.995	.991	.986	.978
18	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.996	.993	.988
19	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.997	.994
20	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.997

Table A3, continued. Poisson distribution

x	λ														
	11	12	13	14	15	16	17	18	19	20	22	24	26	28	30
0	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
1	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
2	.001	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
3	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
4	.015	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
5	.038	.020	.011	.006	.003	.001	.001	.000	.000	.000	.000	.000	.000	.000	.000
6	.079	.046	.026	.014	.008	.004	.002	.001	.001	.000	.000	.000	.000	.000	.000
7	.143	.090	.054	.032	.018	.010	.005	.003	.002	.001	.000	.000	.000	.000	.000
8	.232	.155	.100	.062	.037	.022	.013	.007	.004	.002	.001	.000	.000	.000	.000
9	.341	.242	.166	.109	.070	.043	.026	.015	.009	.005	.002	.000	.000	.000	.000
10	.460	.347	.252	.176	.118	.077	.049	.030	.018	.011	.004	.001	.000	.000	.000
11	.579	.462	.353	.260	.185	.127	.085	.055	.035	.021	.008	.003	.001	.000	.000
12	.689	.576	.463	.358	.268	.193	.135	.092	.061	.039	.015	.005	.002	.001	.000
13	.781	.682	.573	.464	.363	.275	.201	.143	.098	.066	.028	.011	.004	.001	.000
14	.854	.772	.675	.570	.466	.368	.281	.208	.150	.105	.048	.020	.008	.003	.001
15	.907	.844	.764	.669	.568	.467	.371	.287	.215	.157	.077	.034	.014	.005	.002
16	.944	.899	.835	.756	.664	.566	.468	.375	.292	.221	.117	.056	.025	.010	.004
17	.968	.937	.890	.827	.749	.659	.564	.469	.378	.297	.169	.087	.041	.018	.007
18	.982	.963	.930	.883	.819	.742	.655	.562	.469	.381	.232	.128	.065	.030	.013
19	.991	.979	.957	.923	.875	.812	.736	.651	.561	.470	.306	.180	.097	.048	.022
20	.995	.988	.975	.952	.917	.868	.805	.731	.647	.559	.387	.243	.139	.073	.035
21	.998	.994	.986	.971	.947	.911	.861	.799	.725	.644	.472	.314	.190	.106	.054
22	.999	.997	.992	.983	.967	.942	.905	.855	.793	.721	.556	.392	.252	.148	.081
23	1.00	.999	.996	.991	.981	.963	.937	.899	.849	.787	.637	.473	.321	.200	.115
24	1.00	.999	.998	.995	.989	.978	.959	.932	.893	.843	.712	.554	.396	.260	.157
25	1.00	1.00	.999	.997	.994	.987	.975	.955	.927	.888	.777	.632	.474	.327	.208
26	1.00	1.00	1.00	.999	.997	.993	.985	.972	.951	.922	.832	.704	.552	.400	.267
27	1.00	1.00	1.00	.999	.998	.996	.991	.983	.969	.948	.877	.768	.627	.475	.333
28	1.00	1.00	1.00	1.00	.999	.998	.995	.990	.980	.966	.913	.823	.697	.550	.403
29	1.00	1.00	1.00	1.00	1.00	.999	.997	.994	.988	.978	.940	.868	.759	.623	.476
30	1.00	1.00	1.00	1.00	1.00	.999	.999	.997	.993	.987	.959	.904	.813	.690	.548
31	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.996	.992	.973	.932	.859	.752	.619
32	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.995	.983	.953	.896	.805	.685
33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.989	.969	.925	.850	.744
34	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.994	.979	.947	.888	.797
35	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996	.987	.964	.918	.843
36	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.992	.976	.941	.880
37	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.995	.984	.959	.911
38	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.990	.972	.935
39	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.994	.981	.954
40	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996	.988	.968
41	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.992	.978
42	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.995	.985
43	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.990
44	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.994
45	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996
46	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998
47	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
48	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
49	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table A4. Table of Student's **T-distribution**

t_α ; critical values, such that $P\{t > t_\alpha\} = \alpha$

ν (d.f.)	α , the right-tail probability									
	.10	.05	.025	.02	.01	.005	.0025	.001	.0005	.0001
1	3.078	6.314	12.706	15.89	31.82	63.66	127.3	318.3	636.6	3185
2	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60	70.71
3	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92	22.20
4	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610	13.04
5	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.894	6.869	9.676
6	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959	8.023
7	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408	7.064
8	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041	6.442
9	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781	6.009
10	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587	5.694
11	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437	5.453
12	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318	5.263
13	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221	5.111
14	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140	4.985
15	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073	4.880
16	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015	4.790
17	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965	4.715
18	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922	4.648
19	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883	4.590
20	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850	4.539
21	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819	4.492
22	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792	4.452
23	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768	4.416
24	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745	4.382
25	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725	4.352
26	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707	4.324
27	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.689	4.299
28	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674	4.276
29	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.660	4.254
30	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646	4.234
32	1.309	1.694	2.037	2.141	2.449	2.738	3.015	3.365	3.622	4.198
34	1.307	1.691	2.032	2.136	2.441	2.728	3.002	3.348	3.601	4.168
36	1.306	1.688	2.028	2.131	2.434	2.719	2.990	3.333	3.582	4.140
38	1.304	1.686	2.024	2.127	2.429	2.712	2.980	3.319	3.566	4.115
40	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551	4.094
45	1.301	1.679	2.014	2.115	2.412	2.690	2.952	3.281	3.520	4.049
50	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496	4.014
55	1.297	1.673	2.004	2.104	2.396	2.668	2.925	3.245	3.476	3.985
60	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460	3.962
70	1.294	1.667	1.994	2.093	2.381	2.648	2.899	3.211	3.435	3.926
80	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416	3.899
90	1.291	1.662	1.987	2.084	2.368	2.632	2.878	3.183	3.402	3.878
100	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390	3.861
200	1.286	1.653	1.972	2.067	2.345	2.601	2.838	3.131	3.340	3.789
∞	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.290	3.719

13.3 Calculus review

This section is a very brief summary of Calculus skills required for reading this book.

13.3.1 Inverse function

Function g is the **inverse function** of function f if

$$g(f(x)) = x \text{ and } f(g(y)) = y$$

for all x and y where $f(x)$ and $g(y)$ exist.

$$\text{NOTATION } \parallel \text{ Inverse function } \quad g = f^{-1} \parallel$$

(Don't confuse the inverse $f^{-1}(x)$ with $1/f(x)$. These are different functions!)

To find the inverse function, solve the equation

$$f(x) = y.$$

The solution $g(y)$ is the inverse of f .

For example, to find the inverse of $f(x) = 3 + 1/x$, we solve the equation

$$3 + 1/x = y \Rightarrow 1/x = y - 3 \Rightarrow x = \frac{1}{y - 3}.$$

The inverse function of f is $g(y) = 1/(y - 3)$.

13.3.2 Limits and continuity

A function $f(x)$ has a **limit** L at a point x_0 if $f(x)$ approaches L when x approaches x_0 .

To say it more rigorously, for any ε there exists such δ that $f(x)$ is ε -close to L when x is δ -close to x_0 . That is,

$$\text{if } |x - x_0| < \delta \text{ then } |f(x) - L| < \varepsilon.$$

A function $f(x)$ has a **limit** L at $+\infty$ if $f(x)$ approaches L when x goes to $+\infty$. Rigorously, for any ε there exists such N that $f(x)$ is ε -close to L when x gets beyond N , i.e.,

$$\text{if } x > N \text{ then } |f(x) - L| < \varepsilon.$$

Similarly, $f(x)$ has a **limit** L at $-\infty$ if for any ε there exists such N that $f(x)$ is ε -close to L when x gets below $(-N)$, i.e.,

$$\text{if } x < -N \text{ then } |f(x) - L| < \varepsilon.$$

$$\text{NOTATION} \left\| \begin{array}{l} \lim_{x \rightarrow x_0} f(x) = L, \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow x_0 \\ \lim_{x \rightarrow \infty} f(x) = L, \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow \infty \\ \lim_{x \rightarrow -\infty} f(x) = L, \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow -\infty \end{array} \right\|$$

Function f is **continuous** at a point x_0 if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

Function f is **continuous** if it is continuous at every point.

13.3.3 Sequences and series

A **sequence** is a function of a positive integer argument,

$$f(n) \text{ where } n = 1, 2, 3, \dots$$

Sequence $f(n)$ **converges** to L if

$$\lim_{n \rightarrow \infty} f(n) = L$$

and **diverges** to infinity if for any M there exists N such that

$$f(n) > M \text{ when } n > N.$$

A **series** is a sequence of partial sums,

$$f(n) = \sum_{k=1}^n a_k = a_1 + a_2 + \dots + a_n.$$

Geometric series is defined by

$$a_n = Cr^n,$$

where r is called the **ratio** of the series. In general,

$$\sum_{k=m}^n Cr^k = f(n) - f(m-1) = C \frac{r^m - r^{n+1}}{1-r}.$$

For $m = 0$, we get

$$\sum_{k=0}^n Cr^k = C \frac{1 - r^{n+1}}{1-r}.$$

A geometric series converges if and only if $|r| < 1$. In this case,

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n Cr^k = \frac{Cr^m}{1-r} \quad \text{and} \quad \sum_{k=0}^{\infty} Cr^k = \frac{C}{1-r}.$$

A geometric series diverges to ∞ if $r \geq 1$.

13.3.4 Derivatives, minimum, and maximum

Derivative of a function f at a point x is the limit

$$f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x}$$

provided that this limit exists. Taking derivative is called **differentiation**. A function that has derivatives is called **differentiable**.

NOTATION $\left\| f'(x) \quad \text{or} \quad \frac{d}{dx} f(x) \right\|$

Differentiating a function of several variables, we take **partial derivatives** denoted as

$$\frac{\partial}{\partial x_1} f(x_1, x_2, \dots), \quad \frac{\partial}{\partial x_2} f(x_1, x_2, \dots), \quad \text{etc..}$$

The most important derivatives are:

Derivatives

$$\begin{aligned} (x^m)' &= m x^{m-1} \\ (e^x)' &= e^x \\ (\ln x)' &= 1/x \\ C' &= 0 \\ (f+g)'(x) &= f'(x) + g'(x) \\ (Cf)'(x) &= C f'(x) \\ (f(x)g(x))' &= f'(x)g(x) + f(x)g'(x) \\ \left(\frac{f(x)}{g(x)} \right)' &= \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} \end{aligned}$$

for any functions f and g and any number C

To differentiate a composite function

$$f(x) = g(h(x)),$$

we use a **chain rule**,

Chain rule

$$\frac{d}{dx} g(h(x)) = g'(h(x))h'(x)$$

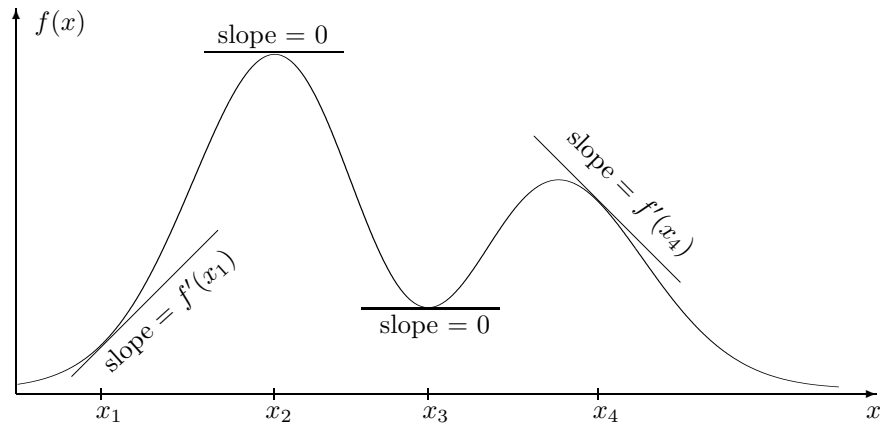


FIGURE 13.1: Derivative is the slope of a tangent line.

For example,

$$\frac{d}{dx} \ln^3(x) = 3 \ln^2(x) \frac{1}{x}.$$

Geometrically, derivative $f'(x)$ equals the slope of a tangent line at point x ; see Figure 13.1.

Computing maxima and minima

At the points where a differentiable function reaches its minimum or maximum, the tangent line is always flat; see points x_2 and x_3 on Figure 13.1. The slope of a horizontal line is 0, and thus,

$$f'(x) = 0$$

at these points.

To find out where a function is maximized or minimized, we consider

- solutions of the equation $f'(x) = 0$,
- points x where $f'(x)$ fails to exist,
- endpoints.

The highest and the lowest values of the function can only be attained at these points.

13.3.5 Integrals

Integration is an action opposite to differentiation.

A function $F(x)$ is an **antiderivative (indefinite integral)** of a function $f(x)$ if

$$F'(x) = f(x).$$

Indefinite integrals are defined up to a constant C because when we take derivatives, $C' = 0$.

$$\text{NOTATION} \quad \left\| \quad F(x) = \int f(x) dx \quad \right\|$$

An **integral** (**definite integral**) of a function $f(x)$ from point a to point b is the difference of antiderivatives,

$$\int_a^b f(x) dx = F(b) - F(a).$$

Improper integrals

$$\int_a^\infty f(x) dx, \quad \int_{-\infty}^b f(x) dx, \quad \int_{-\infty}^\infty f(x) dx$$

are defined as limits. For example,

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

The most important integrals are:

**Indefinite
integrals**

$$\begin{aligned} \int x^m dx &= \frac{x^{m+1}}{m+1} \quad \text{for } m \neq -1 \\ \int x^{-1} dx &= \ln(x) \\ \int e^x dx &= e^x \\ \int (f(x) + g(x)) dx &= \int f(x) dx + \int g(x) dx \\ \int C f(x) dx &= C \int f(x) dx \\ &\text{for any functions } f \text{ and } g \text{ and any number } C \end{aligned}$$

For example, to evaluate a definite integral $\int_0^2 x^3 dx$, we find an antiderivative $F(x) = x^4/4$ and compute $F(2) - F(0) = 4 - 0 = 4$. A standard way to write this solution is

$$\int_0^2 x^3 dx = \frac{x^4}{4} \Big|_{x=0}^{x=2} = \frac{2^4}{4} - \frac{0^4}{4} = 4.$$

Two important integration skills are *integration by substitution* and *integration by parts*.

Integration by substitution

An integral often simplifies when we can denote a part of the function as a new variable (y). The limits of integration a and b are then recomputed in terms of y , and dx is replaced by

$$dx = \frac{dy}{dy/dx} \quad \text{or} \quad dx = \frac{dx}{dy} dy,$$

whichever is easier to find. Notice that dx/dy is the derivative of the *inverse function* $x(y)$.

**Integration
by substitution**

$$\int f(x) dx = \int f(x(y)) \frac{dx}{dy} dy$$

For example,

$$\int_{-1}^2 e^{3x} dx = \int_{-3}^6 e^y \left(\frac{1}{3}\right) dy = \frac{1}{3} e^y \Big|_{y=-3}^{y=6} = \frac{e^6 - e^{-3}}{3} = 134.5.$$

Here we substituted $y = 3x$, recomputed the limits of integration, found the inverse function $x = y/3$ and its derivative $dx/dy = 1/3$.

In the next example, we substitute $y = x^2$. Derivative of this substitution is $dy/dx = 2x$:

$$\int_0^2 x e^{x^2} dx = \int_0^4 x e^y \frac{dy}{2x} = \frac{1}{2} \int_0^4 e^y dy = \frac{1}{2} e^y \Big|_{y=0}^{y=4} = \frac{e^4 - 1}{2} = 26.8.$$

Integration by parts

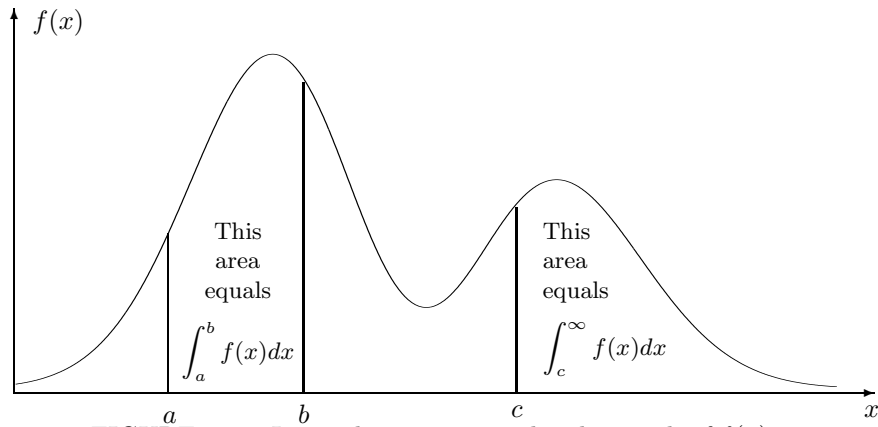
This technique often helps to integrate a *product* of two functions. One of the parts is integrated, the other is differentiated.

**Integration
by parts**

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx$$

Applying this method is reasonable only when function (fg') is simpler for integration than the initial function $(f'g)$.

In the following example, we let $f'(x) = e^x$ be one part and $g(x) = x$ be the other. Then

FIGURE 13.2: Integrals are areas under the graph of $f(x)$.

$f'(x)$ is integrated, and its antiderivative is $f(x) = e^x$. The other part $g(x)$ is differentiated, and $g'(x) = x' = 1$. The integral simplifies, and we can evaluate it,

$$\int x e^x dx = x e^x - \int (1)(e^x) dx = x e^x - e^x.$$

Computing areas

Area under the graph of a positive function $f(x)$ and above the interval $[a, b]$ equals the integral,

$$(\text{area from } a \text{ to } b) = \int_a^b f(x) dx.$$

Here a and b may be finite or infinite; see Figure 13.2.

Gamma function and factorial

Gamma function is defined as

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx \quad \text{for } t > 0.$$

Taking this integral by parts, we obtain two important properties of a Gamma function,

$$\begin{aligned} \Gamma(t+1) &= t\Gamma(t) \quad \text{for any } t > 0, \\ \Gamma(t+1) &= t! = 1 \cdot 2 \cdot \dots \cdot t \quad \text{for integer } t. \end{aligned}$$